Aldehydes and Ketones

DISCLAIMER

"The content provided herein are created and owned by various authors and licensed to Sorting Hat Technologies Private Limited ("Company"). The Company disclaims all rights and liabilities in relation to the content. The author of the content shall be solely responsible towards, without limitation, any claims, liabilities, damages or suits which may arise with respect to the same."

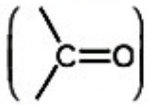
Know about Carbonyl Compounds

Introduction

Aldehyde and ketones, both have the general formula, $C_nH_{2n}O$

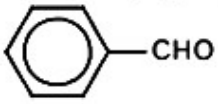
Aldehyde are obtained by replacing the two terminal hydrogen atoms of alkanes by the oxygen atom while ketones are obtained by replacing the two non-terminal hydrogen atoms of alkanes by oxygen. Due to this, aldehyde group becomes chain terminating group and ketonic group occurs within the chain.

Points to remember

The common feature of both is carbonyl  group and, therefore, they are called carbonyl compounds. Aldehydes and ketones are the first oxidation products of primary and secondary alcohols respectively.

Aldehydes

In aldehydes, –CHO group may be attached to a hydrogen atom or alkyl group or aryl group.

Example: HCHO; CH_3CHO; 

Ketone

In ketones,  group may be attached to two alkyl groups, two aryl groups or one alkyl and one aryl group. If both the groups are same, they are simple ketones and if they are different, they are mixed ketones.

Example

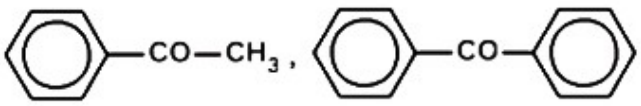
Simple ketones (or) symmetrical ketones: R–CO–R

$CH_3–CO–CH_3$, $C_2H_5–CO–C_2H_5$.

Mixed ketones (or) unsymmetrical ketones: R–CO–R'

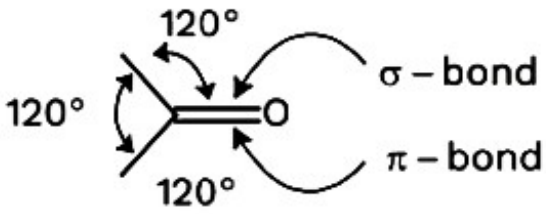
$CH_3–CO–C_2H_5$, $CH_3–CO–C_3H_7$

Aromatic ketones: Ar–CO–R



Nature of Carbonyl Group

Carbonyl compound has sp^2 hybrid orbitals and one unhybridized p-orbital.
One sigma bond and one pi bond including planer shape is suggested.
120° degree bond angle is suitable for nucleophilic upside and downside attack.

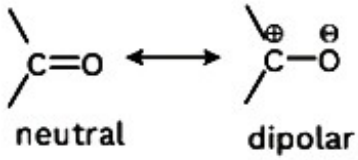


Higher dipole moment of C=O bond is responsible for higher polarization power of carbonyl compound.

In carbonyl compounds carbon act as nucleophilic centre (Lewis acid) and oxygen act as nucleophile (Lewis base).

Points to remember

Resonating structure of C=O bond act as perfect dipole for explaining nature of C=O bond, that's why carbonyl act as nucleophilic centre.

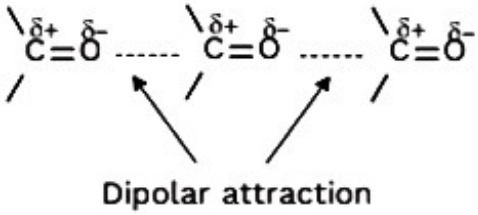


3. Physical properties

Aldehydes are colourless with pungent smell liquid while ketones are pleasant smell liquids, but formaldehyde is gaseous in nature. Lower carbonyl compounds are soluble in water it is due to polarity in carbonyl group.

Solubility

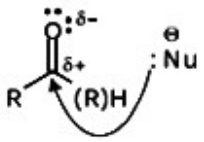
More covalent character in C=O bond is key reason for low solubility in water. Boiling points of both the aldehydes and ketones have somewhat higher than predicted from their molecular formulae. This is because of the dipolar attraction.



Boiling Point

Compare to alcohols carbonyl compounds have lower boiling point because alcohols form H-bonding. Among the isomeric aldehydes and ketones, the later have slightly higher boiling points because they are more polar.

- **Relative Reactivities of Carbonyl Compounds**



HCHO > RCHO > RCOR > PhCHO > $PhCOCH_3$ > PhCOPh

Points to remember

Greater negative share with carbon in C=O bond is due to polarization in carbonyl bond , attack is more feasible clearly indicate electrophilic and nucleophilic nature of respective C and O atoms.

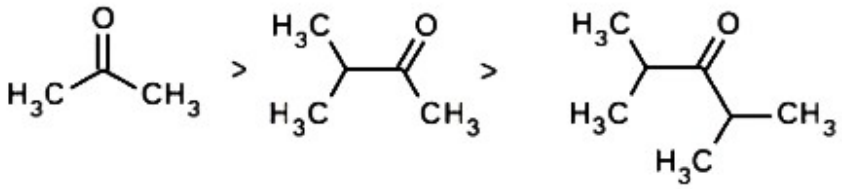
Reactivity

If we compare aldehyde and ketone , aldehyde is more reactive, alkyl group in ketone exert more inductive effect due to which partial negative charge decreases and act as less susceptible for nucleophilic addition reaction.

Q1 **Explain role of steric factor in reactivity of carbonyl compound.**

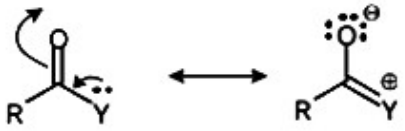
Sol. In aldehyde compare to ketone, H atom is less steric more suspectable for nucleophilic attack. Aldehyde is more reactive due to more electropositive nature of C=O bond and size of carbonyl compound.

- **Relative Reactivities towards Nucleophilic addition**



Q2 **Explain rate of nucleophilic substitution for given sequence.**
Acyl Halides > Acid Anhydride > Ester > Amide

Sol. The carbonyl compounds have a pair of non-bonding electrons on an atom attached to the carbonyl group, these compounds (acid derivatives) has a resonance contributor that aldehydes and ketones do not have. This resonance contributor places some of the positive charge on the atom adjacent to the carbonyl carbon. We have seen that the reactivity of these carbonyl compounds is related to the basicity of the Y^-. The weaker the basicity of Y^-, the more reactive the carbonyl group, because weak bases are better able to withdraw electrons inductively from the carbonyl carbon and are less apt to donate electrons by resonance to the carbonyl carbon.



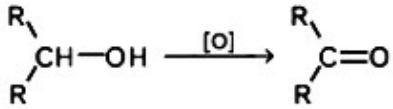
So, aldehydes and ketones are not as reactive as carbonyl compounds in which Y^- is a very weak base (acyl halides and acid anhydrides) but are more reactive than carbonyl compounds in which Y^- is relatively stronger base (carboxylic acids, ester and amides).

General methods of Preparation:

(i) Oxidation of alcohols

1° alcohol on oxidation by using PCC or PDC gives aldehydes and 2° alcohols gives ketone on oxidation by $K_2Cr_2O_7$

$$R-CH_2-OH \xrightarrow{[O]} R-CH=O$$



Note

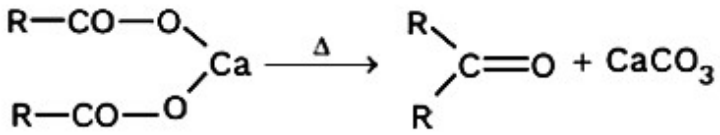
If acidified $K_2Cr_2O_7$ or $KMnO_4$ is used then the aldehydes are further oxidised to give carboxylic acid. It should be remembered that only those alcohols can be oxidised which have got at least one α-H.

(ii) By dry distillation of calcium salts of carboxylic acid:

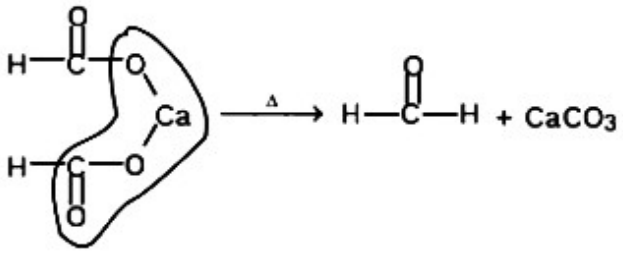
When calcium salts of acids are subjected to dry distillation, then carbonyl compounds are formed. The yield increases on taking Barium Manganese and Thorium salts of carboxylic acids.

Tips and Trick

Dicarboxylic acid gives off CO_2 gas when heated , calcium salt dicarboxylic acid when heated gives $CaCO_3$ and ketone or aldehyde.



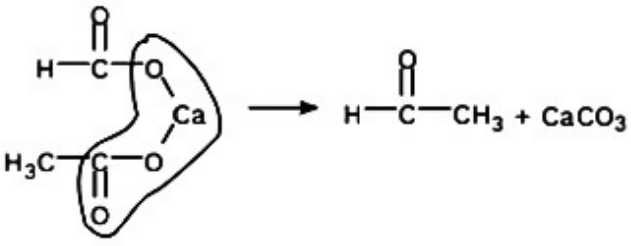
Formaldehyde is formed on taking calcium formate (R=H)



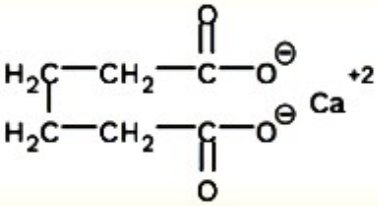
Acetone is formed on taking calcium acetate ($R=CH_3$)

Q3 **Acetaldehyde can be formed by taking a mixture of calcium formate and calcium acetate**

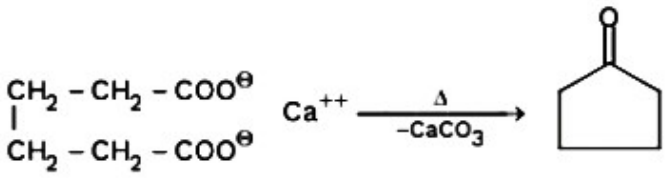
Sol.



Q4

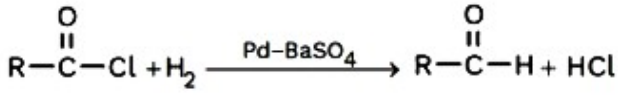


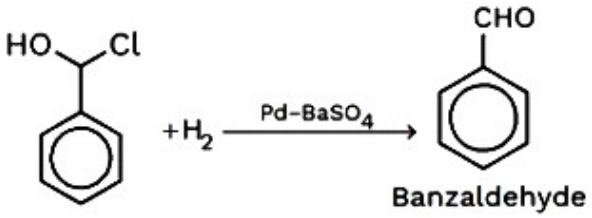
Sol.



(iii) Reduction of acid chloride. (Resenmund reaction)

Acid chloride when reduced with H_2/Pd in presence of $BaSO_4$ poisoned with sulphur in order to deactivate the catalyst Pd, so that the aldehyde formed in the reaction is prevented from further reduction to alcohol.

e.g. 

e.g. 

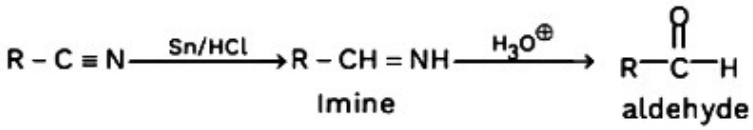
Banzaldehyde

Points to remember

Formaldehyde cannot be prepared by this method because formyl chloride is unstable at normal temperature Ketones cannot be prepared by this method.

(iv) Stephen's Reaction:-

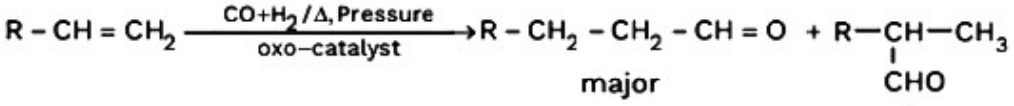
When alkyl cyanide is reduced in presence of Sn/HCl we get imine which upon acidified hydrolysis yields aldehydes.



Imine aldehyde

(v) Oxo process :

Alkenes when reacted with water gas (CO + H_2) at higher temperature and pressure in the presence of oxo-catalyst $[Co(CO)_4]_2$ we get aldehydes.



major minor

(vi) Dialkyl cadmium with acid chloride:

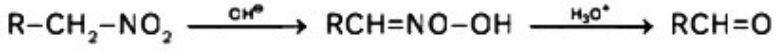
One of the most valuable method for the synthesis of ketones.

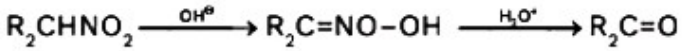




[Here: R' = Ar or 1° alkyl]

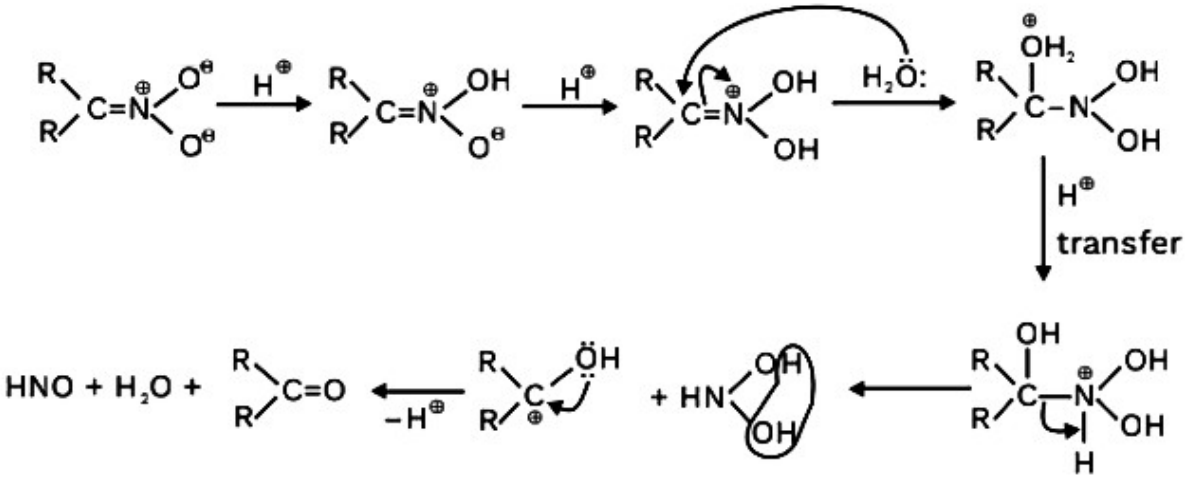
(vii) Oxidation of nitroalkanes: (Nef reaction)



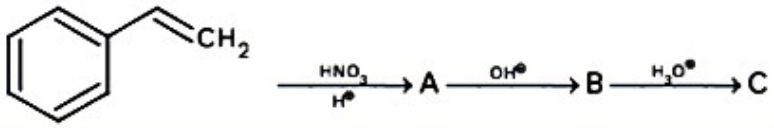


By nef reaction formaldehyde can be obtained from nitromethane, acetaldehyde from nitroethane and acetone from 2-nitropropane.

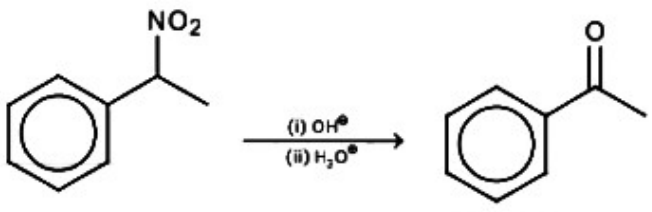
Mechanism



Q5



Sol.



Ketone

(viii) From alkynes

(a) Hydration of alkynes: It is an addition of water in presence of metal ion. Hence metal ion increases the reactivity of alkyne for addition reaction, preparation of carbonyl compounds from alkyne depends upon the –R part

Ex. $R-C\equiv CH \xrightarrow[H_2O]{Hg^{2+}/H_2SO_4} R-\overset{O}{\overset{\|}{C}}-CH_3$

Ex. $CH_3-C\equiv CH \xrightarrow[H_2O]{Hg^{2+}/H_2SO_4} H_3C-\overset{O}{\overset{\|}{C}}-CH_3$

(b) Hydroboration of alkynes:

Ex. $R-C\equiv CH \xrightarrow[(ii)\ H_2O_2/OH^-]{(i)\ B_2H_6} R-CH_2CH=O$

Ex. $CH_3-CH_2-C\equiv C-CH_2-CH_3 \xrightarrow[2.H_2O_2/OH^-]{1.BH_3,THF} CH_3-CH_2-\overset{O}{\overset{\|}{C}}-CH_2-CH_2-CH_3$

(ix) From alkenes:

(a) Wacker process: When an alkene is treated with an acidified aq. solution of $PdCl_2$ and cupric chloride, aldehydes or ketones can be obtained.

Ex. $CH_2=CH_2 + PdCl_2 + H_2O \xrightarrow[H^\oplus]{CuCl_2} CH_3CHO + Pd + 2HCl$

Ex. $CH_3-CH=CH_2 + PdCl_2 + H_2O \xrightarrow[H^\oplus]{CuCl_2} CH_3COCH_3 + Pd + 2HCl$

(b) Reductive ozonolysis of alkenes: It is used to get carbonyl compounds from alkene

Ex. $R-CH=CH-R \xrightarrow[H_2O]{O_3/Zn} 2RCHO + H_2O$

Ex. $CH_3-CH=CH-CH_3 \xrightarrow[H_2O]{O_3/Zn} 2CH_3CHO + H_2O$

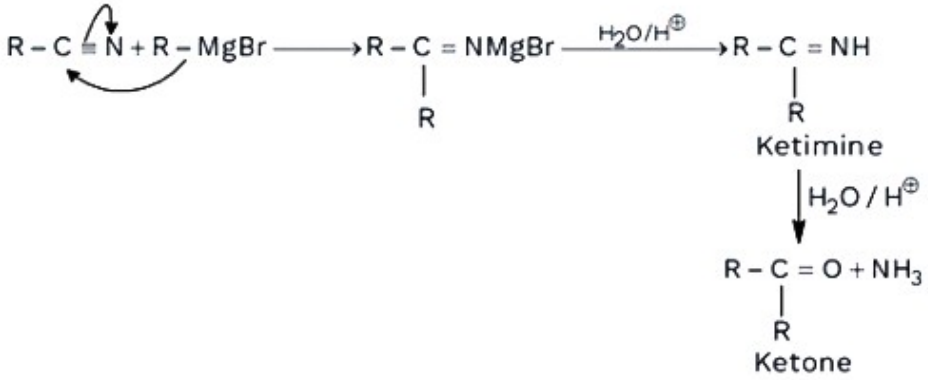
(x) From Grignard reagent:

Grignard reagents give aldehyde with hydrogen cyanide and ketones with alkyl cyanide.

Ex. $H-C\equiv N + R-MgBr \longrightarrow \underset{\underset{R}{|}}{H-C=NMgBr} \xrightarrow{H_2O/H^\oplus} \underset{\underset{R}{|}}{H-C=NH} \xrightarrow{H_2O/H^\oplus} \underset{\text{aldehyde}}{R-CHO} + NH_3$

Aldimine

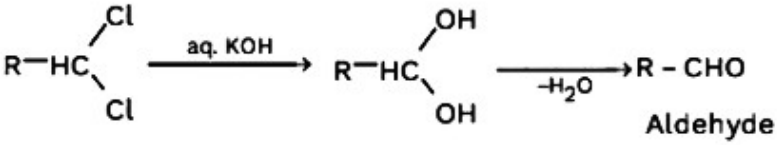
Ex.



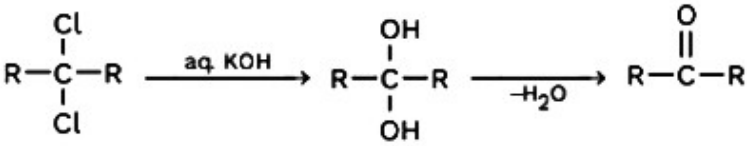
(xi) From gemdihalides:

Terminal gemdihalides will give aldehyde while non-terminal gemdihalides will give ketones.

Ex.



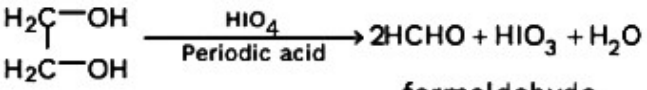
Ex.



(xii) From glycol:

(a) By reaction with HIO_4:

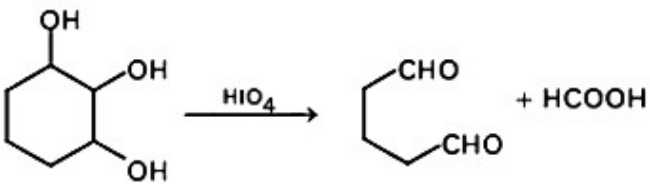
Ex.



Ex.

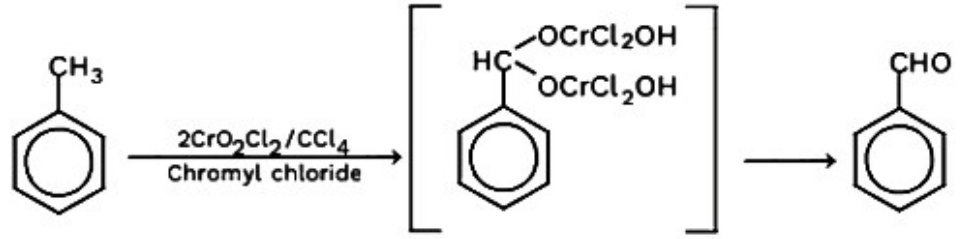


Ex.



(xiii) From Étard's Reaction:

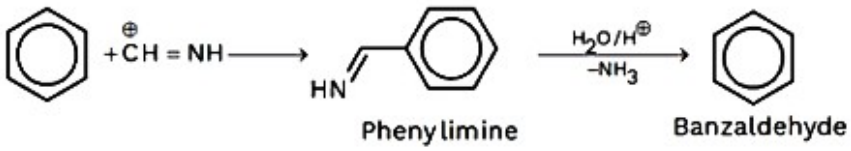
Ex.



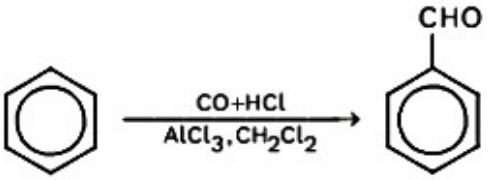
(xiv)

(a) Gattermann reaction:

$$H-C\equiv N+HCl \xrightarrow{AlCl_3} ClCH=NH$$

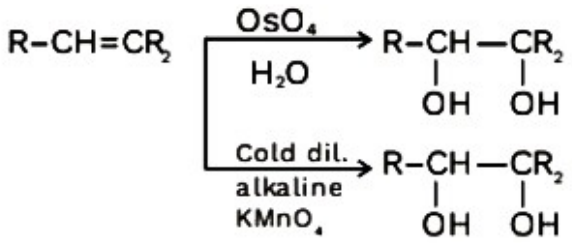


(b) Gattermann–Koch reaction: When benzene is treated carbon monoxide and hydrogen chloride in presence of anhydrous aluminium chloride and methylene chloride gives benzaldehyde



OXIDATION OF ALKENES, ALCOHOLS & CARBONYL COMPOUNDS

(I) OXIDATION OF ALKENES



If we use acidic $KMnO_4$ or warm $KMnO_4$ or too concentrated $KMnO_4$ the oxidative cleavage of Glycol occurs resulting in mixture of Carboxylic acids & Ketones.

$$R-CH=CR_2 \xrightarrow[\Delta]{H^{\oplus},KMnO_4} RCOOH + R_2C=O$$

Points to remember

Cold dil. alkaline $KMnO_4$ is called as Bayer's reagent.
Overall syn addition
Given by alkenes & alkynes
Benzene & Cyclopropane cannot give this reaction.

Hot acidic $KMnO_4$, Hot acidic $K_2Cr_2O_7$ & hot acidic $NaIO_4$ gives same result with alkene. The effect is similar to that of oxidative ozonolysis on alkenes.

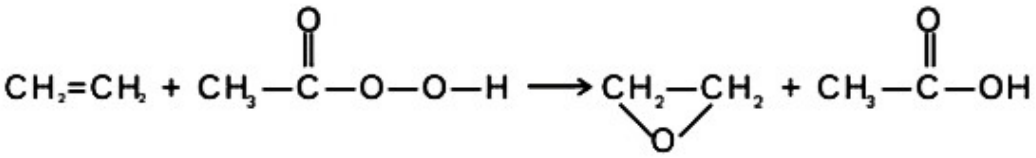
Points to remember

- With the decrease in nucleophilicity of double bond, rate of reaction decreases.
- With the decrease in $e^{\ominus}$ withdrawing substituents in leaving group, rate decreases.

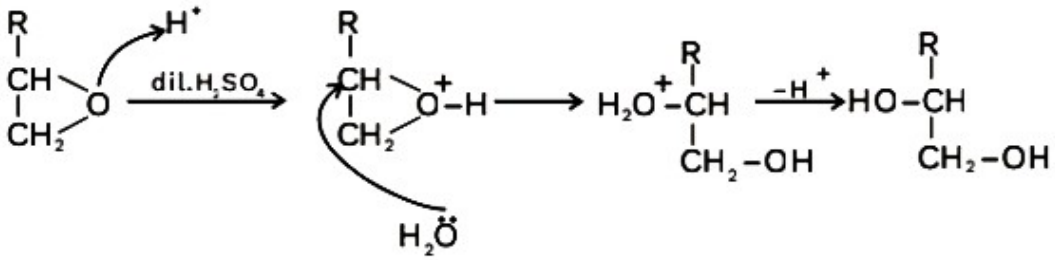
Prelischaive reaction :

Epoxidation of alkenes is reaction of alkenes with peroxyacids.

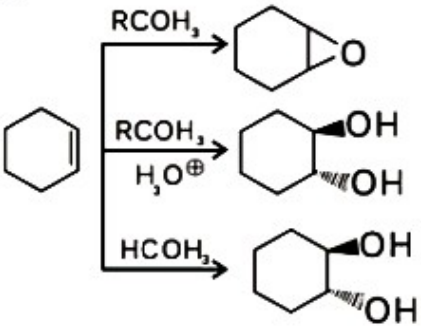
Reaction:



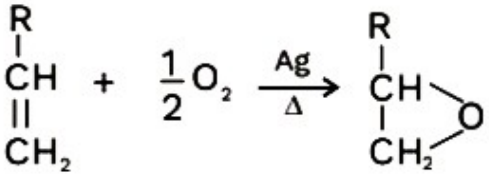
Mehanism:



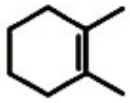
Example-1 :



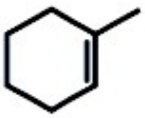
Example-2 :

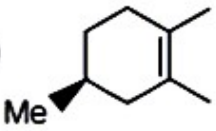
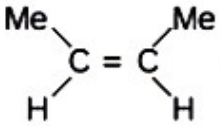


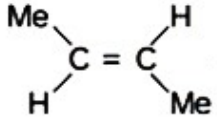
Q6

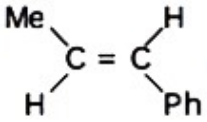
(i)  1% alkaline $KMnO_4$ → (A)

 mCPBA\hydrolysis → (B)

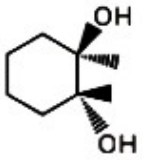
(ii)  mCPBA →

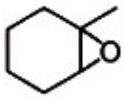
(iii)  (1) mCPBA (2) hydrolysis → (iv)  (1) mCPBA (2) hydrolysis →

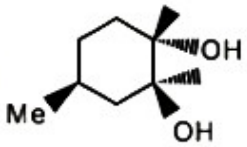
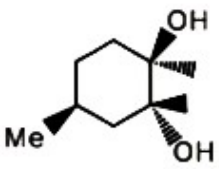
(v)  (1) mCPBA (2) hydrolysis →

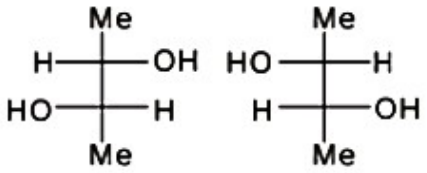
(vi)  Ag_2O or $2Ag + \frac{1}{2}O_2$, Δ →

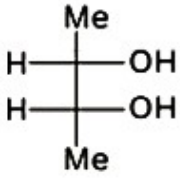
Sol.

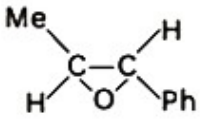
(i) (A)  (B) 

(ii) 

(iii)  

(iv) 

(v) 

(vi) 

Strong Oxidation of Pi Bond

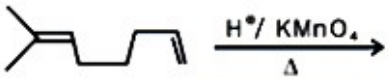
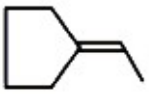
If we use acidic $KMnO_4$ or warm $KMnO_4$ or too concentrated $KMnO_4$ the oxidative cleavage of Glycol occurs resulting in mixture of Carboxylic acids & Ketones.

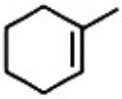
$$R-CH=CR_2 \xrightarrow[\Delta]{H^{\oplus},KMnO_4} RCOOH + R_2C=O$$

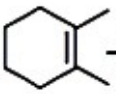
Tips and Trick

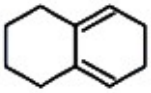
Hot acidic $KMnO_4$, Hot acidic $K_2Cr_2O_7$ & hot acidic $NaIO_4$ gives same result with alkene. The effect is similar to that of oxidative ozonolysis on alkenes.

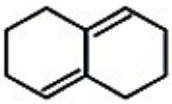
Q7 (i) $CH_2=CH_2 \xrightarrow[\Delta]{H^{\oplus}/KMnO_4}$ (ii) $CH_3-CH=CH_2 \xrightarrow[\Delta]{H^{\oplus}/KMnO_4}$

(iii)  $\xrightarrow[\Delta]{H^{\oplus}/KMnO_4}$ (iv)  $\xrightarrow[\Delta]{H^{\oplus}/KMnO_4}$

(v)  $\xrightarrow[\Delta]{H^{\oplus}/KMnO_4}$ (vi)  $\xrightarrow[\Delta]{H^{\oplus}/KMnO_4}$

(vii)  $\xrightarrow[\Delta]{H^{\oplus}/KMnO_4}$ (viii)  $\xrightarrow[\Delta]{H^{\oplus}/KMnO_4}$

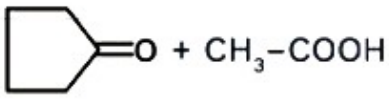
(ix)  $\xrightarrow[\Delta]{H^{\oplus}/KMnO_4}$ (x)  $\xrightarrow[\Delta]{H^{\oplus}/KMnO_4}$

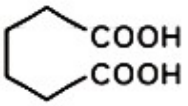
(xi)  $\xrightarrow[\Delta]{H^{\oplus}/KMnO_4}$

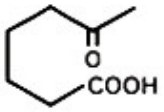
(xii) $C_{10}H_{18} \xrightarrow[\Delta]{H^{\oplus}/KMnO_4} HOOC-CH_2-CH_2-CH_2-\underset{\displaystyle CH_2-COOH}{\underset{|}{CH}}-COOH$

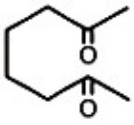
Sol. (i) CO_2 (ii) $CH_3-COOH + CO_2$

(iii)  $+ CO_2 + HOOC-CH_2-CH_2-CH_2-COOH$

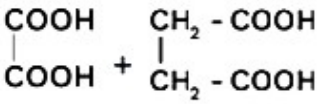
(iv) 

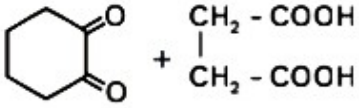
(v) 

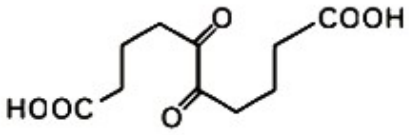
(vi) 

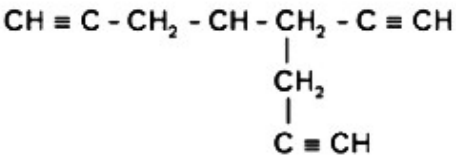
(vii) 

(viii) $HOOC-CH_2-COOH$

(ix) 

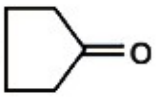
(x) 

(xi) 

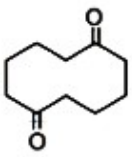
(xii) 

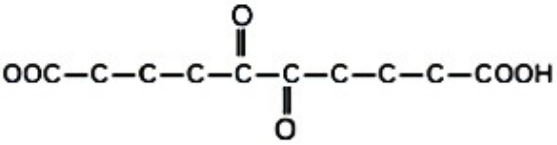
Q8 **A to F alkenes with minimum possible carbon.**

(i) A $\xrightarrow[\Delta]{H^{\oplus}/KMnO_4}$ MeCOOH as the only product

(ii) B $\xrightarrow[\Delta]{H^{\oplus}/\ KMnO_4}$ 

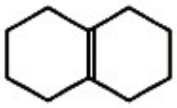
(iii) C $\xrightarrow[\Delta]{H^{\oplus}/KMnO_4}$ $MeCH_2COOH$ as the one of the organic product

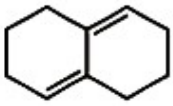
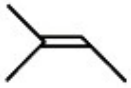
(iv) D $\xrightarrow[\Delta]{H^{\oplus}/\ KMnO_4}$ 

(v) E $\xrightarrow[\Delta]{H^{\oplus}/\ KMnO_4}$ 

(vi) F acetone + ethanoic acid

Sol. (i) $Me-CH=CH-CH_3$ (ii) 

(iii) $Me-CH_2-CH=CH_2$ (iv) 

(v)  (vi) 

(II) OXIDATION OF ALCOHOLS

Different cases in case of oxidation of alcohol

(I) $R-\underset{2°\text{alcohol}}{CH_2-OH} \xrightarrow[\text{agent}]{\text{mild oxidising}} R-\underset{\underset{O}{\|}}{C}-H$ (Aldehyde)

(II) $R-\underset{1°\text{alcohol}}{\overset{\overset{OH}{|}}{CH}-R'} \xrightarrow[\text{agent}]{\text{mild oxidising}} R-\overset{\overset{O}{\|}}{C}-R'$ (Ketone)

(III) $R-\underset{1°\text{alcohol}}{CH_2-OH} \xrightarrow[\text{agent}]{\text{Strong oxidising}} R-\underset{\underset{O}{\|}}{C}-H$

(IV) $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-OH \xrightarrow[\text{agent}]{Cu\ 300°C} \underset{CH_3}{\overset{CH_3}{\diagdown}}\!\!>\!\!C=CH_2$ Dehydration takes place.

Definitions

Oxidising agents

Alcohol are oxidized to convert into carbonyl or carboxylic acid. Oxidation depend upon number of hydrogen atoms present 1 or 2

Reagents used in case of oxidation of alcohol

(1) Cu / 300°C (or Red hot Cu tube)

Properties

Oxidation of alcohol 1 degree in acid , 2 degree in ketone still you can't perform oxidation in case of 3 degree alcohol.

(2) $H^{\oplus}/KMnO_4$, Δ (Strong oxidising agent)

Properties

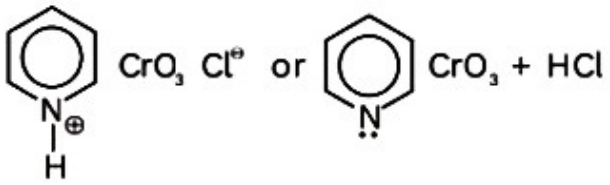
(i) Alcohol oxidation are same as Cu / 300°C (or Red hot Cu tube).
(ii) Oxidation of alkene or alkyne allowed.
(iii) Depend upon number of pi bond.
(iv) It affects carbon-carbon multiple bond and gives carboxylic acid.

(3) $H^{\oplus}/K_2Cr_2O_7$, Δ (Strong oxidising agent)

Properties

(i) It oxidises 1° alcohol into carboxylic acid.
(ii) It oxidises 2° alcohol into ketone.
(iii) It does not affect 3° alcohol.
(iv) It affects carbon-carbon multiple bond and give carboxylic acid.

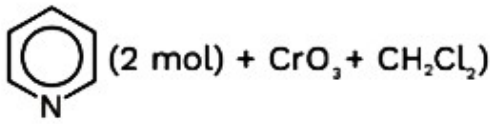
(4) PCC (Pyridinium chloro chromate)



Points to remember

- Further oxidation of aldehyde not possible, aldehyde is final product.
- 2° alcohol changes into ketone, does not oxidises multiple bond not even 3 degree alcohol.
- Oxidation of allylic primary alcohol to corresponding aldehyde.

5) Collin's reagent



Properties

(i) It oxidises 1° alcohol into aldehyde
(ii) It oxidises 2° alcohol into ketone
(iii) Multiple bond is not affected
(iv) 3° alcohol is not affected

(6) Jones reagent (H_2CrO_4 in Anhydrous acetone) or CrO_3 + H_2SO_4 in acetone.
Sufficiently mild so that it oxidises alcohols without oxidising or rearranging double bonds.

Properties
(i) It oxidised 1° alcohol into carboxylic acid
(ii) It oxidised 2° alcohol into ketone
(iii) Multiple bond is not affected
(iv) 3° alcohol is not affected

Q9 (i) $CH_3-CH_2-CH_2-OH \xrightarrow[\text{or } KMnO_4/H^+,\Delta]{KMnO_4/OH^-,\Delta} ?$

$CH_3-CH_2-CH_2-OH \xrightarrow[\text{or conc. } HNO_3,\Delta]{K_2Cr_2O_7/H^+,\Delta} ?$

(ii) $CH_3-\underset{\displaystyle OH}{\underset{|}{CH}}-CH_2-CH_3 \xrightarrow[\text{or } K_2Cr_2O_7,\ H^+]{KMnO_4,\ H^+} ?$

Sol. (i) $CH_3-CH_2-COO^{\ominus}$, CH_3-CH_2-COOH

(ii) $CH_3-\overset{\displaystyle O}{\overset{\|}{C}}-CH_2-CH_3$

(7) TsCl + DMSO + $NaHCO_3$

$RCH_2 \xrightarrow{Ts\text{-}Cl} RCH_2OTs \xrightarrow[NaHCO_3]{DMSO} RCHO$

$R_2CHOH \xrightarrow{Ts\text{-}Cl} R_2CH-OTs \xrightarrow[NaHCO_3]{DMSO} R_2CO$

$R_3COH \xrightarrow{Ts\text{-}Cl} R_3C-OTs \xrightarrow[NaHCO_3]{DMSO} \times$

(8) MnO_2 Oxidises only allylic or benzylic–OH.i.e.

1° Allylic or benzylic OH $\xrightarrow{MnO_2}$ Aldehyde

2° Allylic or benzylic OH $\xrightarrow{MnO_2}$ Ketone

No effect on 3° ROH and on Carboncarbon multiple bond.

Tips and Trick

Tips and Tricks
(i) It oxidised 1° alcohol into aldehyde
(ii) It oxidised 2° alcohol into ketone
(iii) Multiple bond is not affected
(iv) 3° alcohol is not affected

Q10 Which one of the following alcohols are oxidised by MnO_2?

(A) $C_6H_5 - CH_2 - CH_2-OH$

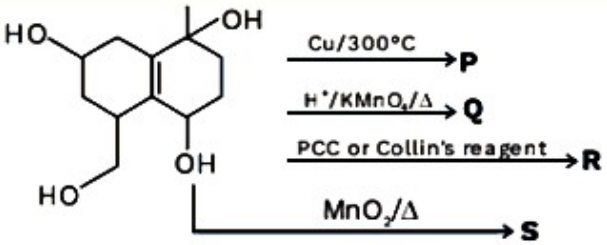
(B) $CH_2=CH-CH_2-\underset{\displaystyle OH}{\underset{|}{CH}}-CH_3$

(C) $CH_3-CH=CH-\underset{\displaystyle OH}{\underset{|}{CH}}-CH_3$

(D) $CH_3-CH_2 - CH_2 -OH$

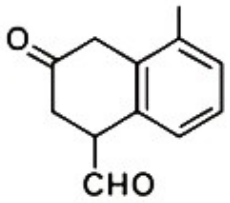
Sol. (C) Alkyl positioned OH is oxidizable.

Q11

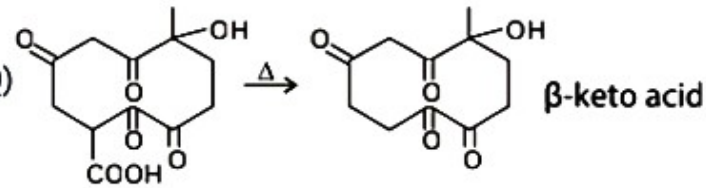


Sol.

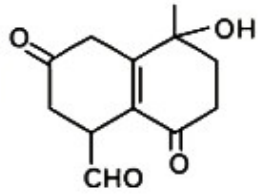
(P)



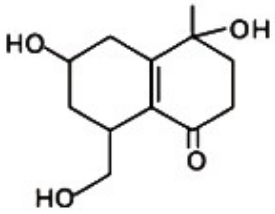
(Q)



(R)



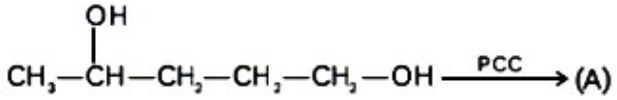
(S)

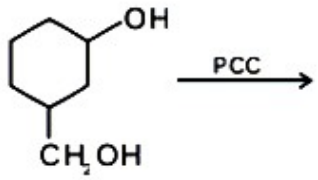


Q12

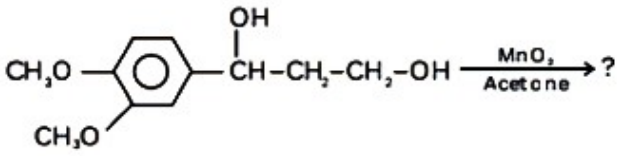
(i) $CH_2 = CH - (CH_2)_3 - CH_2 - OH \xrightarrow{PCC}$

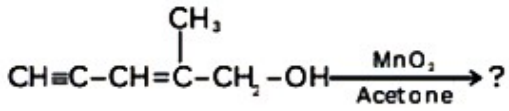
(ii) $C_6H_5 - CH = CH - CH_2 - OH \xrightarrow{PCC}$

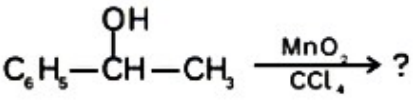
(iii) 

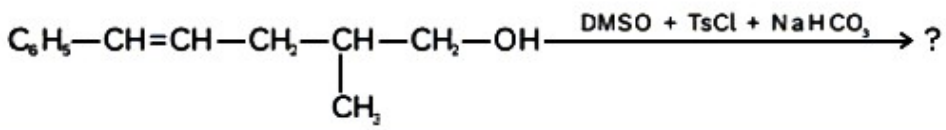
(iv) 

(v) $CH_2 = CH - CH_2 - OH \xrightarrow{MnO_2} ?$

(vi) 

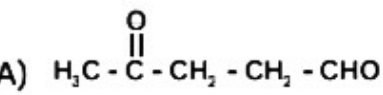
(vii) 

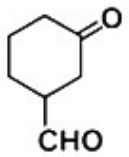
(viii) 

(ix) 

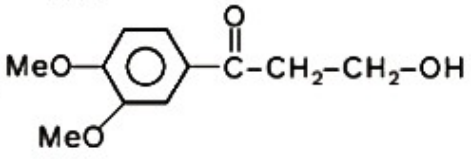
Sol.

(i) $CH_2 = CH - (CH_2)_3 - CHO$ (ii) $Ph - CH = CH - CHO$

(iii) (A) 

(iv) 

(v) $H_2C = CH-CHO$

(vi) 

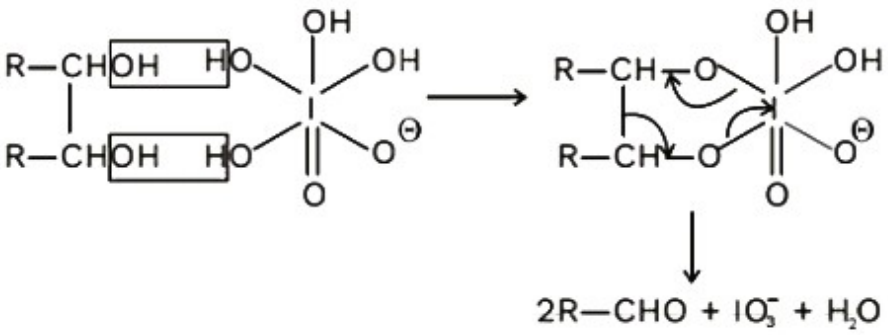
(vii) $CH \equiv C - CH = \overset{\overset{CH_3}{|}}{C} - CHO$

(viii) $Ph - \overset{\overset{O}{\|}}{C} - CH_3$

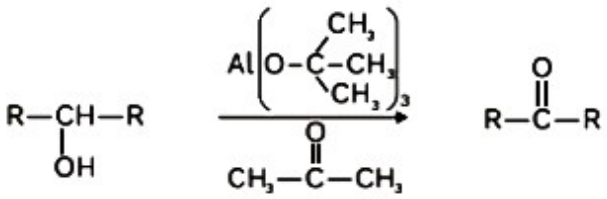
(ix) $Ph - CH = CH - CH_2 - \overset{\overset{CH_3}{|}}{CH} - CHO$

(9) Periodic cleavage

- Reagents Used -H_5IO_6 or HIO_4. $2H_2O$ (α-hydroxy carbonyl compounds, 1,2-diketones is key product)
- Oxidation of adjacent OH bond due to periodic cleavage is formation of C=O bond.

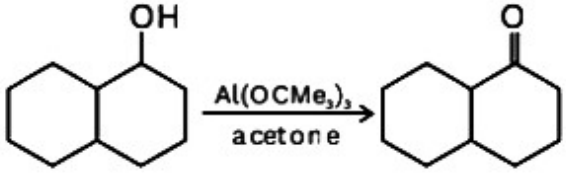


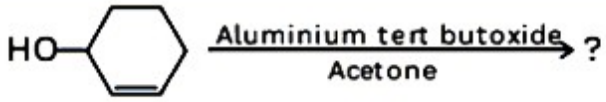
(10) Openaur oxidation

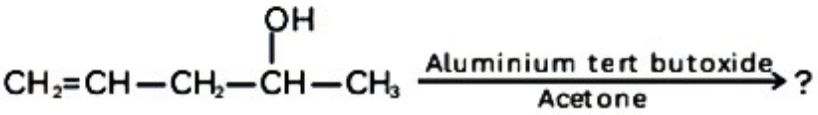


Oxidation of alcohol with aluminium tertiary butoxide is Openaur oxidation.

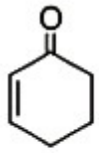
Example :

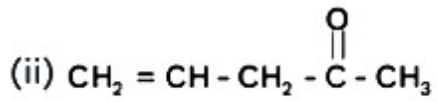


Q13 (i) 

(ii) 

Sol.

(i) 

(ii) 

(11) NBS

It oxidises 1° alcohol into aldehyde.
It oxidises 2° alcohol into ketone.
3° alcohol is not affected
In case of alkene allylic position is brominated.

(III) OXIDATION OF CARBONYL COMPOUNDS.

1. $RCHO + [Ag(NH_3)_2]OH \xrightarrow{\Delta} RCOOH + NH_3 + Ag \downarrow$

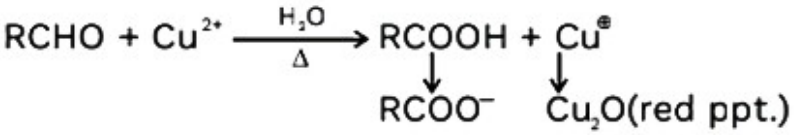
Oxidation of aldehyde is possible due to tollens reagent . Tollen's test require gentle heating for 20 to 25 mins.

2. Fehling's Solutions

Fehling's A	**Fehling's B**
aq. $CuSO_4$	Alk. solution of Roschelye salt (sodium potassium tartrate) It acts a carrier for Cu^{2+} as it make reversible complex with Cu^{2+}

This test is also used in Blood and Urine test.

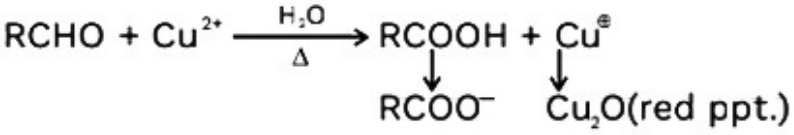
Example :



3. Benedict's solution

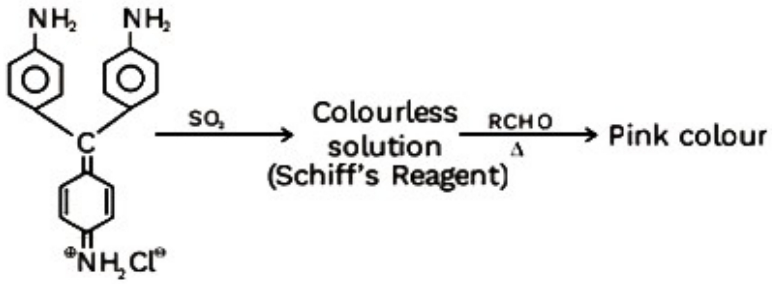
Sodium Citrate + NaOH + $NaHCO_3$ + $CuSO_4$

Example :



4. Schiff's Reagent

Schiff's Reagent is aq. solution of following base decolourised by passing SO_2. Aldehyde restore pink colour of Schiff's reagent.

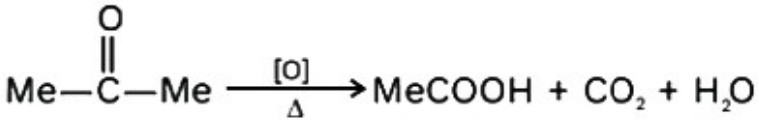


p-Rosaniline Hydrochloride
Magenta colour (Fuschin)

Let's understand

Ketons are not easy to oxidize so they do not give these 5 tests. These five tests can be used to distinguish aldehyde and ketones. Both gives 2,4 DNP test

Ketones are difficult to Oxidize

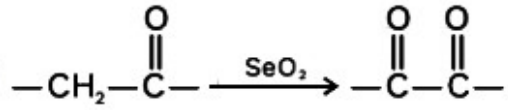


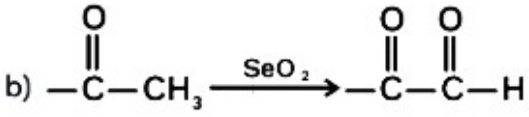
Oxidation by using SeO_2

SeO_2 is a selective oxidizing agent with converts $-CH_2-$ group adjacent to carbonyl group into carbonyl group. The reagent, in general, oxidises active methylene and methyl groups to ketonic and aldehydic groups respectively.

Points to remember

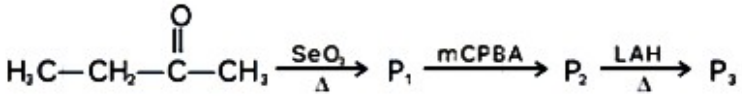
Popoff's rule states that oxidation of carbonyl compound is according electron rich part of alkene. Lower part of alkene will be remain with carbonyl part of oxidation while more electron donating group / part will be oxidized.

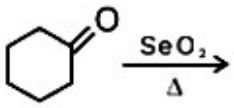
a) 

b) 

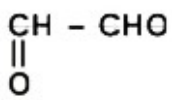
Q14 (a) $CH_3 - CHO \xrightarrow[\Delta]{SeO_2}$

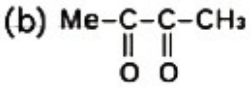
(b) $Me_2CO \xrightarrow[\Delta]{SeO_2}$

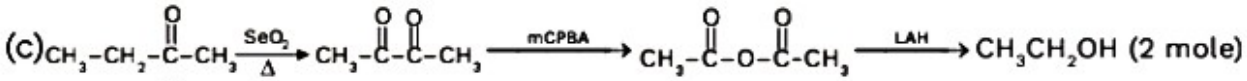
(c) 

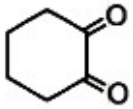
(d) 

Sol.

(a) 

(b) 

(c) 

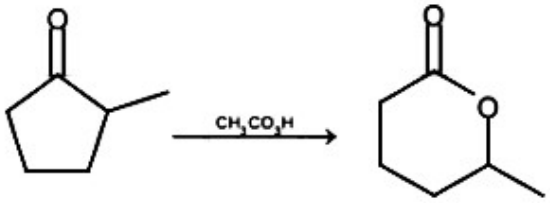
(d) 

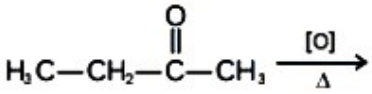
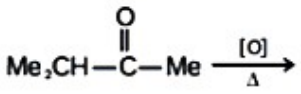
The Baeyer-Villager Oxidation of Aldehydes and Ketones.

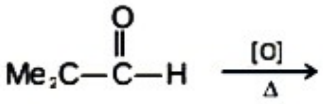
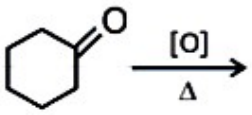
Oxidation of Aldehydes and Ketones

The Baeyer-Villager oxidation is a useful method for conversion of aldehydes and ketones to esters by the insertion of an oxygen atom from a peroxycarboxylic acid (RCO_3H).

Example :



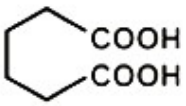
Q15 (a)  (b) 

(c)  (d) 

Sol. (a) $2CH_3 - COOH$

(b) $Me_2CO + MeCOOH$

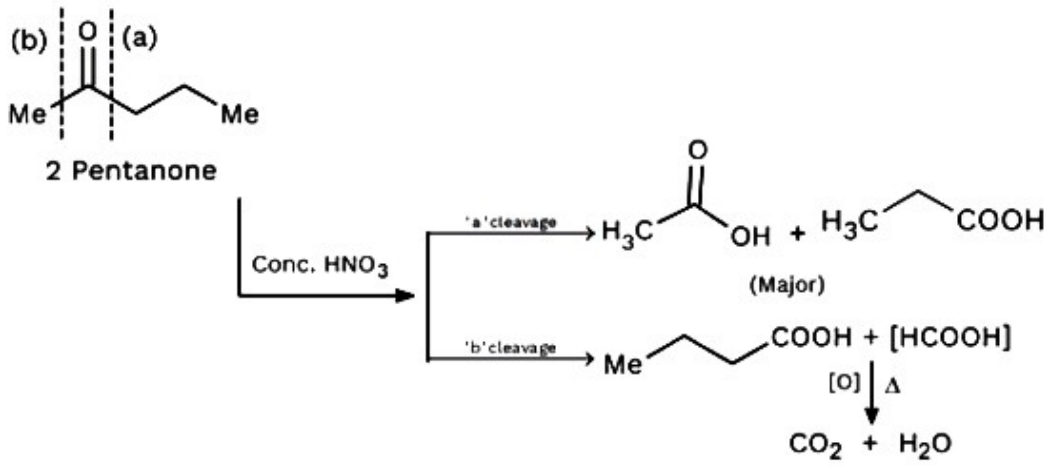
(c) $Me_3C-COOH + CO_2 + H_2O$

(d) 

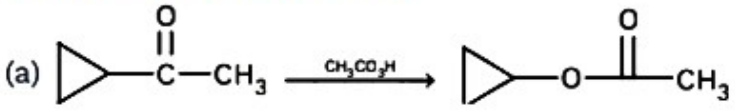
Oxidation of Ketones

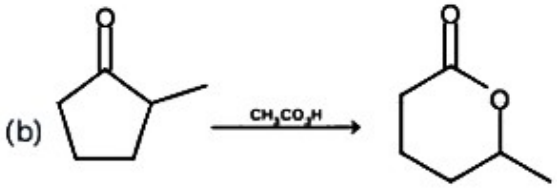
Ketones cannot be oxidised by weak oxidising agent (O.A.) such as Tollens and Fehling's solutions. However, they can be oxidised with powerful O.A., such as conentrated HNO_3, acidic oxidised with powerful O.A., such as concentrated HNO_3, acidic $KMnO_4$, acidic $K_2Cr_2O_7$, etc. During oxidiation, (C–C) bonds break giving a mixture of acids, each containing lesser number of C atoms than the original ketone. In case of unsymmetrical ketones, such as 2-butanone, 2-pentanone, etc., the keto group stays preferentially with the smaller alkyl group, and this is called popoff's rule.

Popoff's rule



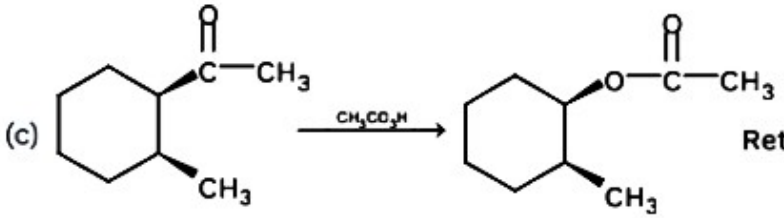
Examples of Popoff's Rule:



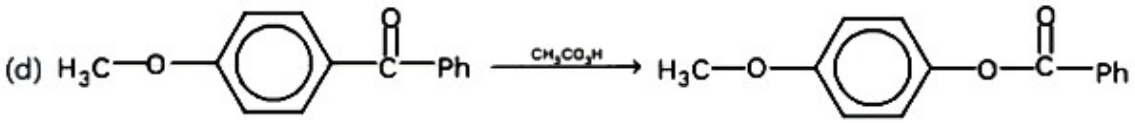


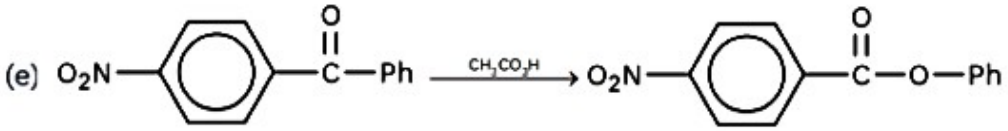
Note

In case of 2-pentanone, the major mode of cleavage is path 'a' in which the keto group stays with the smaller methyl group.



Retention of configuration





Reducing agents and their role

Group	Product	LAH in ether	$LiAlH(OCMe_3)_3$ in THF	$NaBH_4$ in EtOH	$LiAlH_4$ + $AlCl_3$	B_2H_6 in THF	H_2^* **catalyst
–CHO	$-CH_2OH$	+	–	+	+	+	+
>C=O	>CH–OH	+	–	+	+	+	+
$-CO_2H$	$-CH_2OH$	+	–	–	+	+	+
$-CO_2R$	$-CH_2OH$	+	–	–	+	+	+
–COCl	$-CH_2OH$	+	+*	+	+	–	+
$-CONH_2$	$-CH_2NH_2$	+	–	–	+	+	+
$(RCO)_2O$	RCH_2OH	+	–	–	+	+	+
–CN	$-CH_2NH_2$	+	–	–	+	+	+
>C=NOH	$-CH_2NH_2$	+	–	–	+	–	+
>C=C<	>CH–CH<	–	–	–	–	+	+
–C≡C–	–CH=CH–	–	–	–	+	+	+
1° RX	RH	+	–	–	+	–	+

* Product is RCHO

** Catalyst : Ni / Pd / Pt / Ru

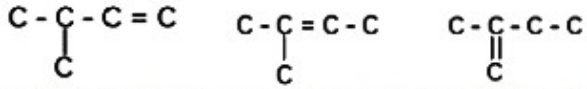
Q16 **How many alkene on catalytic reduction give normal butane as product.**

(i) (A) $\xrightarrow{H_2/Pt}$ nbutane **(ii)** (B) $\xrightarrow{H_2/Pt}$ Isopentane

(iii) (C) $\xrightarrow{H_2/Pt}$ Neopentane **(iv)** (D) $\xrightarrow{H_2/Pt}$ Cyclopentane

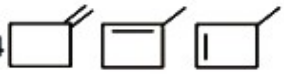
(v) (E) $\xrightarrow[Pt]{H_2}$ 

Sol. (i) cis & trans 2butene & 1butene;

(ii) 

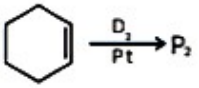
(iii) zero (Neopentane can not be prepared by catalytic hydrogenation of alkene);

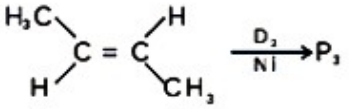
(iv) One

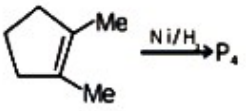
(v) Including optical = 4  , Excluding optical = 3

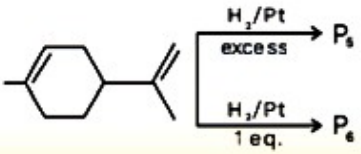
Q17 **Give the expected major product for each reaction, including stereochemistry where applicable.**

(a) $CH_3-CH_2-CH=CH_2 \xrightarrow[Pt]{H_2} P_1$

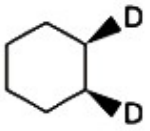
(b) 

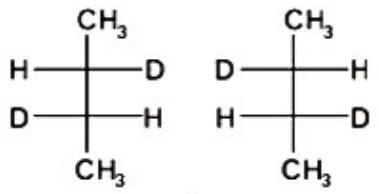
(c) 

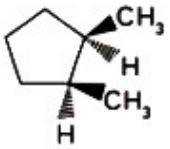
(d) 

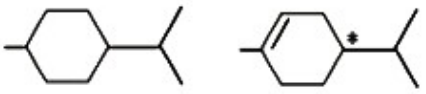
(e) 

Sol. (a) 

(b) 

(c) 

(d) 

(e) 

2 G.I.

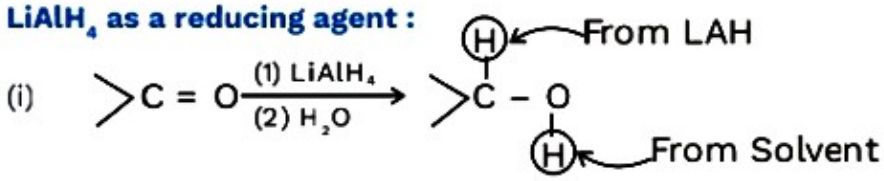
Q18 (i) $CH_2 = CH-CH_2 - CH = O \xrightarrow[\text{excess}]{H_2/Pt} P_1$

(ii) $CH_2=CH-CH_2-C\equiv N \xrightarrow[\text{excess}]{H_2/Pt} P_2$

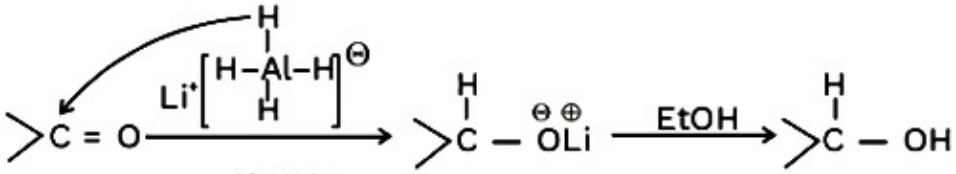
Sol. (i) P_1 $CH_3-CH_2-CH_2-CH_2-OH$

(ii) P_2 $CH_3-CH_2-CH_2-CH_2-NH_2$

$LiAlH_4$ as a reducing agent :

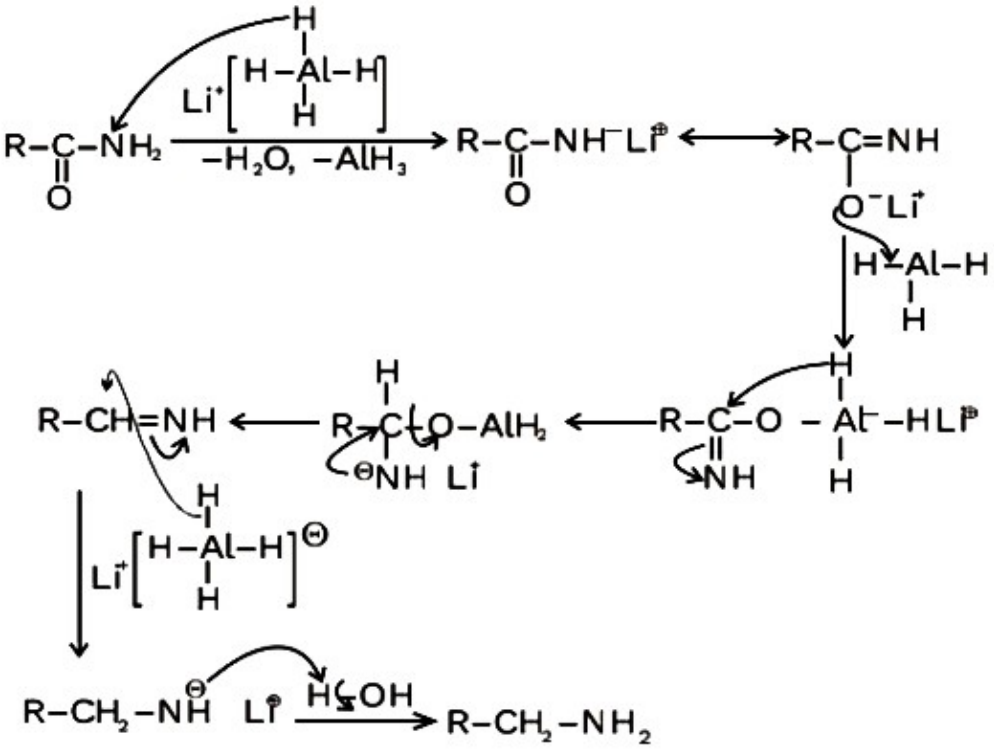
(i) 

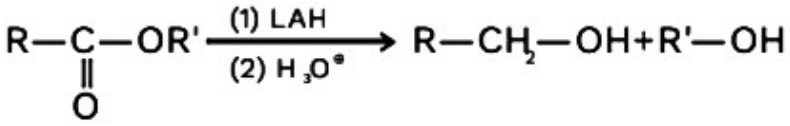
Mechanism



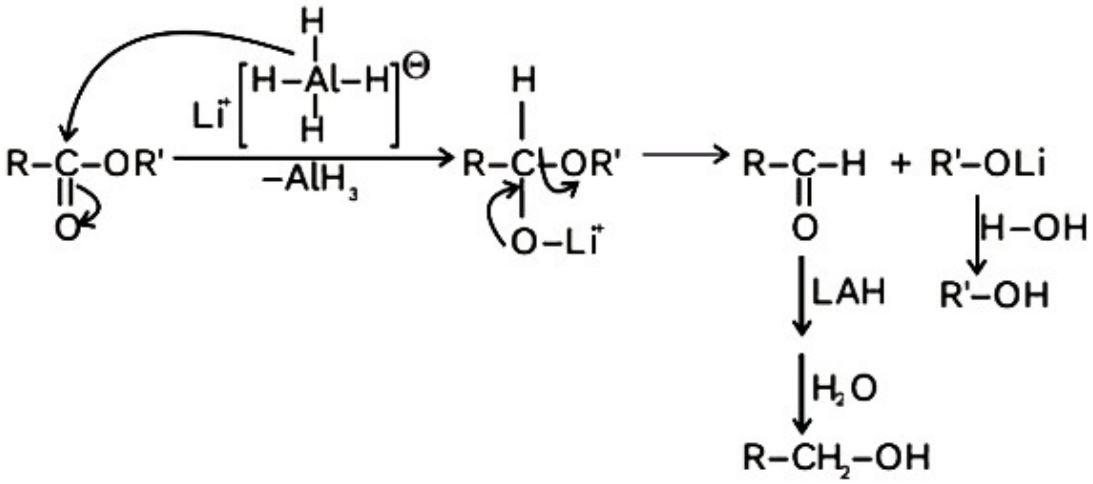
(ii) $R-\underset{\underset{O}{\|}}{C}-NH_2 \xrightarrow[(2)\ H_2O]{(1)\ LiAlH_4} R-CH_2-NH_2$

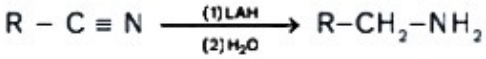
Mechanism



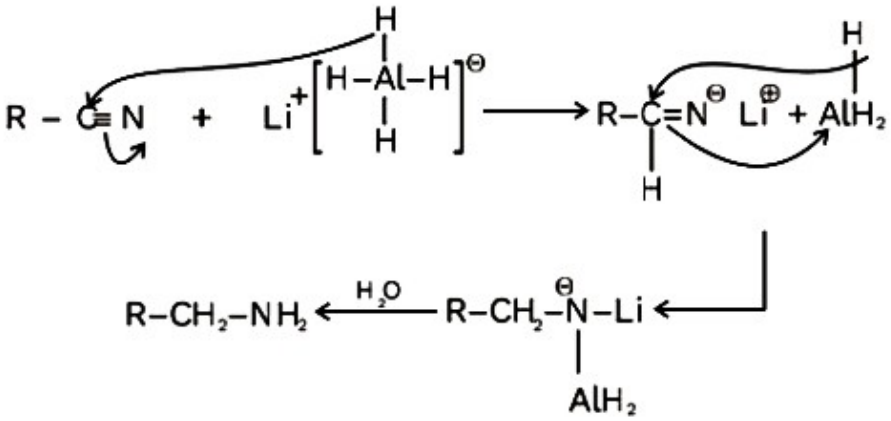
(iii) 

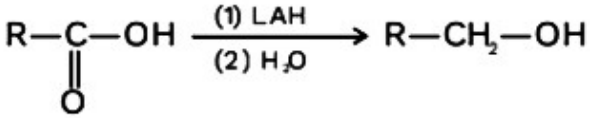
Mechanism



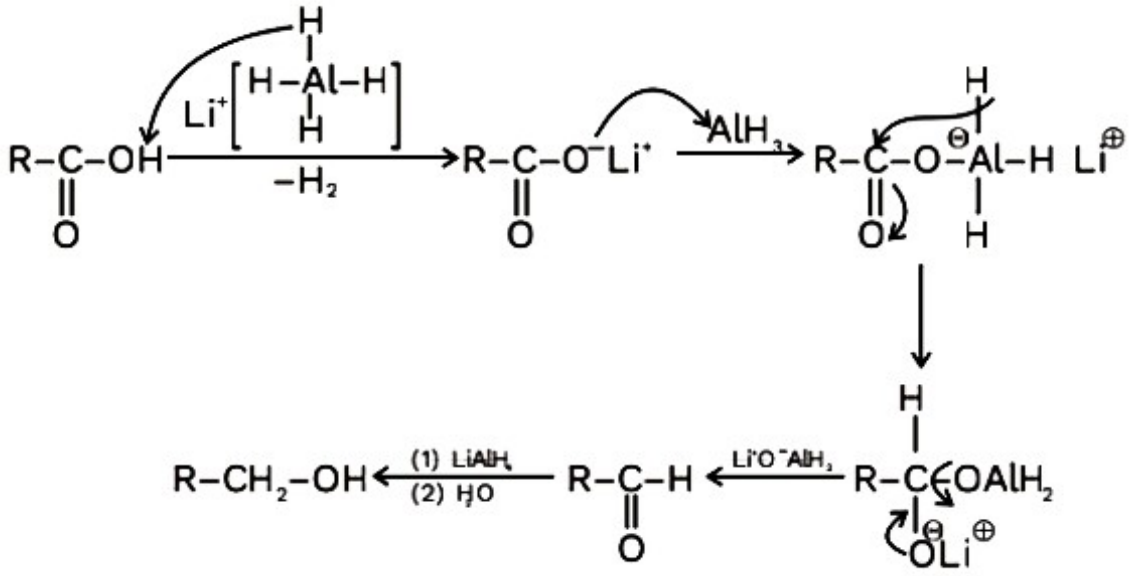
(iv) 

Mechanism



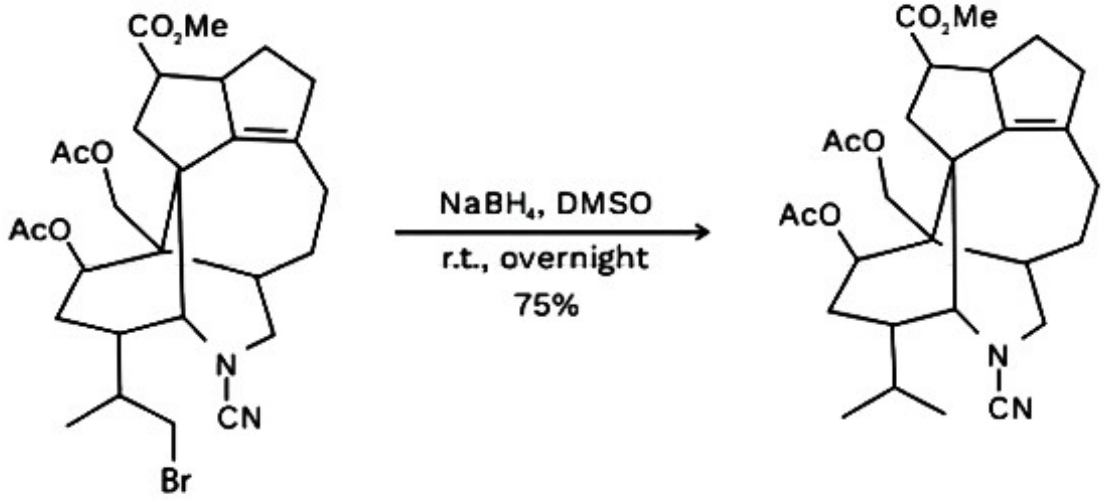
(v) 

Mechanism

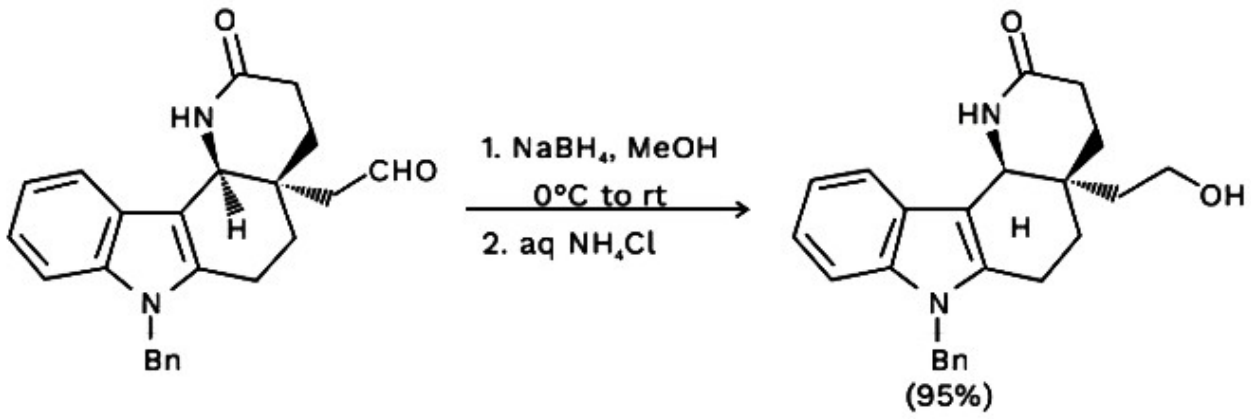


$NaBH_4$:

Sodium borohydride, a representative borohydride reagent, behaves as an effective source of nucleophilic hydride in an aprotic polar solvent, such as DMSO, sulfolane, HMPA, DMF or diglyme, and is used for the reduction of alkyl halides primary and secondary iodides, bromides and chlorides are converted to hydrocarbons at temperatures between 25 and 100 °C using sodium borohydride. Vicinal dihalides, such as 1,2- dibromooctane, are smoothly converted to the corresponding saturated hydrocarbons, Sodium borohydride reduction offers a significant advantage in synthetic applications. The method allows the reductive removal of halides selectively without affecting other functional groups, such as ester, carboxylic acid, nitrile and sulfone



Sodium borohydride is a relatively selective reducing agent. Ethanolic solutions of sodium borohydride reduce aldehydes and ketones in the presence of epoxides, esters, lactones, acids, nitriles, or nitro groups.



Q19 **Identify the product?**

(i) $\xleftarrow{NaBH_4}$ Me–CHO $\xrightarrow{LiAlH_4}$ **(ii)** $\xleftarrow{NaBH_4}$ Me_2CO $\xrightarrow{LiAlH_4}$

(iii) $\xleftarrow{NaBH_4}$ Me–COCl $\xrightarrow{LiAlH_4}$ **(iv)** $\xleftarrow{NaBH_4}$ Me–COOEt $\xrightarrow{LiAlH_4}$

(v) $\xleftarrow{NaBH_4}$ Me–COOH $\xrightarrow{LiAlH_4}$ **(vi)** $\xleftarrow{NaBH_4}$ Me–COOMe $\xrightarrow{LiAlH_4}$

(vii) $\xleftarrow{NaBH_4}$ $Me–CONH_2$ $\xrightarrow{LiAlH_4}$ **(viii)** $\xleftarrow{NaBH_4}$ Me–CONH–Me $\xrightarrow{LiAlH_4}$

(ix) $\xleftarrow{NaBH_4}$ $Me–CONMe_2$ $\xrightarrow{LiAlH_4}$ **(x)** $\xleftarrow{NaBH_4}$ Me–CH=NH $\xrightarrow{LiAlH_4}$

(xi) $\xleftarrow{NaBH_4}$ $CH_3–CH = CH_2$ $\xrightarrow{LiAlH_4}$

Sol.

(i) $MeCH_2–OH$, $Me–CH_2OH$ (ii) $Me_2CH–OH$, $Me_2CH–OH$

(iii) $MeCH_2–OH$, $Me–CH_2OH$ (iv) No reaction , $Me–CH_2–OH$ + EtOH

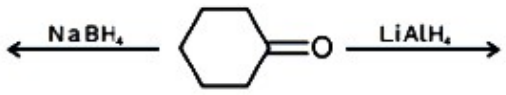
(v) No reduction, $MeCH_2–OH$ (vi) No reaction, $MeCH_2–OH$ + MeOH

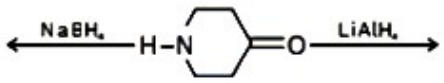
(vii) No reduction, $MeCH_2NH_2$ (viii) No reaction, $Me–CH_2–NH–Me$

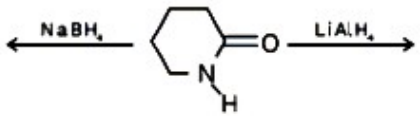
(ix) No reduction, $MeCH_2–N(Me)(Me)$

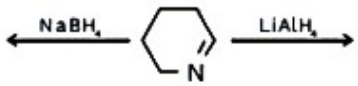
(x) $Me–CH_2–NH_2$, $Me–CH_2–NH_2$ (xi) No reaction, No reaction

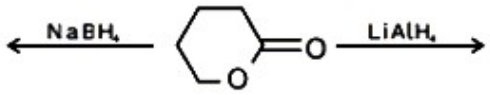
Q20 **Give product in following reactions.**

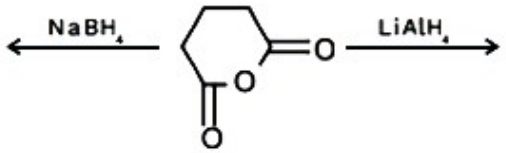
(i) 

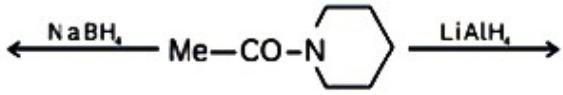
(ii) 

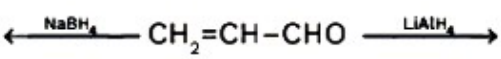
(iii) 

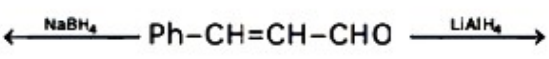
(iv) 

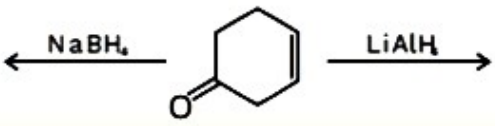
(v) 

(vi) 

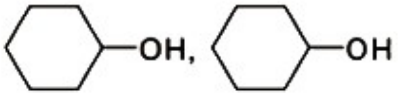
(vii) 

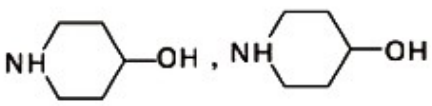
(viii) 

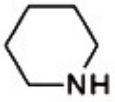
(ix) 

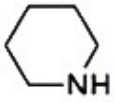
(x) 

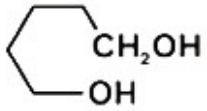
Sol.

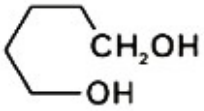
(i) 

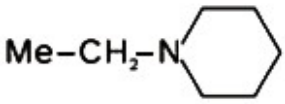
(ii) 

(iii) No reaction, 

(iv) No reaction , 

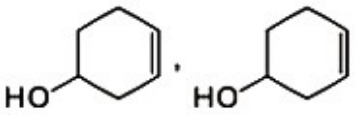
(v) No reaction, 

(vi) No reaction, 

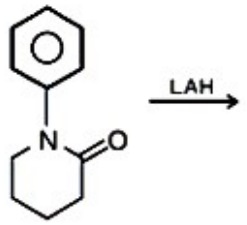
(vii) No reaction, 

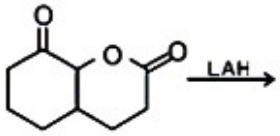
(viii) CH_2= CH–CH_2–OH, CH_2=CH–CH_2–OH

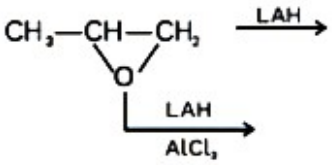
(ix) Ph–CH=CH–CH_2–OH, Ph–CH_2–CH_2–CH_2–OH

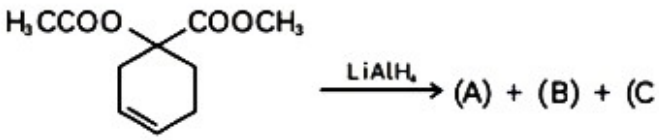
(x) 

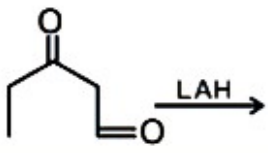
Q21 **Give product in following reactions.**

(i) 

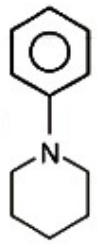
(ii) 

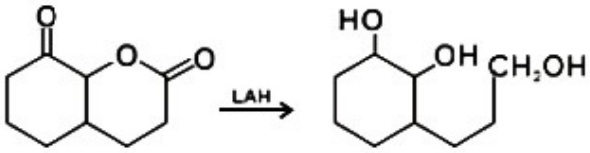
(iii) 

(iv) 

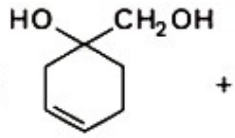
(v) 

Sol.

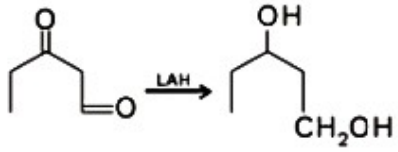
(i) 

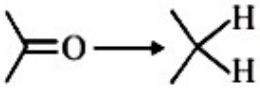
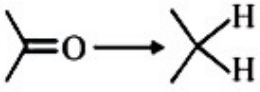
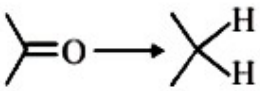
(ii) 

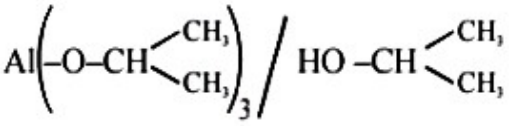
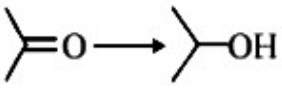
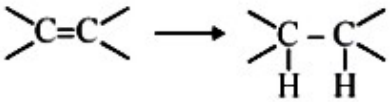
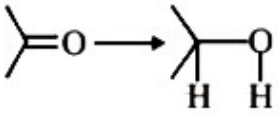
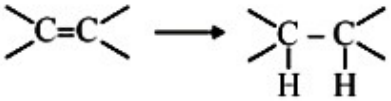
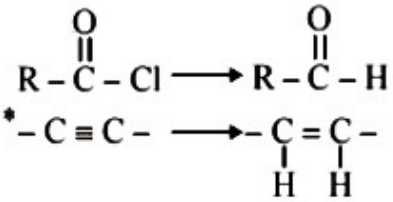
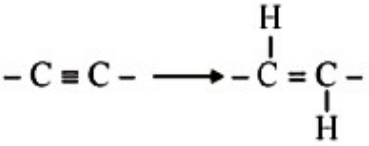
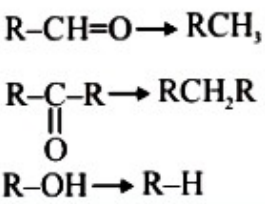
(iii) $CH_3-\underset{\underset{OH}{|}}{CH}-CH_3$, $CH_3-CH_2-CH_2OH$

(iv) 

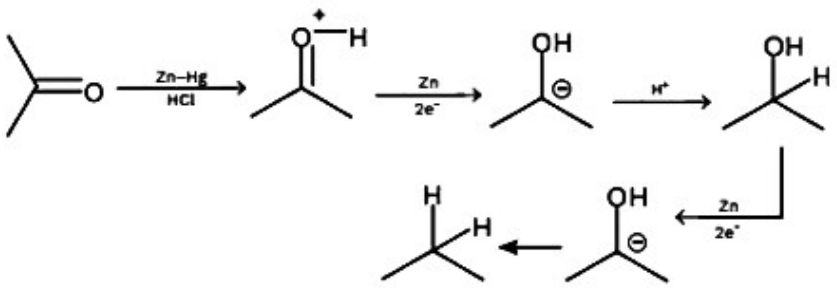
(A) + MeOH (B) + Et–OH (C)

(v) 

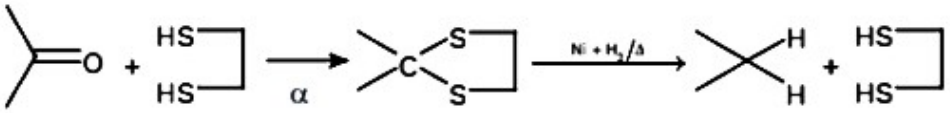
Name	Reagent	Function
Wolf Kishner Reduction	(i) N_2H_4 / (ii) KOH, Δ	
Clemmensen Reduction	Zn – Hg / HCl	
Mozingo Reduction	 Dry HCl followed by Raney Ni	
Stephen's Reduction	$SnCl_2$ / HCl followed by H_3O+	R–C≡N→R–CH=O

Name	Reagent	Function
MPV Reduction		
Hydroboration Reduction	(i) B_2H_6 (ii) AcOH, H_2O	 
Bouvoult Blank Reduction	Na in EtOH	$R-COO-R \rightarrow RCH_2OH + ROH$
Transfer Hydrogenation	N_2H_4 / H_2O_2	
Rosenmund Reduction	H_2, Pd – $BaSO_4$	
Birch Reduction	Na in Liq. NH_3	
Red phosphorus in presence of HI	Red P + HI	

Clemmensen reduction

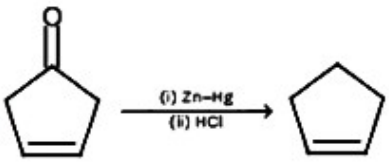


Mosingo Reduction

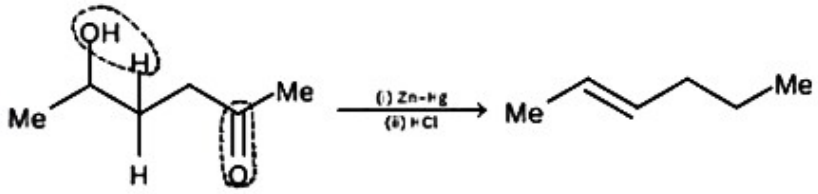


H.O.H. α Unstability

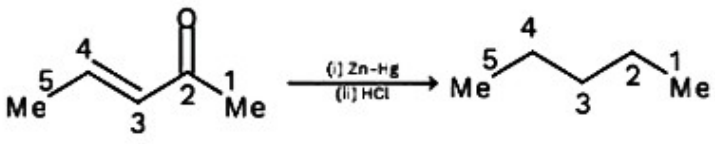
Example :



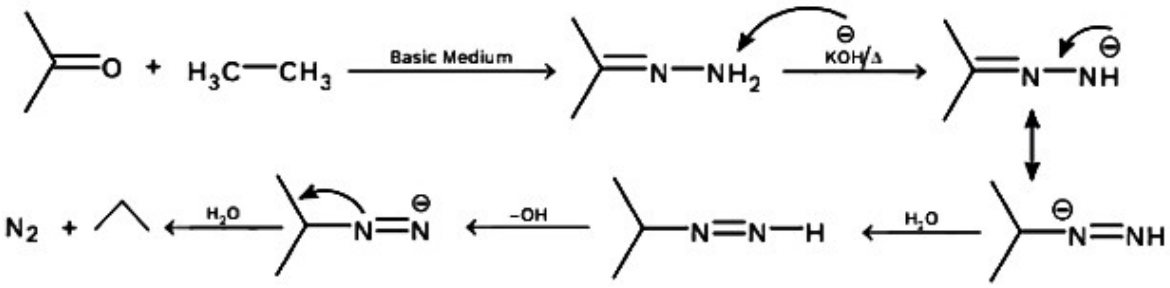
Example :



Example : α, β-Unsaturated carbonyl compounds undergo reduction of both the olefinic and carbonyl groups.

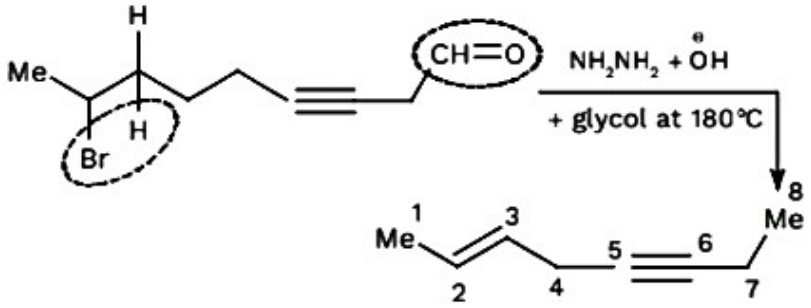


Wolff-Kishner Reduction



Trick

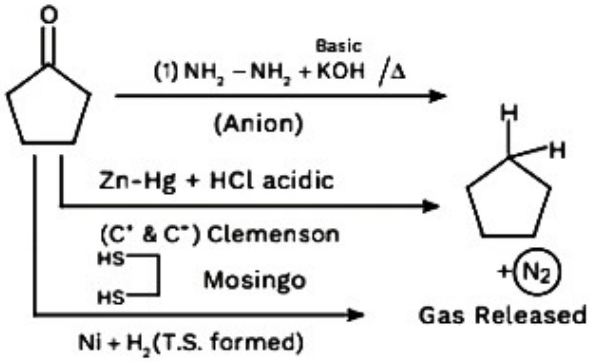




NOTE

(NH_2NH_2 + glycol + KOH, NH_2OH + glycol + KOH, $PhNHNH_2$ + glycol + KOH). It reduces aldehydes and ketones to alkanes, but does not reduce (C=C) and (C≡C) bonds. This is a base catalysed reaction, so simultaneously dehydrohalogenation (–HX) of RX also takes place.

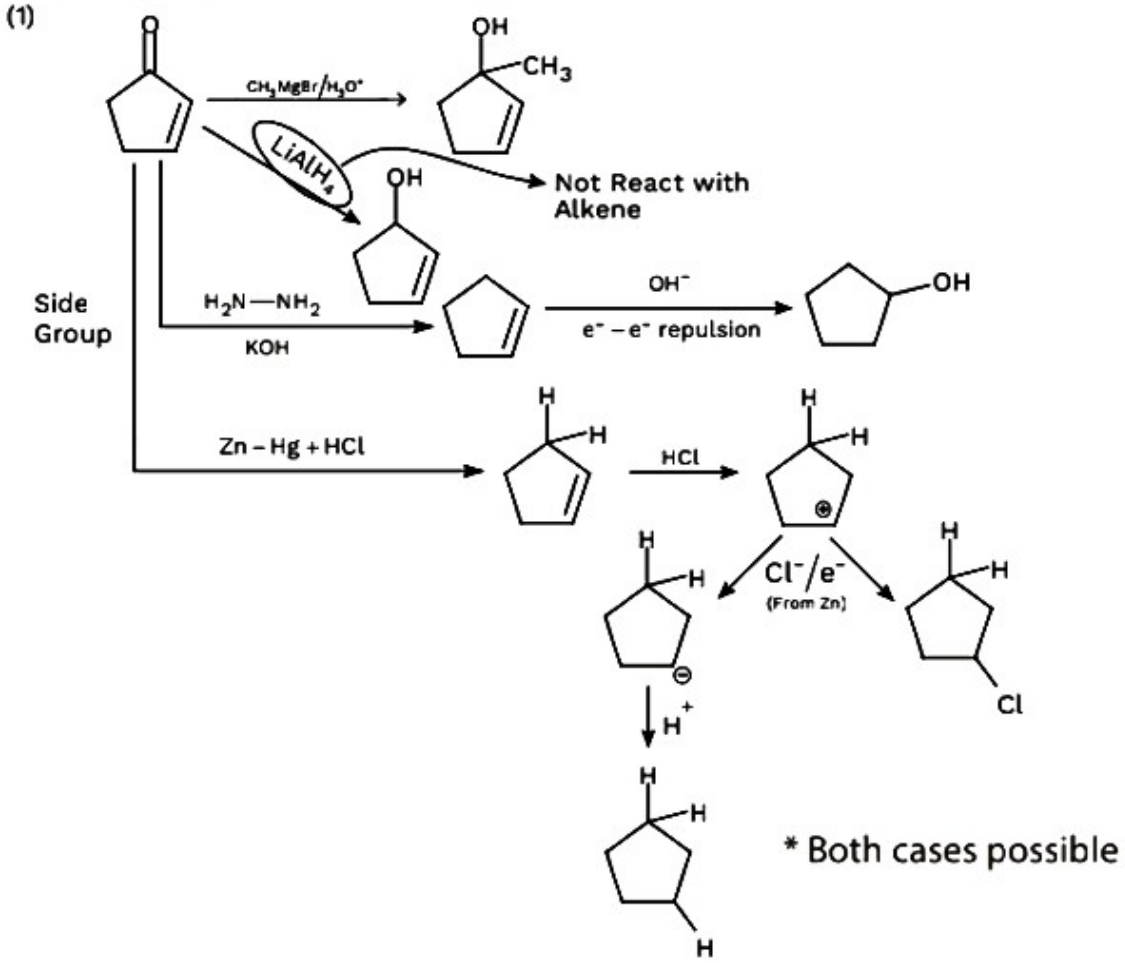
Solved Examples



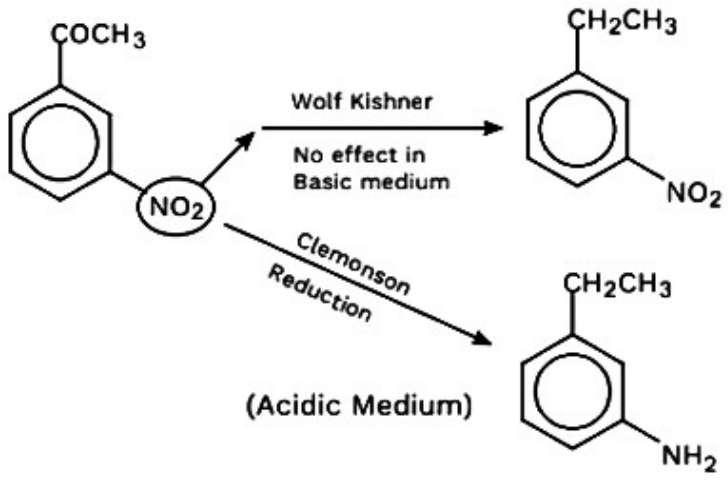
(Same product)

Solved Example :

(1)



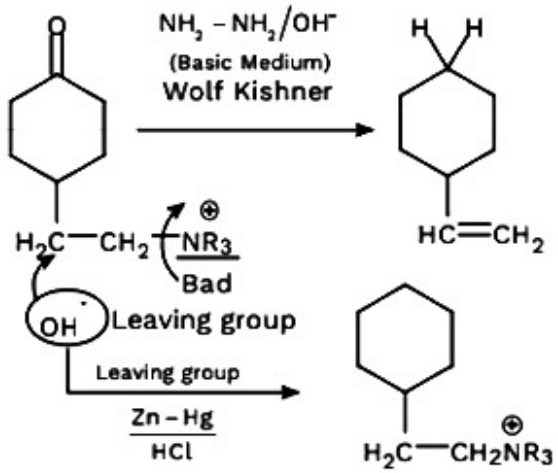
Solved Example :



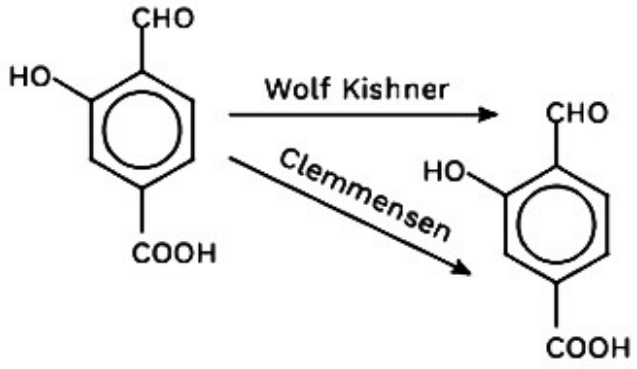
Solved Example :



Solved Example :



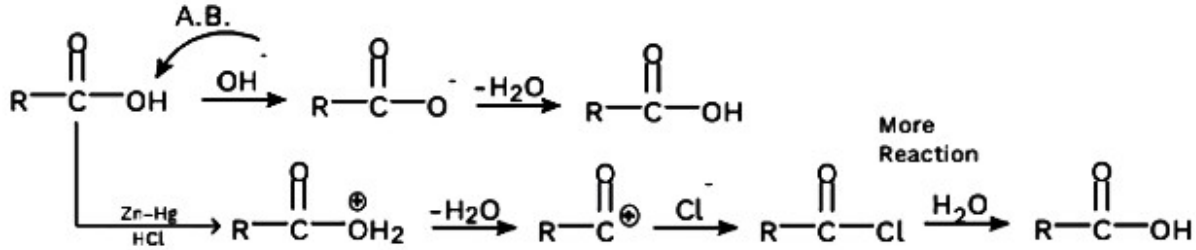
Solved Example :



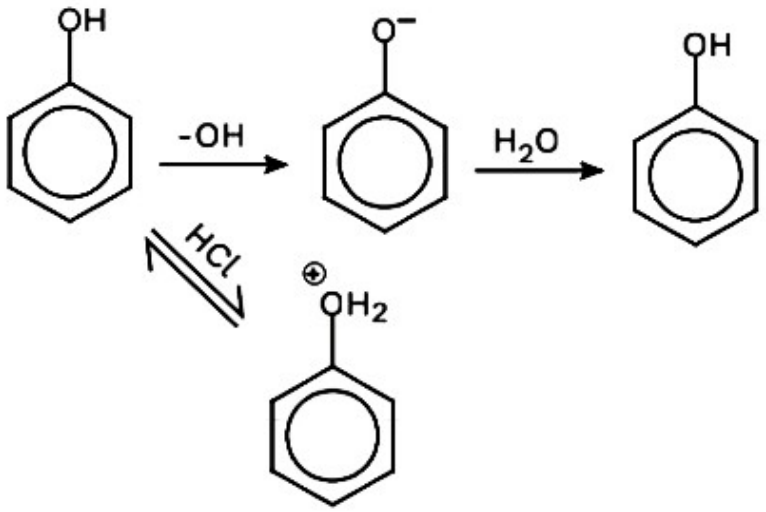
NOTE

Acid and Phenol are uncreative during in wolf kishner & Clemmensen

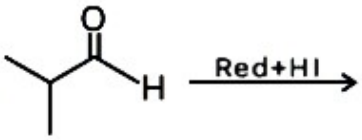
Solved Example :

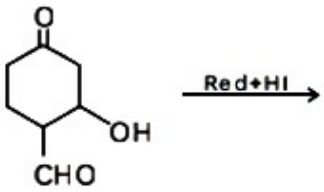


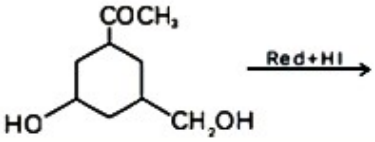
Solved Example :



Q22 **Give product in following reactions.**

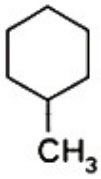
(a) 

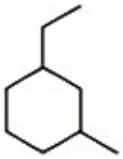
(b) 

(c) 

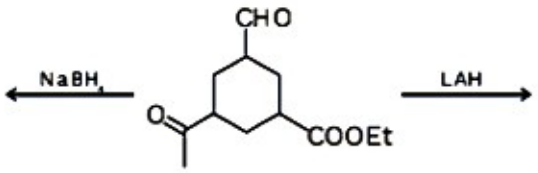
Sol.

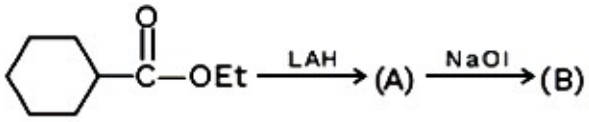
(a) 

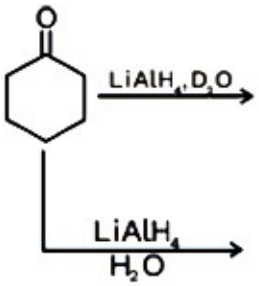
(b) 

(c) 

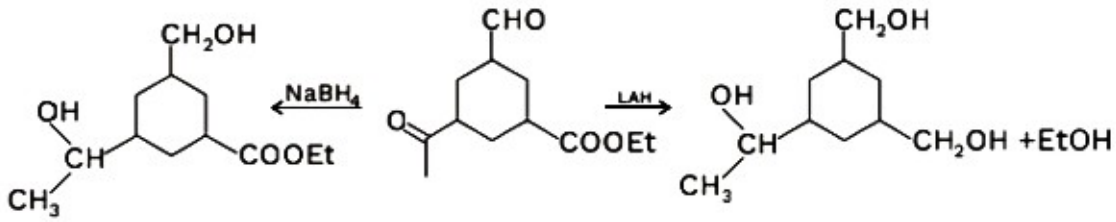
Q23 **Give product in following reactions.**

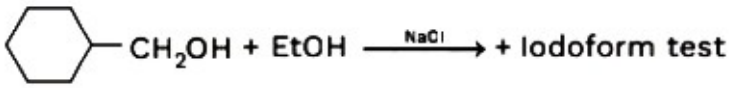
(i) 

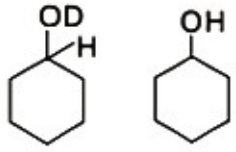
(ii) 

(iii) 

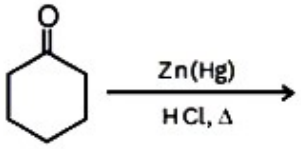
Sol.

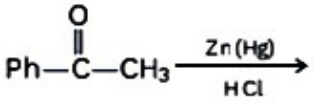
(i) 

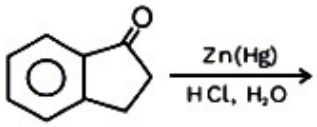
(ii) 

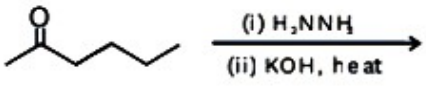
(iii) 

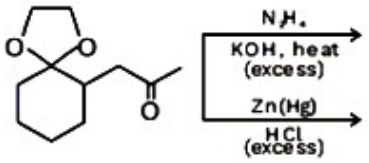
Q24 **(a)** 

(b) 

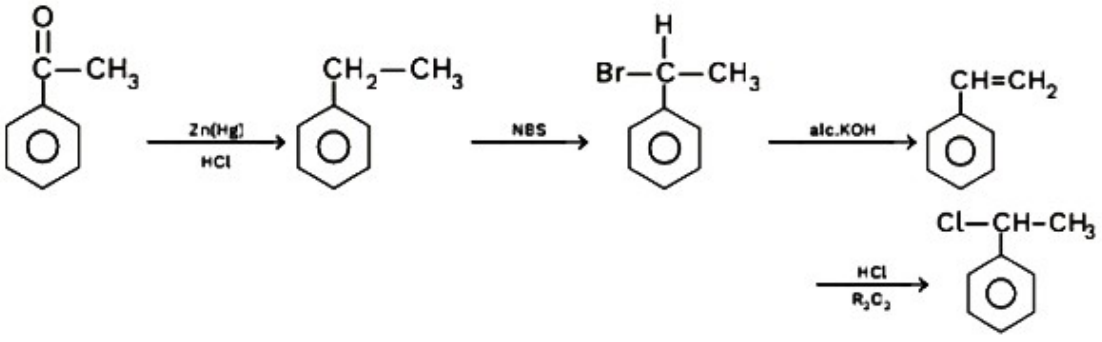
(c) 

(d) 

(e) 

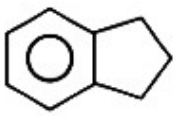
(f) 

Sol.

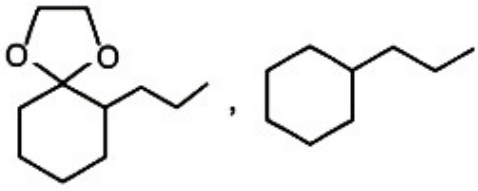
(a) 

(b) 

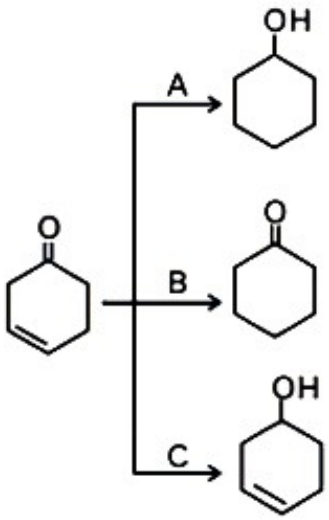
(c) $Ph-CH_2-CH_3$

(d) 

(e) 

(f) 

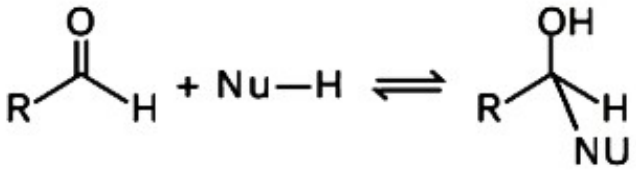
Q25 **Suggest appropriate reagents for following conversion.**



Sol. (A) Ni will reduce alkene, aldehyde and all it is not specific ;
(B) NH_2-NH_2/H_2O_2 ; (C) $LiAlH_4$

Nucleophilic Addition to the Carbon–Oxygen Double Bond:

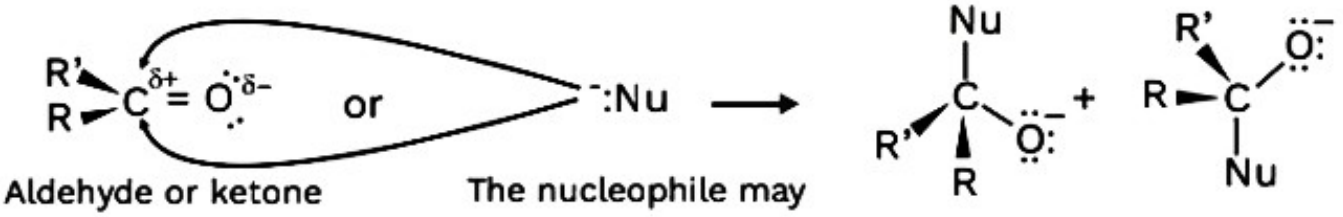
General Reaction :



Aldehydes and ketones are especially susceptible to nucleophilic addition because of the structural features

Tips and Trick

The most characteristic reaction of aldehydes and ketones is nucleophilic addition to the carbon–oxygen double bond



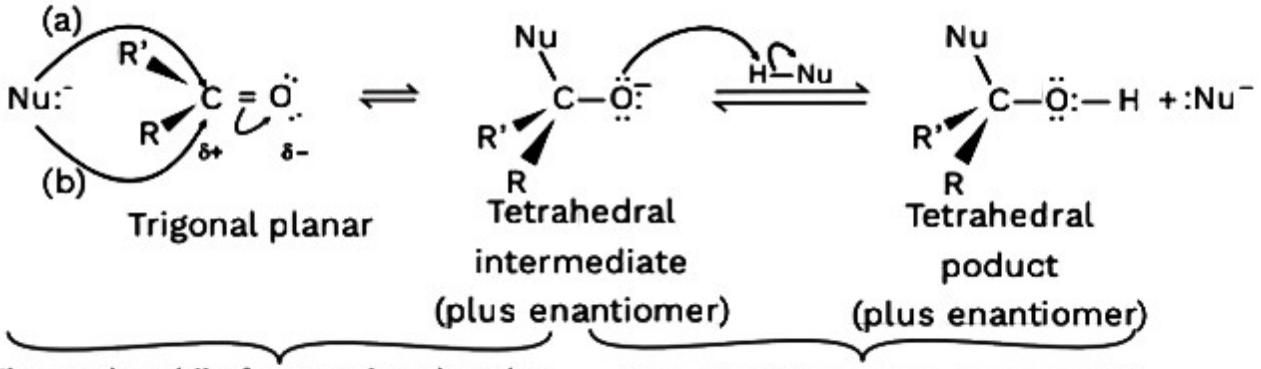
Aldehyde or ketone (R or R' may be H)

The nucleophile may attack from above or below.

Characteristics of CO bond:
The trigonal planar arrangement of groups around the carbonyl carbon atom means that the carbonyl carbon atom is relatively open to attack from above or below the plane of the carbonyl group (see above).
The partial positive charge on the carbonyl carbon makes it susceptible to nucleophilic attack.
Unshared electron pairs on the oxygen allow protonation to make the carbonyl carbon more electrophilic under acidic conditions

Nucleophilic addition to the carbon–oxygen double bond occurs, therefore, in either of two general ways.

1. **When the reagent is a strong nucleophile**
   Mechanism :
   Addition of a Strong Nucleophile to an Aldehyde or Ketone



The nucleophile forms a bond to the carbon by donating an electron pair to the top or bottom face of the carbonyl group [path (a) or (b)]. An electron pair shifts out to the oxygen.

The alkoxide oxygen, because it is strongly basic, removes a proton from H—Nu or some other acid.

2. **When an acid catalyst is present and the nucleophile is weak :**
   When carbonyl compounds are treated with *strong acids* in the presence of *weak nucleophiles*. In the first step the acid donates a proton to an electron pair of the carbonyl oxygen atom. The resulting protonated carbonyl compound, an oxonium cation, is highly reactive toward nucleophilic attack at the carbonyl carbon atom because the carbonyl carbon atom carries more positive charge than it does in the unprotonated compound.

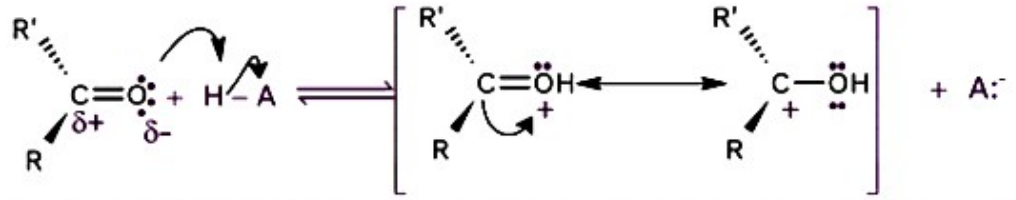
Points to remember

Reaction of the carbonyl oxygen with the acid enhances electrophilicity of the carbonyl group.

Mechanism :

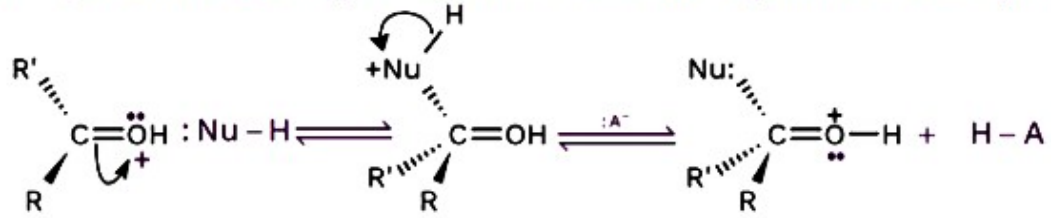
Acid-Catalyzed Nucleophilic Addition to an Aldehyde or Ketone :

Step-1



Under acidic conditions, an electrons pair the carbonyl oxygen accepts a proton from the acid(or associates with a Lewis acid.) The carbon of the protonted carbonyl is now even more susceptible to nucleophillic attack.

Step-2



The protonated carbonyl is attacked the nucleophile,and a base removes a proton from the positively charged atom,regenerating the acid catalyst.

Some Facts about Nucleophilic addition in CO bond

Reversibility of Nucleophilic Additions to the Carbon–Oxygen Double Bond :

Many nucleophilic additions to carbon–oxygen double bonds are reversible; the overall results of these reactions depend, therefore, on the position of an equilibrium.

Relative Reactivity: Aldehydes versus Ketones:

In general, aldehydes are more reactive in nucleophilic additions than are ketones. Both steric and electronic factors favour aldehydes.

Steric Factors :

In aldehydes, where one group is a hydrogen atom, the central carbon of the tetrahedral product formed from the aldehyde is less crowded and the product is more stable. With ketones the two alkyl substituents at the carbonyl carbon cause greater steric crowding in the tetrahedral product and make it less stable. Therefore, a smaller concentration of the product is present at equilibrium.

Electronic Factors:

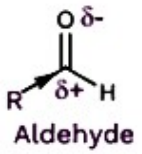
Aldehydes have only one electron-releasing group to partially neutralize, and thereby

stabilize, the positive charge at their carbonyl carbon atom.

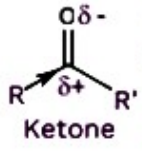
Ketones have two electron-releasing groups and are more stabilized.

Result

Greater stabilization of the ketone (the reactant) relative to its product means that the equilibrium constant for the formation of the tetrahedral product from a ketone is smaller and the reaction is less favourable.



Aldehyde

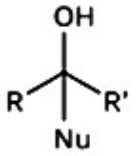


Ketone

The ketone carbonyl carbon is less positive because it has two electron-releasing alkyl groups.

Different products obtained with addition reaction at CO bond (carbonyl)

Nucleophilic addition at  bond lead to formation of stable product. Two product may formed in case of asymmetric carbon.



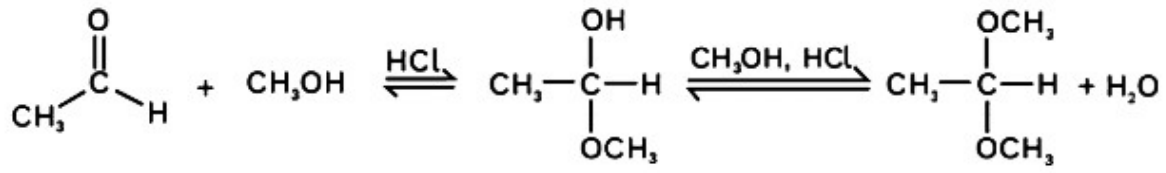
The Addition of Alcohols: Formation of Hemiacetals and Acetals

Aldehydes and ketones react with alcohols to form **hemiacetals** and **acetals** by an equilibrium reaction.

The product formed when one equivalent of an alcohol adds to an aldehyde is called a **hemiacetal**.

The product formed when a second equivalent of alcohol is added is called an **acetal**.

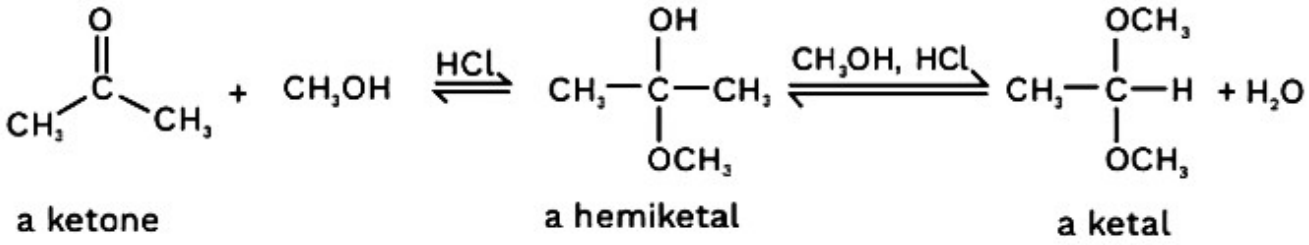
Alcohol is a poor nucleophile, so an acid catalyst is required for the reaction to take place at a reasonable rate.



an aldehyde — a hemiacetal — an acetal

Points to remember

When the carbonyl compound is a ketone instead of an aldehyde, the addition products are called a hemiketal and a ketal.

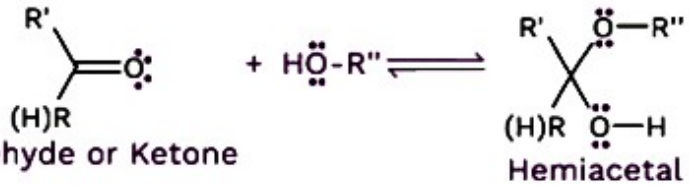


a ketone — a hemiketal — a ketal

Hemiacetals:

A hemiacetal results by nucleophilic addition of an alcohol oxygen to the carbonyl carbon of an aldehyde or ketone.

The formation of hemiacetals is catalyzed by acids and bases.



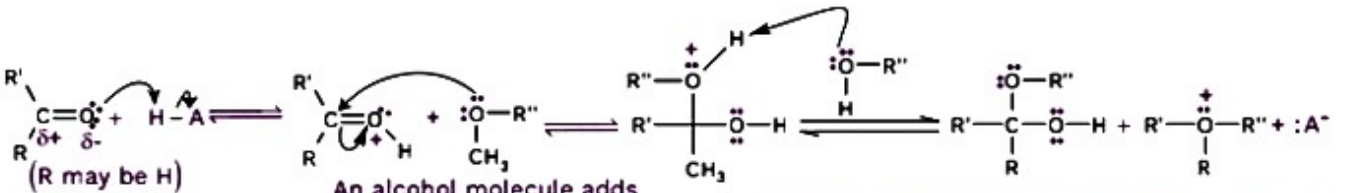
Aldehyde or Ketone — Hemiacetal

Introduction

The essential structural features of a **hemiacetal** are an –OH and an –OR group attached to the same carbon atom.

Mechanism :

Acid-Catalyzed Hemiacetal Formation

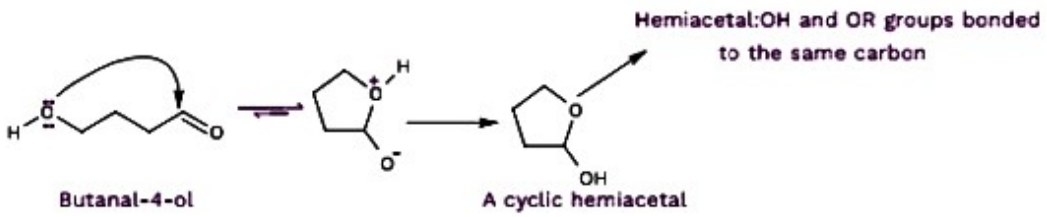


(R may be H)
A strong acid catalyst protonates the carbonyl oxygen

An alcohol molecule adds to the carbon carbonyl.

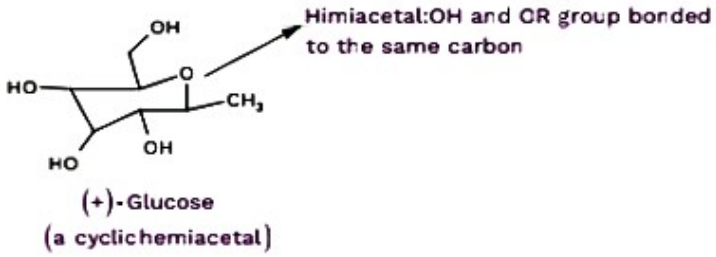
Removal of a proton by some proton acceptor (e.g., another alcohol molecule) leads to the hemiacetal

Cyclic hemiacetals with five- or six-membered rings form very readily and can be:



Hemiacetal:OH and OR groups bonded to the same carbon

Butanal-4-ol — A cyclic hemiacetal



(+)-Glucose
(a cyclichemiacetal)

Acetals :

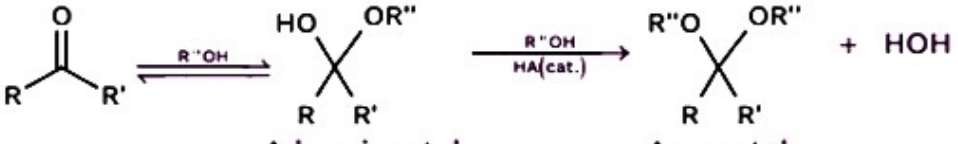
An **acetal** has two –OR groups attached to the same carbon atom.

An acetal forms when an aldehyde or ketone is treated with an excess of alcohol in the presence of an acid catalyst. Although a hemiacetal is an intermediate, it usually cannot be isolated because it continues on to become an acetal.

The mechanism involves acid-catalyzed formation of a hemiacetal, then acid-catalyzed elimination of water followed by addition of a second alcohol group.

All steps involved in the formation of an acetal from an aldehyde are reversible.

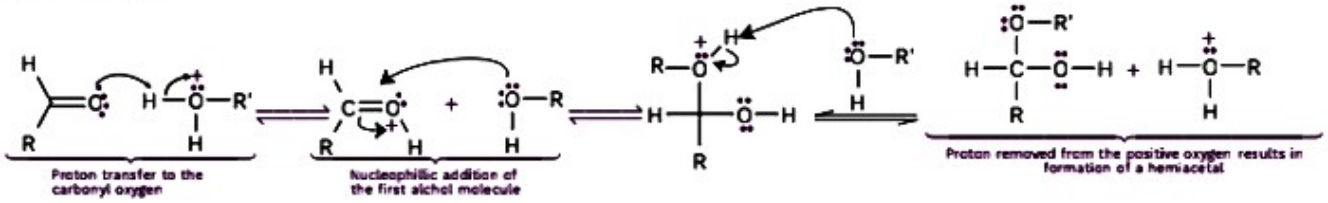
Acid-Catalysed Acetal Formation



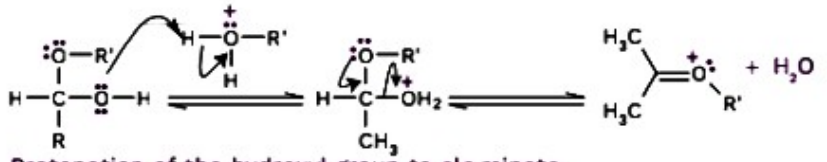
A hemiacetal (R'maybe H)

An acetal (R'maybe H)

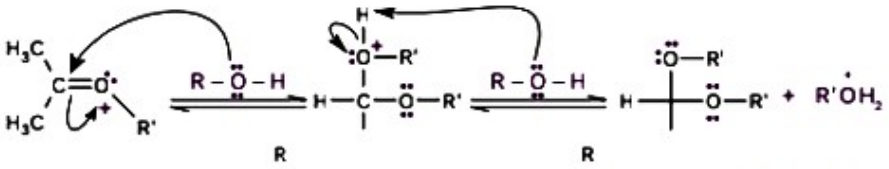
Mechanism :



Regeneration of carbonyl compound incase of hemiacetal



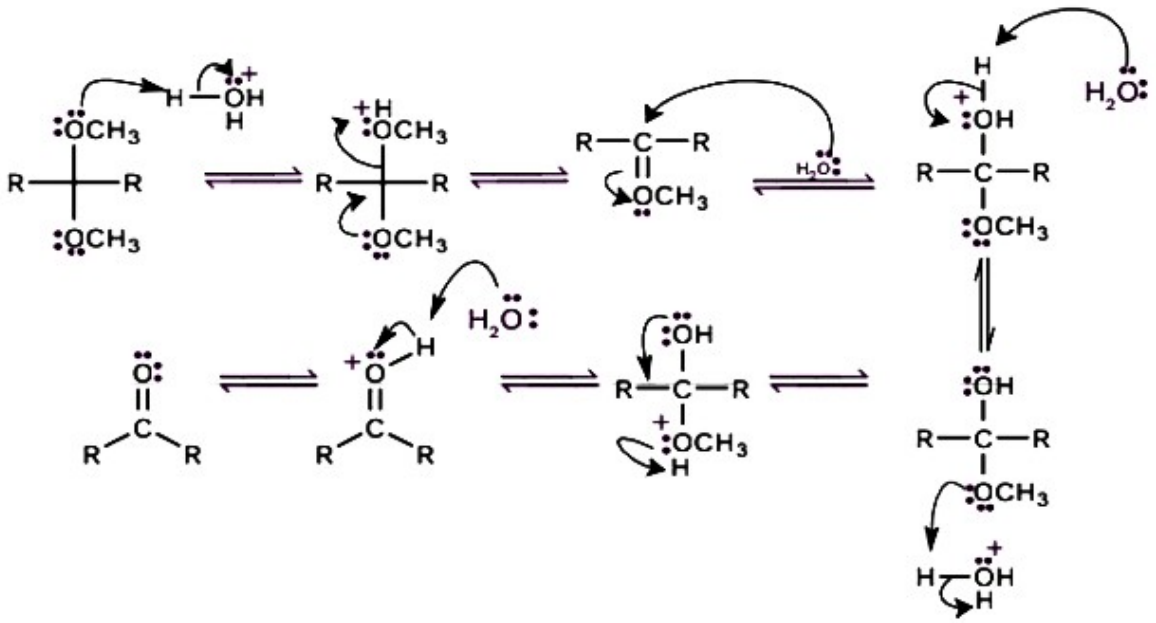
Protonation of the hydroxyl group to eliminate water and formation of a highly reactive oxonium cation



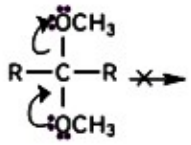
Attack on the carbon of the oxonium ion by a second molecule of the alcohol, followed by removal of a proton, leads to the acetal

Q26 **Explain why acetals and ketals are hydrolysed back to the aldehyde or ketone in acidic aqueous solutions but are stable in basic aqueous solutions.**

Sol. If we place an acetal in water and add a catalytic amount of acid, all of the step's reverse. Under these conditions (an excess of water), the equilibrium favors the formation of the aldehyde. The acetal undergoes hydrolysis.

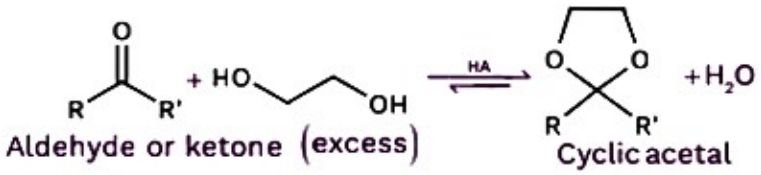


In a basic solution, the CH_3O group cannot be protonated. Therefore, the group that would have to be eliminated to reform the ketone (or aldehyde) would be the very basic CH_3O^-group. A CH_3O^- group is too basic to be eliminated by the other CH_3O group, which has little driving force because of the positive charge that would be placed on its oxygen atom if elimination were to occur.

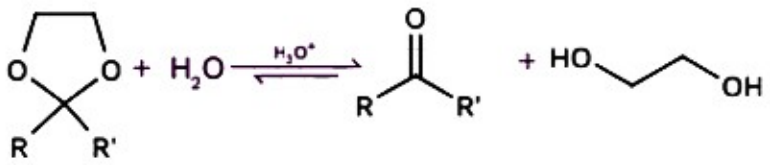


Cyclic Acetals :

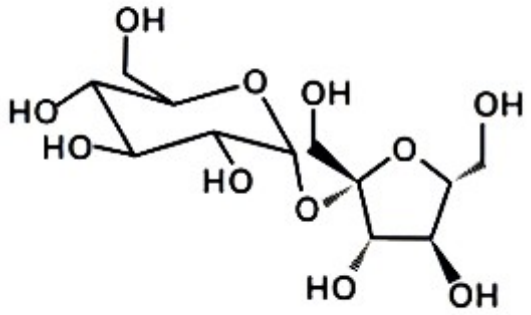
Cyclic acetal formation is favoured when a ketone or an aldehyde is treated with an excess of a 1,2-diol and a trace of nonaqueous acid :



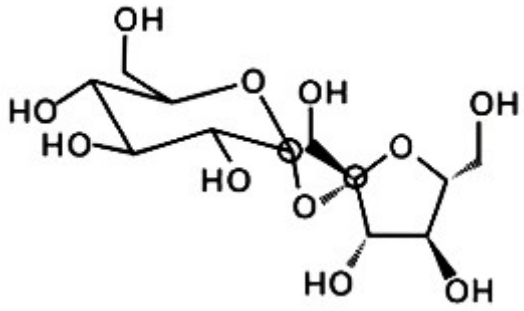
The reaction can be reversed by treating the acetal with aqueous acid :



Q27 **Shown below is the structural formula for sucrose (table sugar). Sucrose has two acetal groupings. Identify these.**



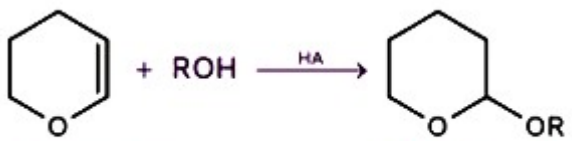
Sol.



Why don't acetals react with nucleophiles?

Q28 **Product of dihydropyran with alcohol in presence of anhydrous HCl or sulphuric acid will be?**

Sol. Dihydropyran reacts with an alcohol in the presence of a trace of anhydrous HCl or H_2SO_4 to form a tetrahydropyranyl (THP) ether :



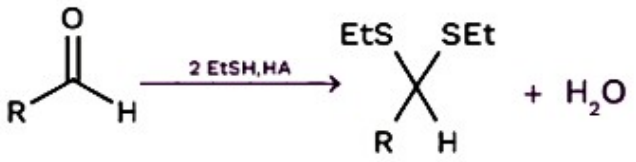
Dihydropyran — Tetrahydrioyranyl ether

Formation of Thioacetals (sulphur instead of oxygen)

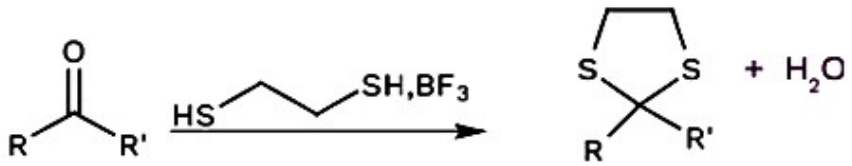
Aldehydes and ketones react with thiols under acid catalysis to form thioacetals.

Thioacetals are important in organic synthesis because they react with hydrogen and Raney nickel to yield hydrocarbons. Raney nickel is a special nickel catalyst that contains adsorbed hydrogen.

Thioacetal formation with subsequent "desulfurization" with hydrogen and Raney nickel gives us an additional method for converting carbonyl groups of aldehydes and ketones to $-CH_2-$ groups:



Thioacetal



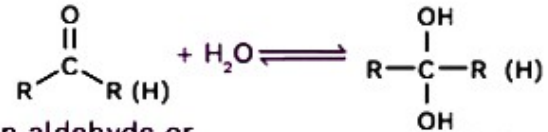
Q29 **Which reaction is slow (P) or (Q) ?**



Sol. P > Q (Rate of Reaction)

Addition of Water :
(Diols formation)

Hydrates of aldehydes or ketones are generally too unstable to be isolated because the tetrahedral carbon is attached to two oxygen atoms.



an aldehyde or a ketone

a gem-diol a hydrate

Points to remember

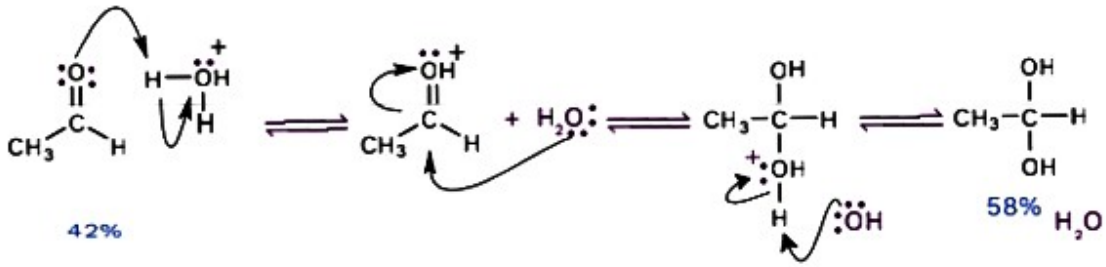
Water adds to an aldehyde or a ketone to form a hydrate. A hydrate is a molecule with two OH groups on the same carbon. Hydrates are also called gem-diols (gem comes from geminus, Latin for "twin").

Note :

Catalyst has no effect on the position of the equilibrium. A catalyst affects the rate at which the equilibrium is achieved. In other words, the catalyst affects the rate at which an aldehyde or a ketone is converted to a hydrate; it has no effect on the amount of aldehyde or ketone converted to hydrate.

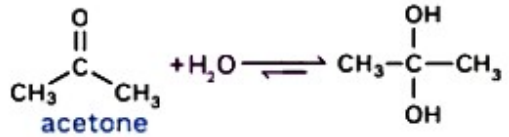
Tips and Trick

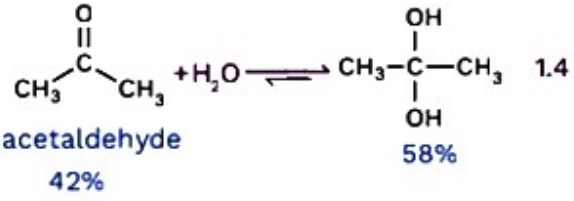
Water is a poor nucleophile and therefore adds relatively slowly to a carbonyl group. The rate of the reaction can be increased by an acid catalyst.

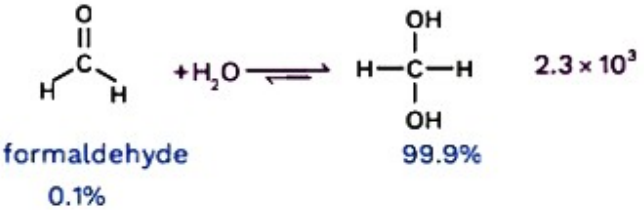


Concept :

The extent to which an aldehyde or a ketone is hydrated in an aqueous solution depends on the aldehyde or ketone:



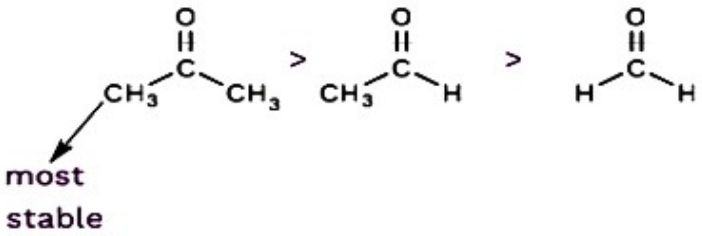




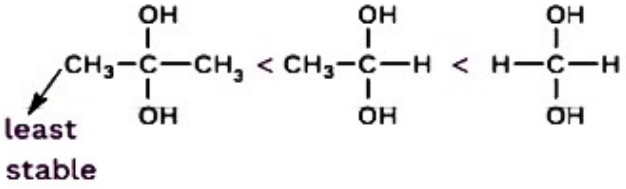
Equilibrium constant for a reaction

The equilibrium constant for a reaction depends on the relative stabilities of the reactants and products.

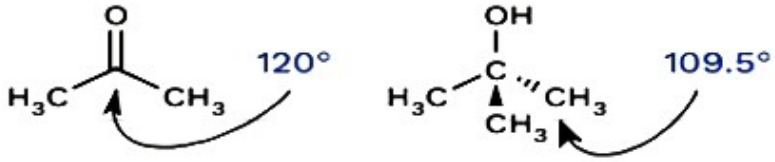
Electron-donating alkyl groups make a carbonyl compound *more stable* (less reactive)



Alkyl groups make the hydrate less stable because of steric interactions between the alkyl groups.



The electron clouds of the alkyl substituents do not interfere with each other in the carbonyl compound because the bond angles are 120°. However, the bond angles in the tetrahedral hydrate are 109.5°, so the alkyl groups are closer to one another.



Conclusion :

The percentage of hydrate present in solution at equilibrium depends on both electronic and steric effects.

Electron donation and bulky substituents *decrease* the percentage of hydrate present at equilibrium, whereas electron withdrawal and small substituents *increase* it.

Points to remember

For faster Reaction

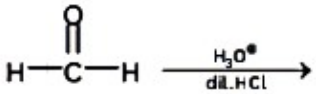
(i) More δ+ magnitude

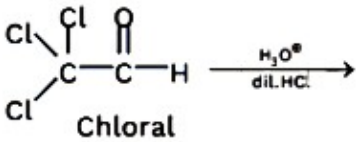
(ii) Less steric

(iii) H-bonding stabilised product

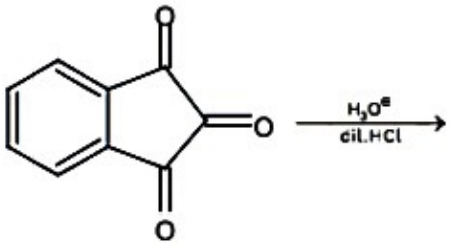
(iv) Angle strain decreases with formation of diol

Q30 **Comment about Water addition products of below compounds and draw products**

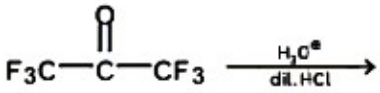
(A) 

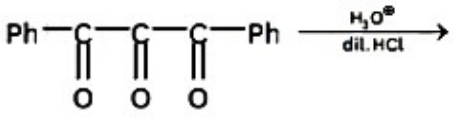
(B) 

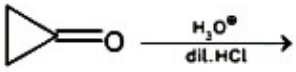
Chloral

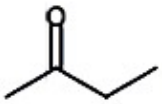
(C) 

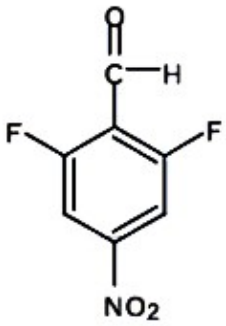
Ninhydrin

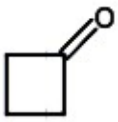
(D) 

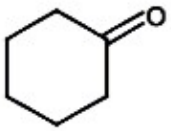
(E) 

(F) 

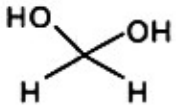
(G) 

(H) 

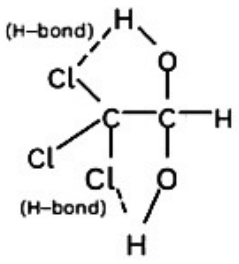
(I) 

(J) 

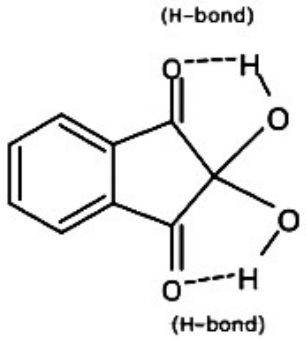
Sol. (A) Froms stable geminal diol due to less hindered carbonyl.



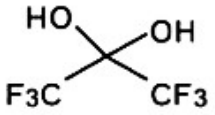
(B) Froms stable geminal diol due to intromolecular H-bonding.



(C)

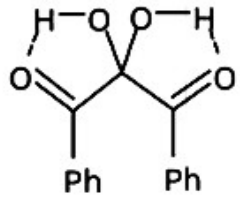


(D)



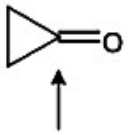
Stable product due to high electrophilicity and intramolecular H-bonding.

(E)



(Intramolecular H-bonding)

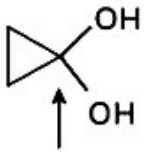
(F)



sp^2, $\theta_{expected}$ = 120°

θ_{actual} = 60°

Angular strain = 60°

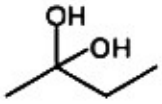


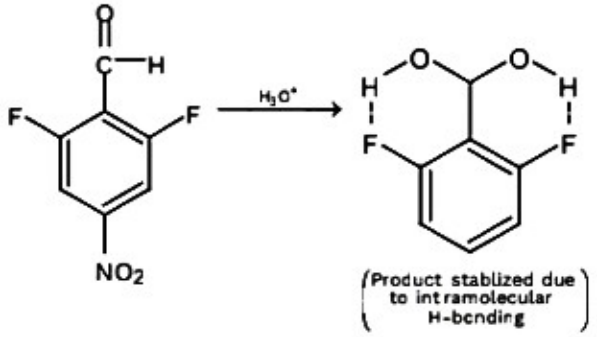
sp^2, $\theta_{expected}$ = 109°

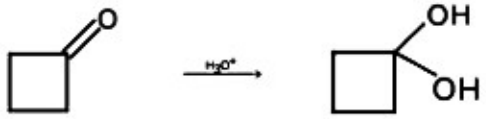
θ_{actual} = 60°

Angular strain = 49°

The product is formed and stable due to relief in angular strain from reactant.

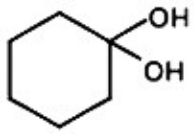
(G)  $sp^3 = 109°28'$

(H) 

(I) 

Strain = 30° Strain = 19°

Product formed due to relief in angular strain.

(J) 

Less stable

The Wittig Reaction :

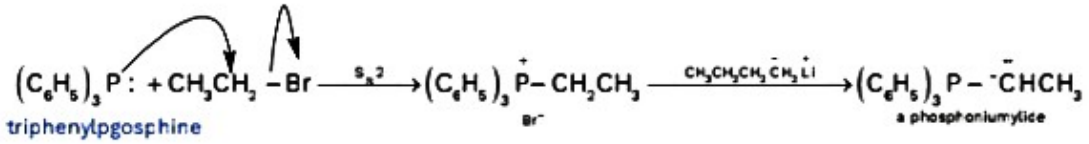
An aldehyde or a ketone reacts with a phosphonium ylide (pronounced "ILL-id") to form an alkene. An **ylide** is a compound that has opposite charges on adjacent covalently bonded atoms with complete octets. The ylide can also be written in the double bonded form because phosphorus can have more than eight valence electrons.

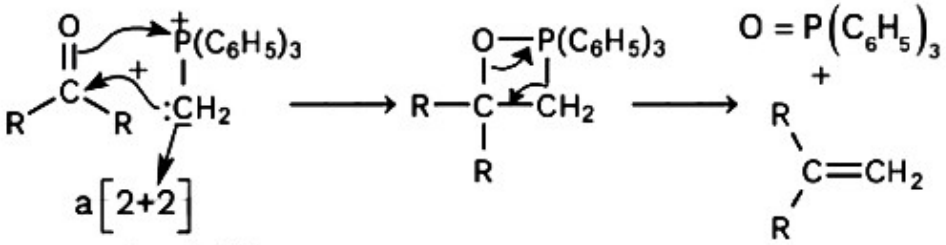
$$(C_6H_5)_3\dot{P}-\bar{C}H_2 \longleftrightarrow (C_6H_5)_3P=CH_2$$

A phosphonium ylide

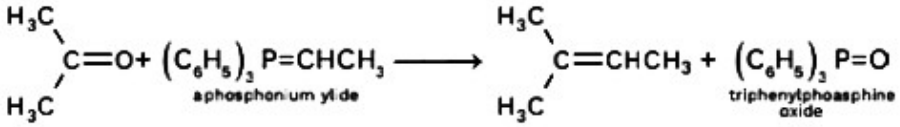
The reaction of an aldehyde or a ketone with a phosphonium ylide to form an alkene is called a **Wittig reaction**. The overall reaction amounts to interchanging the double-bonded oxygen of the carbonyl compound and the double-bonded carbon group of the phosphonium ylide.

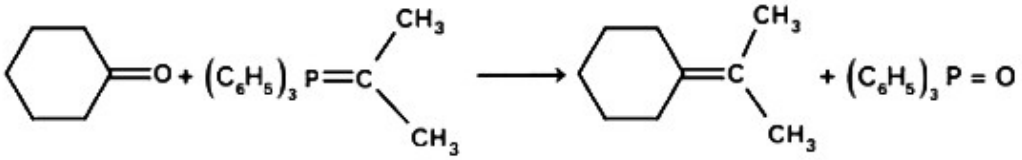
Mechanism :



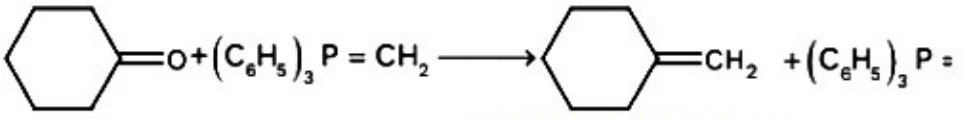


Example:



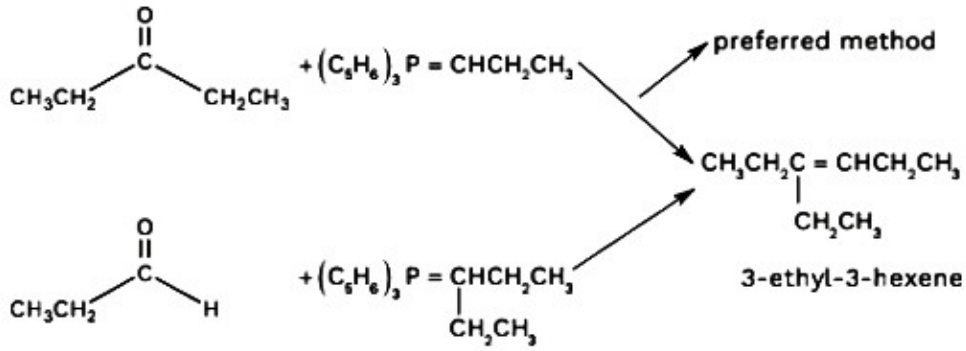


The Witting reaction is a very powerful way to make an alkene because the reaction is completely regioselective—the double bond will be in only one place.



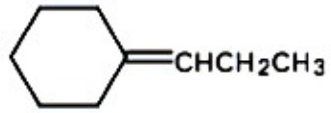
methylenecyclohexane

It is better to use a three-carbon alkyl halide and a five-carbon carbonyl compound than a five-carbon alkyl halide and a three-carbon carbonyl compound for the synthesis of 3-ethyl-3-hexene because it would be easier to form an ylide from 1-bromopropane than from 3-bromopentane.

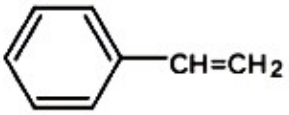


Q31 **What carbonyl compound and what phosphonium ylide are required for the synthesis of the following alkenes ?**

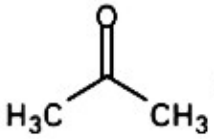
(A) $CH_3CH_2CH_2CH{=}C(CH_3)$ with CH_3 below

(B) 

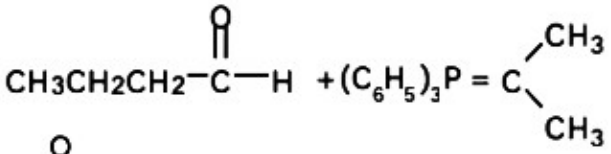
(C) $(C_6H_5)_2C{=}CHCH_3$

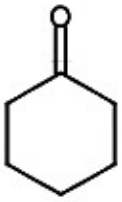
(D) 

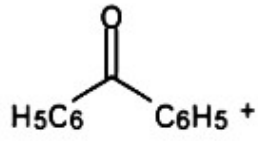
Sol.

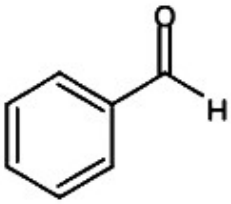
(A)  $+ (C_6H_5)_3P = CHCH_2CH_2CH_3$ (Preferred method)

OR



(B)  $+ (C_6H_5)_3P = CHCH_2CH_3$

(C)  + $(C_6H_5)_3P = CHCH_3$

(D)  + $(C_6H_5)_3P = CH_2$

Q32 **What alkyl halide is required to prepare each of the phosphonium ylides?**

(A) $CH_3CH_2CH_2CH{=}P(C_6H_5)_3$ **(B)** $(Ph)_3P = CHCH_2CH_3$

(C) $(Ph)_3P = CHCH_3$ **(D)** $(C_6H_5)_3P = CH_2$

Sol.

(A) $(C_6H_5)_3P + CH_3CH_2CH_2CH_2{-}Br \xrightarrow{RLi} (Ph)_3P = CHCH_2CH_2CH_3$

(B) $(C_6H_5)_3P + CH_3CH_2CH_2Br$

(C) $(C_6H_5)_3P + CH_3CH_2Br$

(D) $(C_6H_5)_3P + CH_3Br$

Addition Reaction of Hydrogen Cyanide

First Step

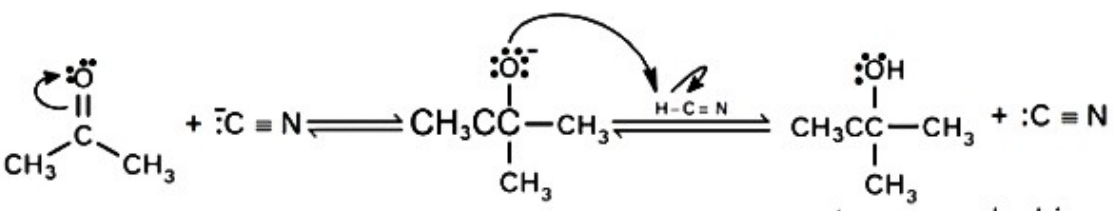
The cyanide ion attacks the carbonyl carbon.

Second Step

The alkoxide ion then accepts a proton from an undissociated molecule of hydrogen cyanide.

Introduction

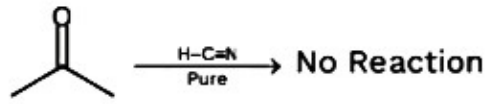
Hydrogen cyanide adds to aldehydes and ketones to form **cyanohydrins**. This reaction forms a product with one more carbon atom than the reactant

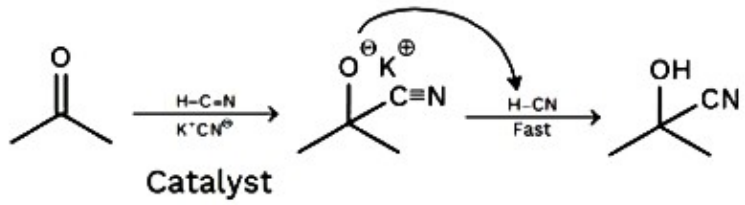


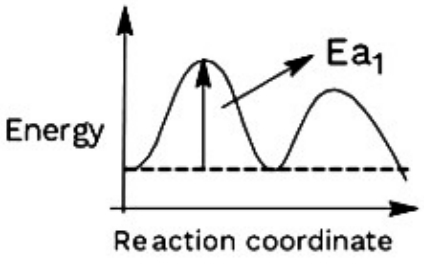
acetone

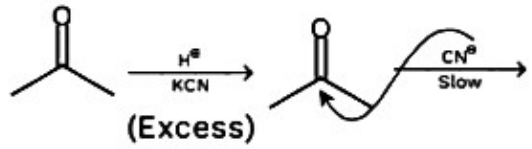
acetone cyanohydrin

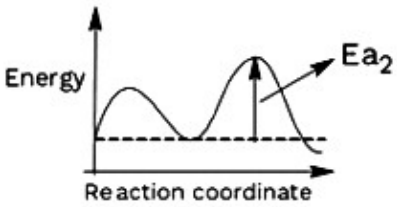
Facts about reaction

(A) 

(B) 



(C) 



Result : $Ea_1 > Ea_2$

$r_2 > r_1$

Conclusion :

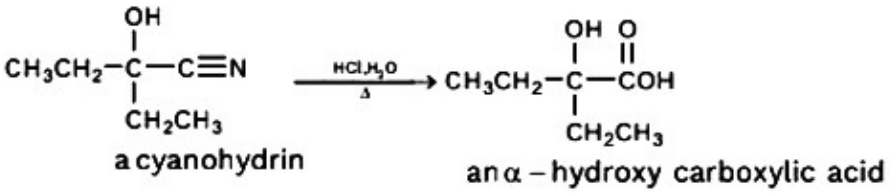
First protonation than addition of nucleophile will be easy.

Q33 **Why HCl is added in reaction to form cyanohydrin (addition product of HCN)**

Sol. Because hydrogen cyanide is a toxic gas, the best way to carry out this reaction is to generate hydrogen cyanide during the reaction by adding HCl to a mixture of the aldehyde or ketone and excess sodium cyanide. Excess sodium cyanide is used in order to make sure that some cyanide ion is available to act as a nucleophile.

Q34 **Why addition of HCN is not possible in case of esters.**

Sol. Cyanide ion does not react with esters because the cyanide ion is a weaker base than an alkoxide ion, so the cyanide ion would be eliminated from the tetrahedral intermediate.
The addition of hydrogen cyanide to aldehydes and ketones is a synthetically useful reaction because of the subsequent reactions that can be carried out on the cyanohydrin.



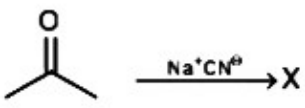
a cyanohydrin — an α – hydroxy carboxylic acid

Q35 **Explain why aldehydes and ketones react with a weak acid such as hydrogen cyanide in the presence of –C≡N but do not react with strong acids such as HCl or H_2SO_4 in the presence of Cl^- or HSO_4^-**

Sol. Strong acids have weak conjugate bases which makes them good leaving groups. HCN is instead a weaker acid making it less likely to be eliminated from the intermediate.

Q36 **Can a cyanohydrin be prepared by treating a ketone with sodium cyanide?**

Sol.



Without protonation reaction not possible.

Q37 **How can the following compounds be prepared, starting with a carbonyl compound with one fewer carbon atoms than the desired product?**

(A) $HOCH_2CH_2NH_2$ **(B)** $CH_3CH(OH)COH$

Sol.

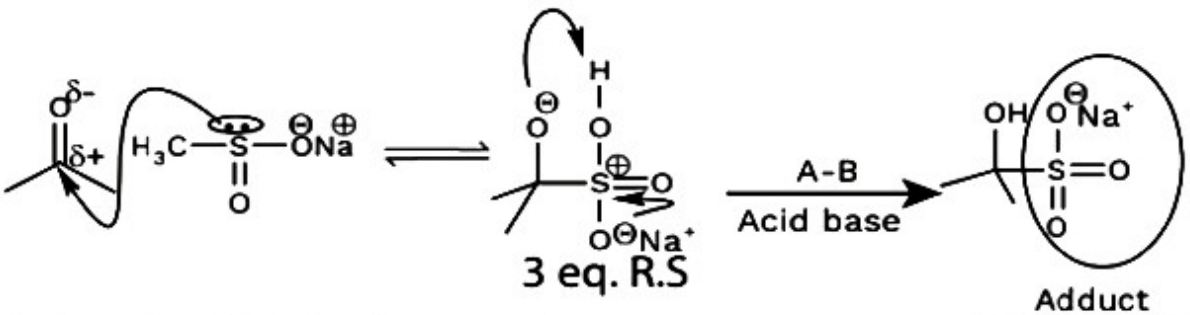
(A) $H-CHO \xrightarrow[KCN]{HCN} HO-CH_2-C\equiv N \xrightarrow[Ni]{H_2} HO-CH_2-CH_2-NH_2$

(B) $H_3C-CHO \xrightarrow[KCN]{HCN} H_3C-CH(OH)-C\equiv N \xrightarrow{H_3O^{\oplus}} H_3C-CH(OH)-COOH$

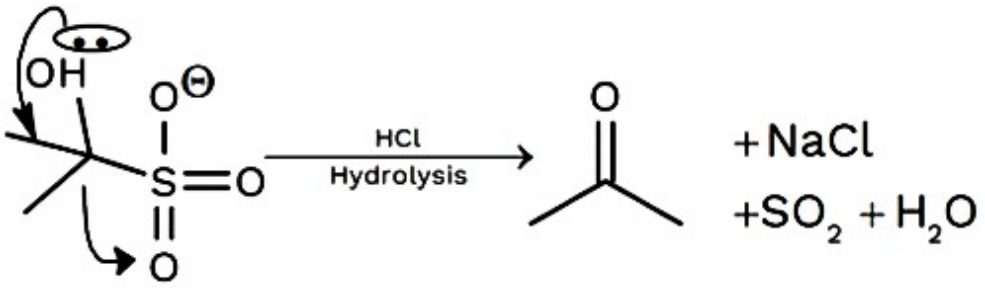
$NaHSO_3$ (White crystal)

Aldehyde and methyl ketone contain one Me group), when treated with saturated solution of sodium disulphite, add a molecule of sodium bisulphite to form bisulphity addition product.
Crystaline solid and addition product reaction is reversible.
Used in purification of carbonyl compound with non carbonyl/compounds.

(1)



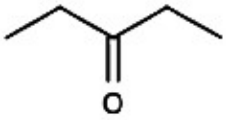
(2) Regeneration of Carbonyl compound



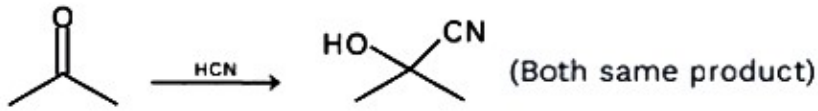
Solved Example :

Isomers of $C_5H_{10}O$ do not show adduct formation Reaction with $NaHSO_3$ (aq)

Ketone do not show adduct formation with $NaHSO_3$(aq). Except methyl ketones.

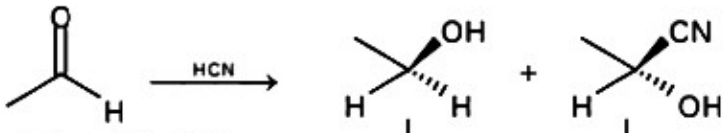


Acetone gives one product with HCN where acetaldehyde gives two compounds that are difficult to separate why?



(Both same product)

(Acetone)



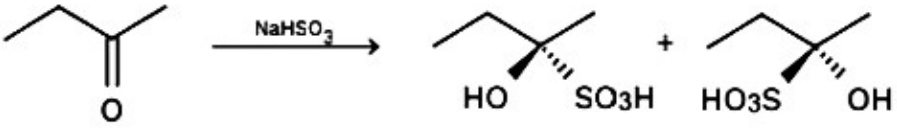
(Acetal adehyde)

(Enantiomers)

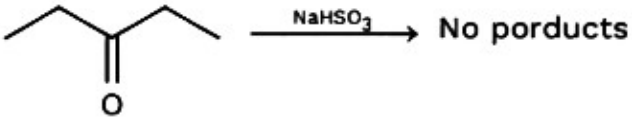
Note: Enantiomers are hard to separate because they have same boiling point.

Solved Example :

Butan-2-one gives sodium bisulphite addition product where pentan-3-one does not why?

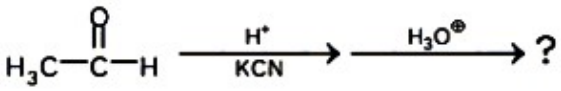


Due to less bulky methyl group formation of product is possible.

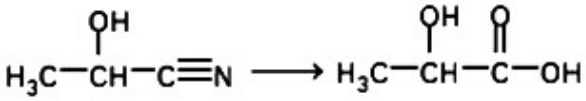


While in pentan-3-one due to bulky ethyl group product fromation is not possible.

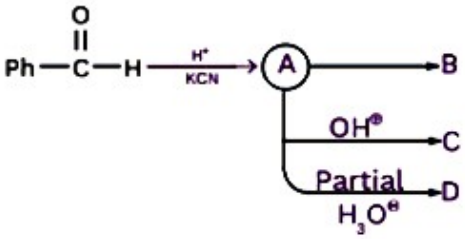
Q38



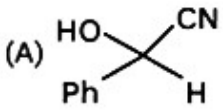
Sol.

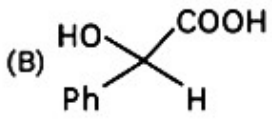


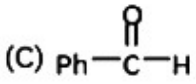
Q39 **Draw all products A, B, C and D**

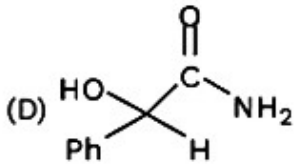


Sol.

(A) 

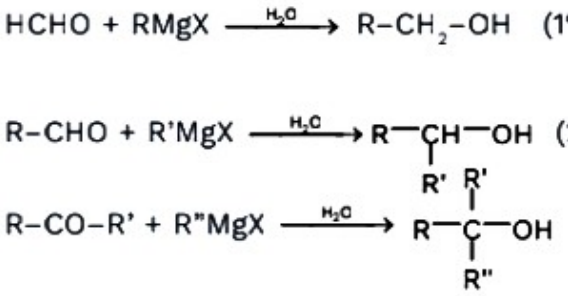
(B) 

(C) 

(D) 

Addition of Grignard Reagent:

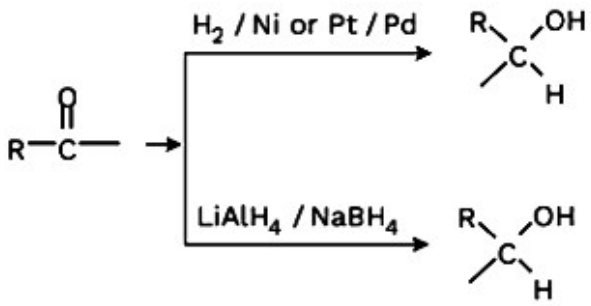
Aldehydes and ketones react with Grignard reagent to form and adduct which on hydrolysis yields the corresponding alcohols



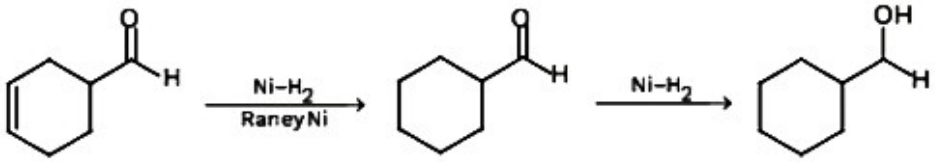
Reduction of carbonyl compounds:

(a) Reduction to alcohols :

Example

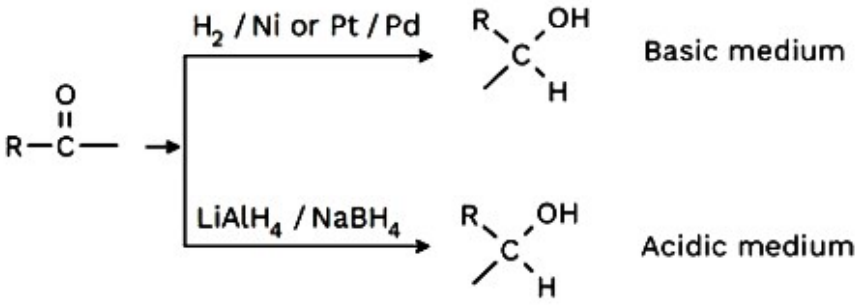


Example a)



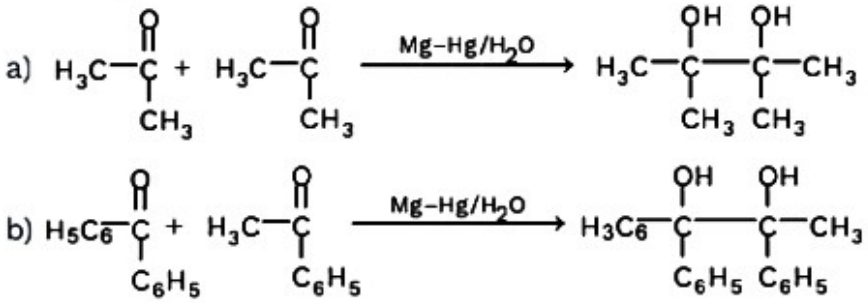
b) $CH_3 - CH_2 - CH_2 - \overset{O}{\overset{\|}{C}} - CH_3 \xrightarrow[2.\ H_3O^+]{1.\ NaBH_4} CH_3 - CH_2 - CH_2 - \overset{OH}{\underset{|}{C}}H - CH_3$

(b) Reduction to alkane :



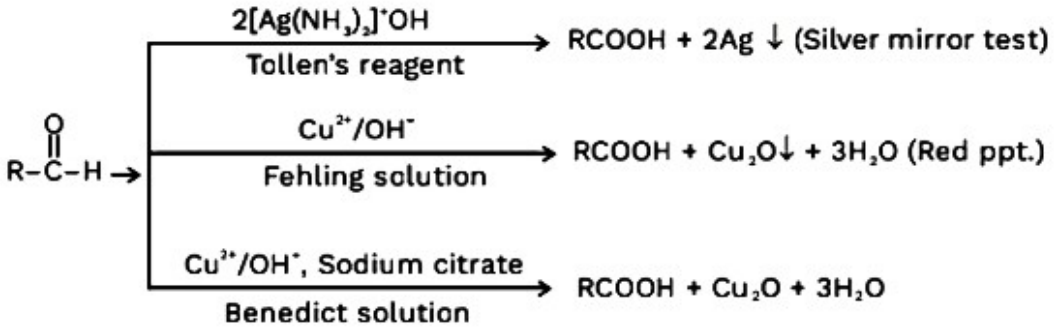
(c) Reduction to pinacols:

Example



Oxidation of carbonyl compounds:

a. Aldehydes are oxidised to carboxylic acids with mild oxidising agent

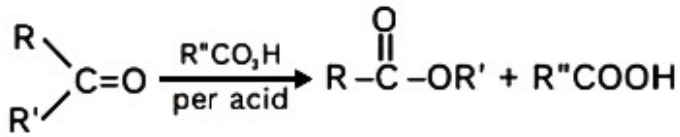


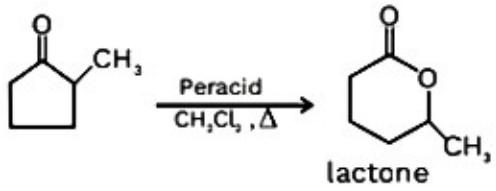
b Schiff's reagent:

It is an aqueous solution of magenta or pink coloured rosaniline hydrochloride which has been decolourised by passing SO_2. Aldehyde restores red colour when treated with schiff's reagent. This reaction is used as a test for aldehydes, because ketones do not restore the pink colour of schiff's reagent.

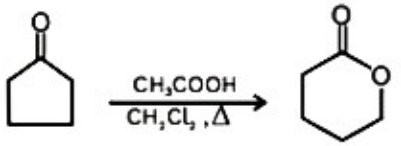
c. Baeyer villiger oxidation:

Both aldehyde and ketones are oxidized by peroxy acids. This reaction is called baeyer-villiger oxidation. This reaction is specially useful with ketones, because it converts to esters.



Ex. 

lactone

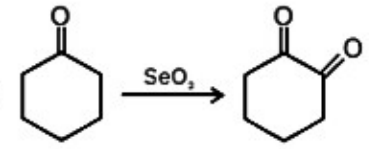
Ex. 

d. Oxidation by selenium dioxide (SeO_2):

Carbonyl compounds containing a methylene group adjacent to a carbonyl group are oxidised by SeO_2 to a dicarbonyl compound.

Ex. $CH_3CHO \xrightarrow{SeO_2} \begin{matrix} CHO \\ | \\ CHO \end{matrix}$ (Glyoxal)

Ex. $CH_3-COCH_2-CH_3 \xrightarrow{SeO_2} CH_3-\overset{O}{\overset{\|}{C}}-\overset{O}{\overset{\|}{C}}-CH_3$

Ex. 

Oxidation by strong oxidising agents:

Strong oxidising agents are $KMnO_4/O\overset{\ominus}{H}/\Delta$, $K_2CrO_7/H^+/\Delta$, Con. HNO_3/Δ

(a) Oxidation of aldehydes:

$RCHO \xrightarrow{[O]} RCOOH$

$C_6H_5CHO \xrightarrow{[O]} C_6H_5COOH$

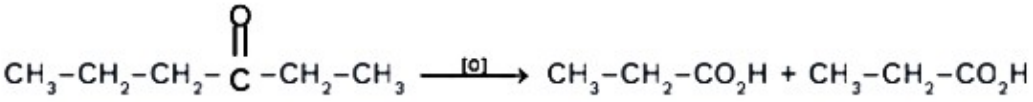
(b) Oxidation of ketones:

Ketones undergo oxidation only in drastic condition:

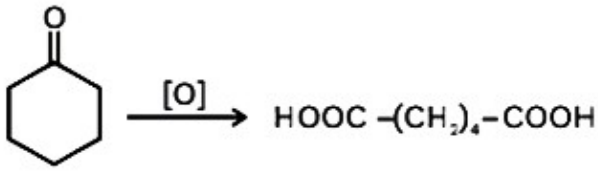
Example

$CH_3-CH_2-CH_2-\overset{O}{\overset{\|}{C}}-CH_2-CH_2-CH_3 \xrightarrow{[O]} CH_3-CH_2-CH_2-CO_2H + CH_3-CH_2-CO_2H$

Example



Example

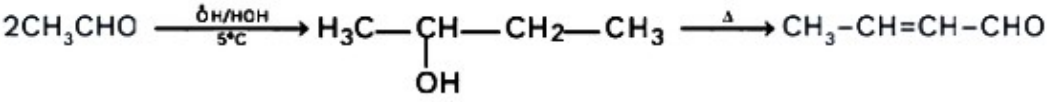


Introduction

In aqueous base, two acetaldehyde molecules react to form a β-hydroxy aldehyde called aldol. The reaction is called Aldol condensation.

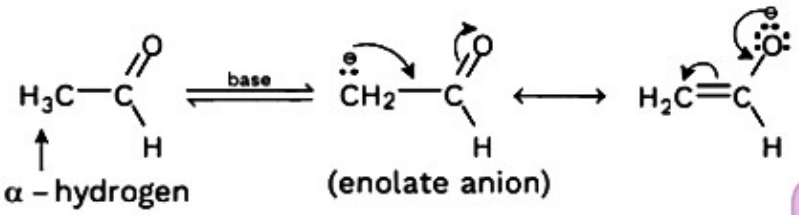
Aldol Reaction:

The enolate ion is the intermediate in the aldol condensation of aldehyde and ketone. Acetaldehyde for instance, forms a dimeric product aldol in presence of a dilute base (»10% NaOH).



β-hydroxy butyraldehye (Aldol)

Acidic Nature of Hydrogen :



α – hydrogen (enolate anion)

Points to remember

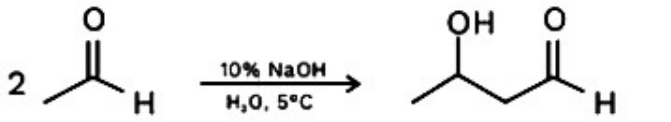
The alpha-hydrogen of a carbonyl compound is acidic due to the fact that the anion (also known as the enolate anion) is stabilized by resonance.

Aldol Addition Reaction :

Aldol additions and aldol condensations together represent an important class of c–c bond-forming reactions.

An aldol reaction begins with addition of an enolate or enol to the carbonyl group of an aldehyde or

ketone, leading to a β-hydroxy aldehyde or ketone as the initial product

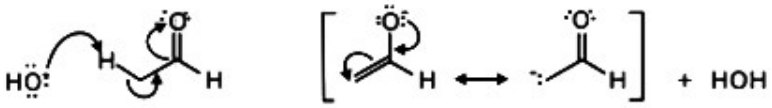


3-Hydroxybutanal
(50%)

Aldol Addition Reactions :

An aldol addition is an equilibrium reaction when it is conducted in a protic solvent with a base such as hydroxide or an alkoxide.

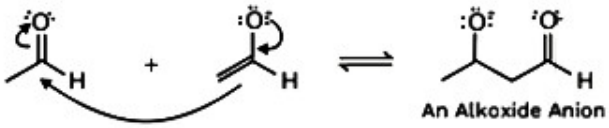
Step 1 *Enolate formation*



Enolate anion

In this step the base (a hydroxide ion) removes a proton from the α carbon of one molecule of acetaldehyde to give a resonance-stabilized enolate.

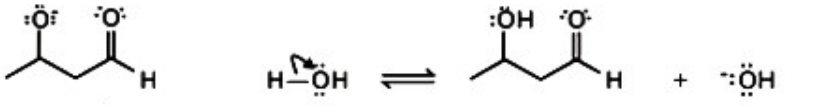
Step 2 *Addition of the enolate*



An Alkoxide Anion

The enolate then acts as a nucleophile and attacks the carbonly carbon of a second molecule of acetaldehyde, proucing an alkoixde anion.

Step 3 *Protonation of the alkoxide*



Stronger base — The aldol product — Weaker base

The alkoxide anion now removes a proton from a molecule of water to from the aldol product.

Dehydration of the Aldol Addition Product:

Dehydration of an aldol addition product leads to a conjugated α, β-unsaturated carbonyl system. The overall process is called an aldol condensation.

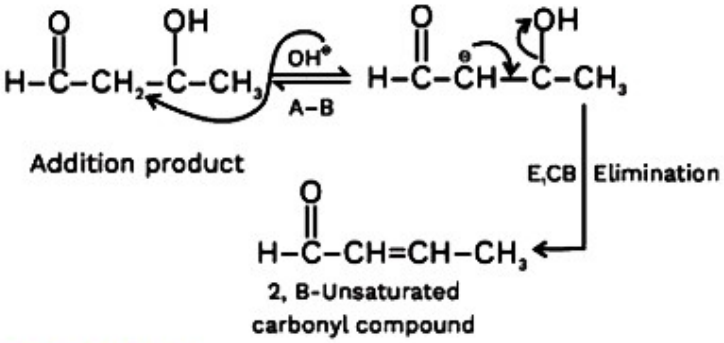
Mechanism :

Dehydration of the Aldol Addition product :

Introduction

The stability of the conjugated enal or enone system means that the dehydration equilib-rium is essentially irreversible.

Mechanism :



Some Facts :

(i) In aldol reaction atletst one α-H required and in aldol condensation, two α-H required.

(ii) In aldol reaction dil base in required.

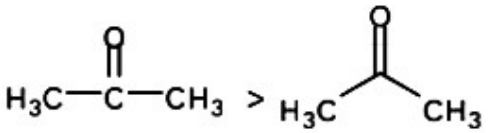
dil KOH | NaOH | $Ca(OH)_2$ | $Mg(OH)_2$

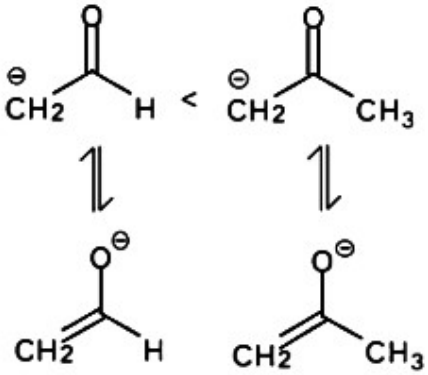
*Ag $NaHCO_3 \rightarrow NaOH + N_2CO_3$

S.B. W.A.

(iii) Aldol is a reversible reaction and E_1 CB is a driving force to forward.

(iv) Less steric reactant and stable enolate ion give more the rate of reaction.

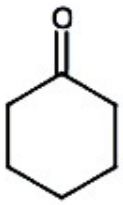
(A) 

(B) 

Q40 **Write the product and mechanism for given reactions.**

(i) $H_3C-CH_2-\underset{\underset{O}{\|}}{C}-CH_3 \xrightarrow{\text{Dil.NaOH}} (A) \xrightarrow{\Delta} (B)$

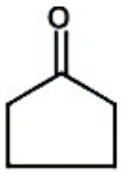
(ii)



$\xrightarrow{\text{Dil.NaOH}} (C)$

(iii) $C_6H_5-CH_2-CHO \xrightarrow{\text{Dil.NaOH}} (D)$

(iv)

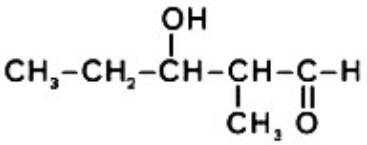


$\xrightarrow{\text{Dil.NaOH}} (E) \xrightarrow{\Delta} (F)$

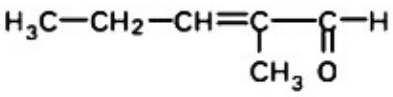
(v) $C_6H_5-\underset{\underset{O}{\|}}{C}-CH_3 \xrightarrow{\text{Dil.NaOH}} (G) \xrightarrow{\Delta} (H)$

Sol.

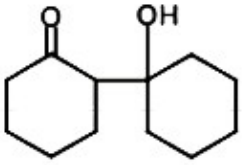
(I) (A)



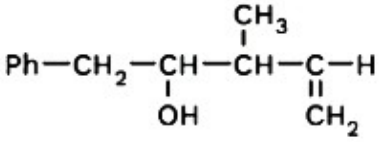
(B)



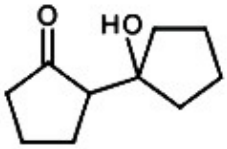
(II) (C)



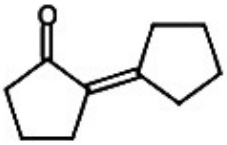
(III) (D)



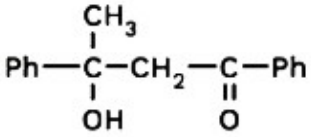
(IV) (E)



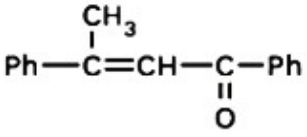
(F)



(V) (G)

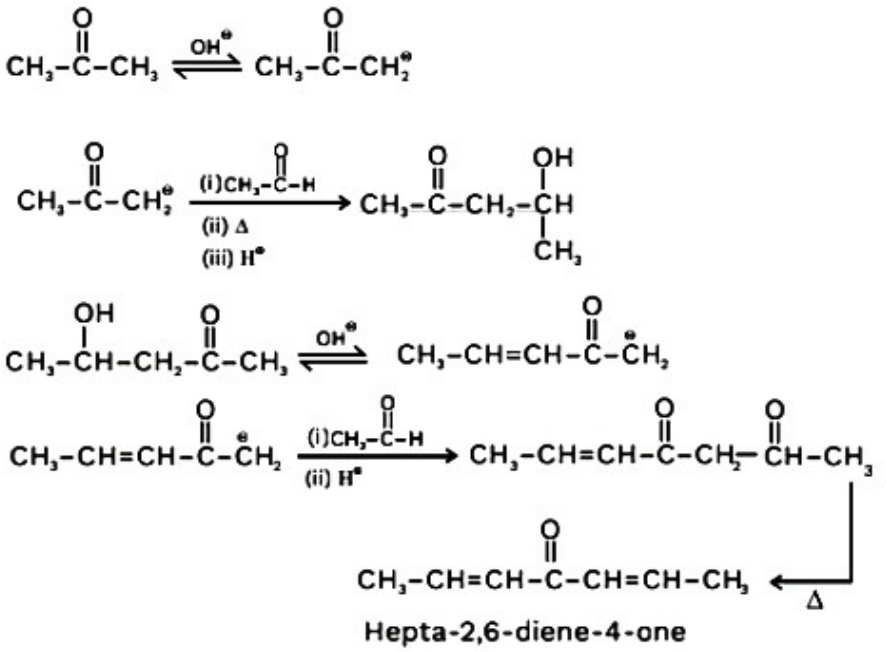


(H)



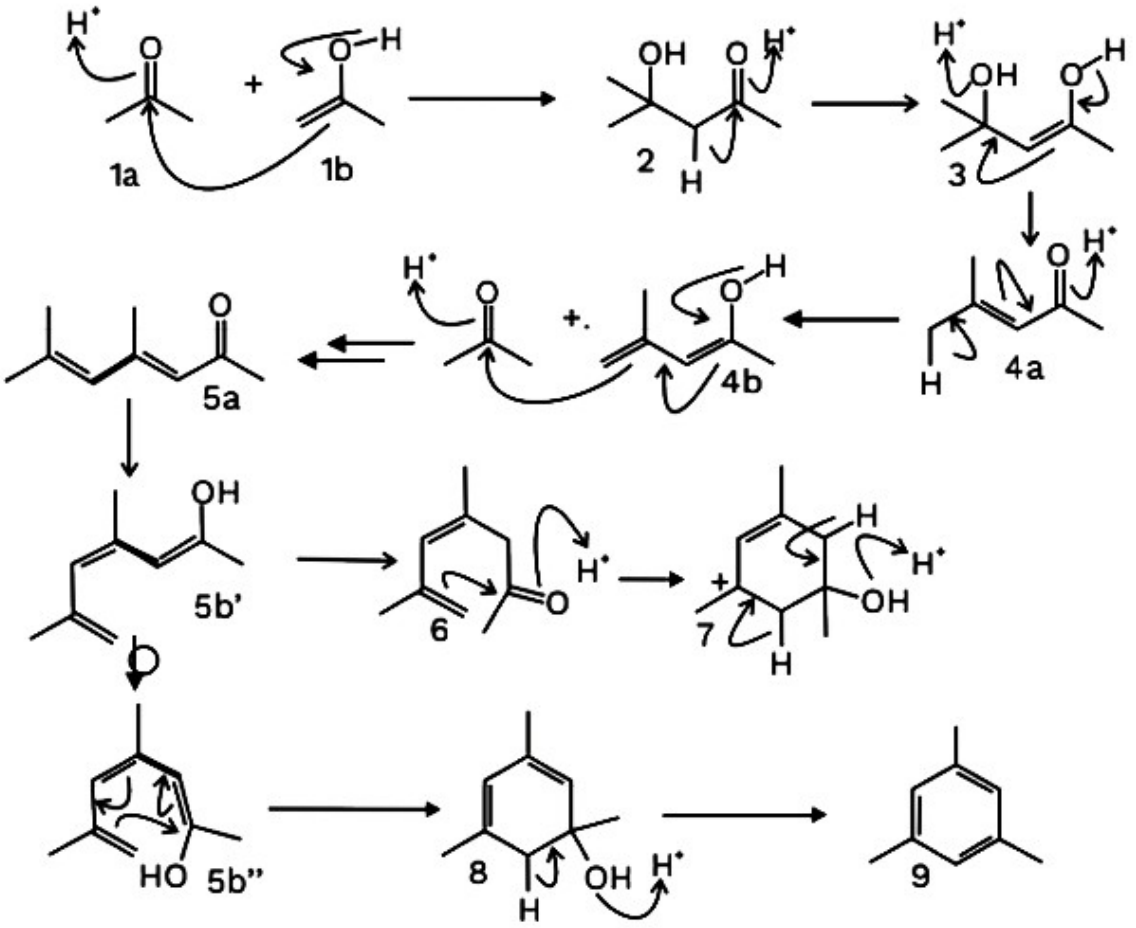
Q41 **The acid-catalyzed aldol condensation of acetone (just shown) also products some 2,6-dimethylhepta-2,5-dien-4-one. Give a mechanism that explains the formation of this product.**

Sol.



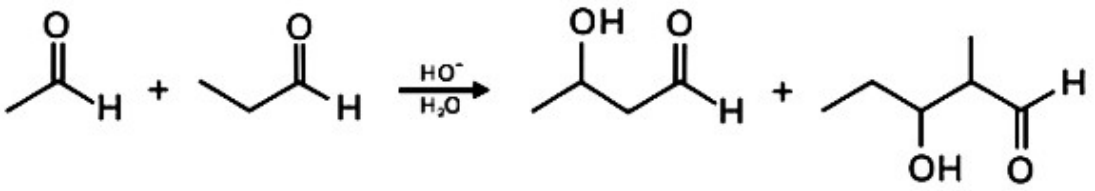
Q42 **Heating acetone with H_2SO_4 leads to the formation of mesitylene (1,3,5-trimethylbenzene). Propose a mechanism for this reaction.**

Sol.

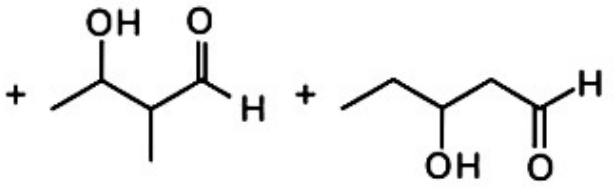


Crossed Aldol Condensations :

An **aldol reaction** that starts with two different carbonyl compounds is called a **crossed aldol reaction**. Unless specific conditions are involved, a crossed aldol reaction can lead to a mixture of products from various pairings of the carbonyl reactants.



Ethanal Propanal 3-Hydroxybutanal (from two molecules of ethanal) 3-Hydroxy-2-methylpentanal (from two molecules of propanal)



3-Hydroxy-2-methylbutanal 3-Hydroxypentanal
(from one molecule of ethanal and one molecule of propanal)

Q43 **Find out the total number of possible aldol products (including and excluding stereo products)**

(I) $CH_3-CHO + CH_3-CH_2-CHO \xrightarrow[5°C]{NaOH/HOH}$

(II) $C_6H_5-CHO + CH_3-CHO \xrightarrow[5°C]{NaOH/HOH}$

(III) $CH_3-CHO + CH_3-\underset{\underset{O}{\|}}{C}-CH_3 \xrightarrow[5°C]{NaOH/HOH}$

(IV) $CH_3-CH=O + CH_3-CH_2-\underset{\underset{O}{\|}}{C}-CH_3 \xrightarrow[5°C]{NaOH/HOH}$

(V) $C_6H_5-CHO + CH_3-CH_2-\underset{\underset{O}{\|}}{C}-CH_3 \xrightarrow[5°C]{NaOH/HOH}$

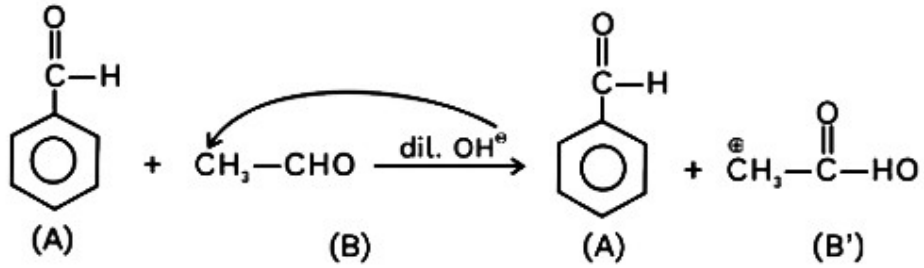
Sol.

	Excluding Stereo	Including Stereo
(I)	4	12
(II)	2	4
(III)	4	6
(IV)	6	16
(V)	4	12

Claisen–Schmidt condensations :

Claisen schmidt :

One compound can't have α-H so that self aldol is not possible.



A + B' = $CH_3-CH=CH-\overset{O}{\overset{\|}{C}}-H$

Crotonaldehyde

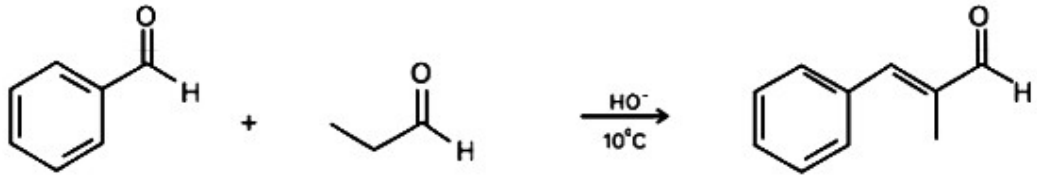
B + B' = $Ph-CH=CH-\overset{O}{\overset{\|}{C}}-H$

Cinnamaldehyde

The reactant with no α hydrogen is placed in base

The reactant with an α hydrogen is added slowly

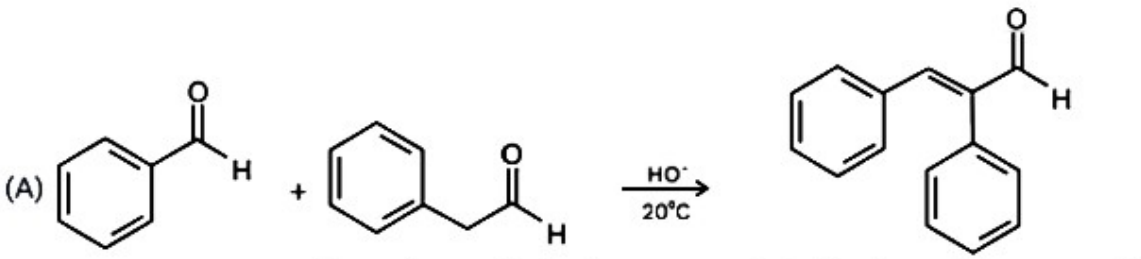
Product



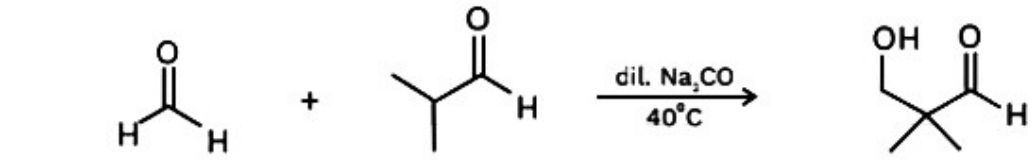
Benzaldehyde

Propanal

2-Methyl-3-phenyl-2-propenal
(α-methylcinnamaldehyde)
(68%)



Benzaldehyde Phenylacetaldehyde 2,3-Dipehenyl-2-propenal

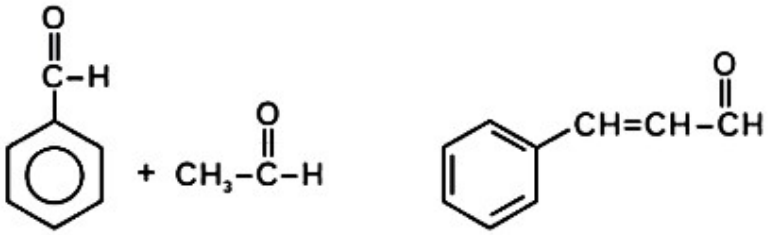


(B)Formaldehyde 2-Methylpropanal 3-Hydroxy-2,2-dimethylpropanal (>64%)

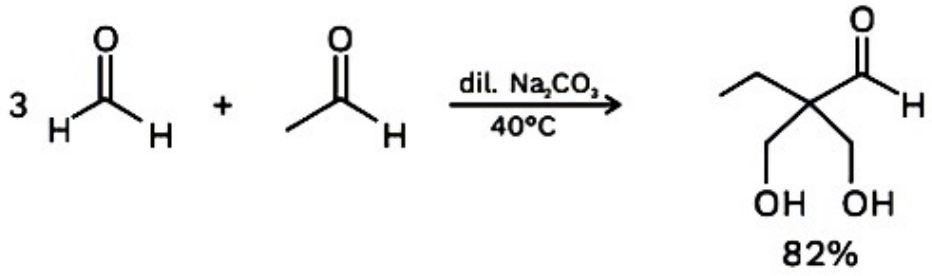
Q44 **Outlined below is a practical crossed aldol reaction that can be used for the synthesis of cinnamaldehyde (the essence of cinnamon, used in cooking). Provide the missing ingredients for this recipe.**



Sol.

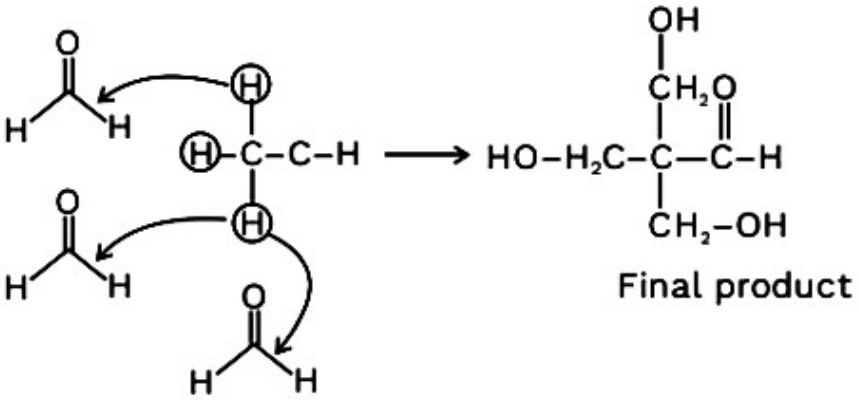


Q45 **When excess formaldehyde in basic solution is treated with ethanal, the following reaction takes place :**

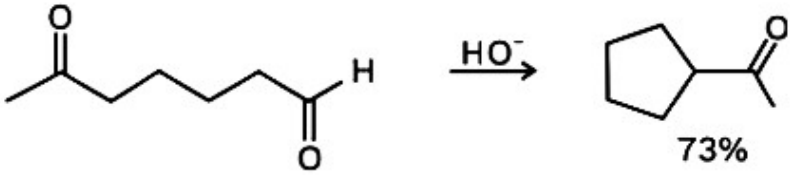


Write a mechanism that accounts for the formation of the product.

Sol.



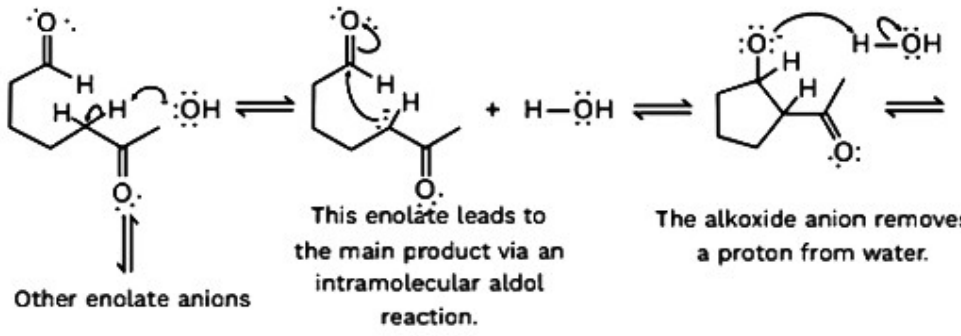
The Aldol Cyclization Intramolecular Aldol reaction

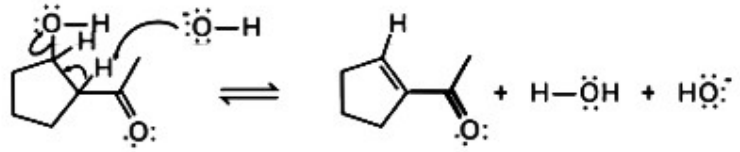


Points to remember

The carbonyl carbon atom of a ketone is less positive (and therefore less reactive toward a nucleophile) because it bears two electron-releasing alkyl groups; it is also more sterically hindered

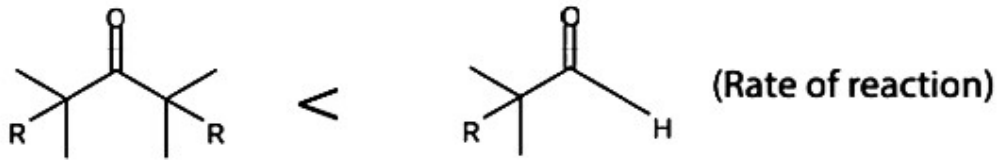
Mechanism :

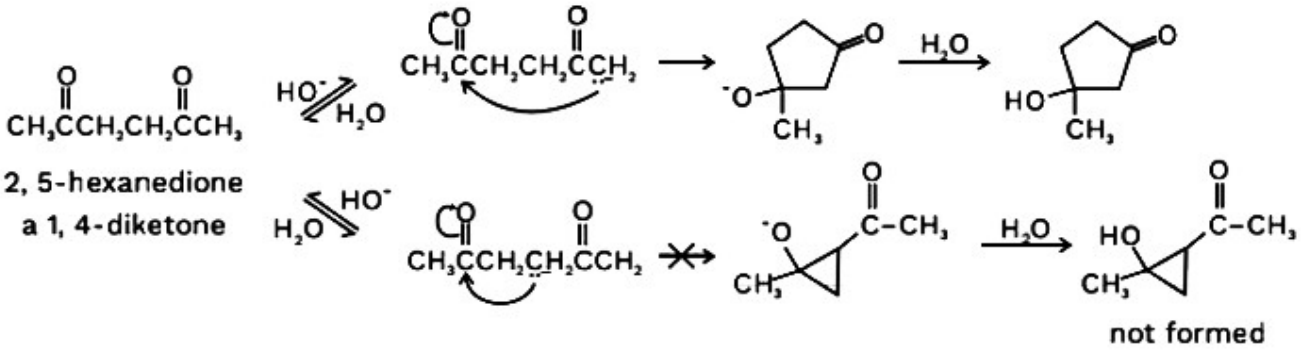


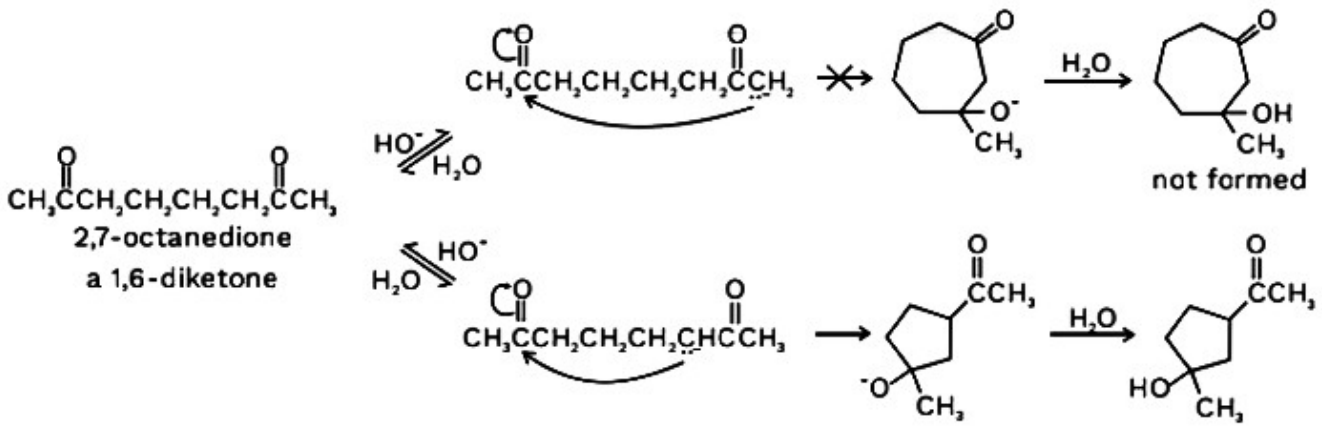


The reason the aldehyde group undergoes addition preferentially may arise from the greater reactivity of aldehydes toward nucleophilic addition generally.

Example:







Q46 **Identify the intramolecular aldol product?**

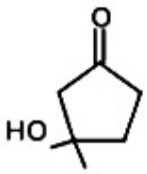
(I) $CH_3-\overset{O}{\overset{\|}{C}}-CH_2-CH_2-\overset{O}{\overset{\|}{C}}-CH_3 \xrightarrow{\text{Dil NaOH}} (A) \xrightarrow{\Delta} (B)$

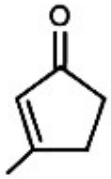
(II) $CH_3-\overset{O}{\overset{\|}{C}}-CH_2-CH_2-CH_2-CH_2-\overset{O}{\overset{\|}{C}}-CH_3 \xrightarrow{\text{Dil NaOH}} (C)$

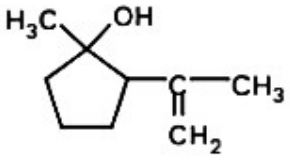
(III) $CH_3-\overset{O}{\overset{\|}{C}}-CH_2-CH_2-CH_2-\overset{O}{\overset{\|}{C}}-CH_3 \xrightarrow{\text{Dil NaCH}} (D)$

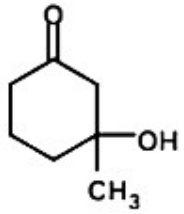
(IV) $CH_3-\overset{O}{\overset{\|}{C}}-CH_2-CH_2-CH_2-CH_2-\overset{O}{\overset{\|}{C}}-CH_3 \xrightarrow[\Delta]{\text{Dil NaOH}} (G)$

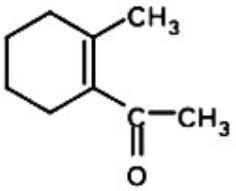
Sol.

(I) (A) 

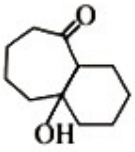
(B) 

(II) (C) 

(III) (D) 

(IV) (G) 

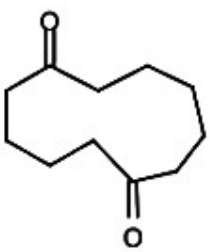
Q47 Identify the structure of substrate?

(I) A $\xrightarrow[5^{\circ}C]{DINaOH}$ 

A = ?

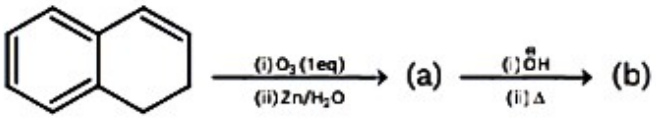
(II) $\underset{(2\,mole)}{A} \xrightarrow[5^{\circ}C]{DINaOH} CH_3-\overset{\overset{O}{\|}}{C}-CH_2-\overset{\overset{OH}{|}}{\underset{\underset{CH_3}{|}}{C}}-CH_3$ A = ?

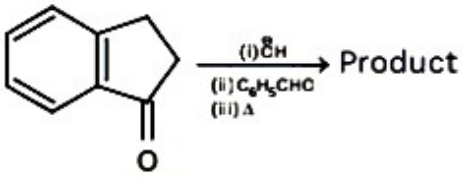
Sol.

(I) 

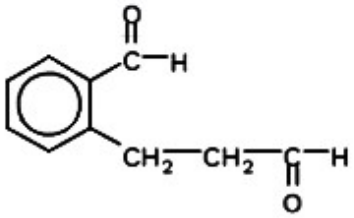
(II) $H_3C-\overset{\overset{O}{\|}}{C}-CH_3$

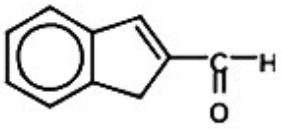
Q48 **Complete reaction sequence:**

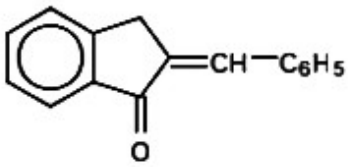
(i) 

(ii) 

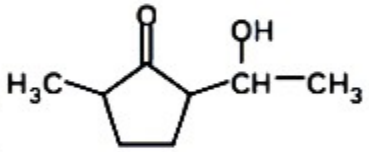
Sol.

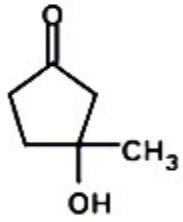
(i) (a) 

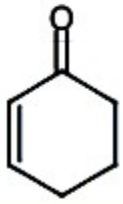
(b) 

(ii) 

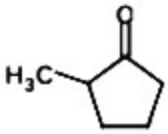
Q49 Identify reactants used in following aldol reactions.

(i) (X) + (Y) $\xrightarrow{\ominus OH}$ 

(ii) (X) $\xrightarrow{\ominus OH}$ 

(iii) [X] $\xrightarrow[(ii)\ \Delta]{(i)\ \ominus OH}$ 

Sol.

(i) X =  Y = $H_3C-\overset{\overset{O}{\|}}{C}-H$

(ii) X = $H_3C-\overset{\overset{O}{\|}}{C}-CH_2-CH_2-\overset{\overset{O}{\|}}{C}-CH_3$

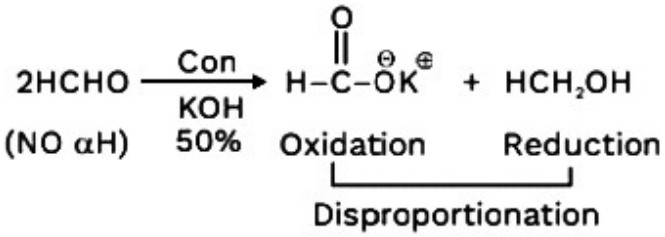
(iii) X = $H_3C-\overset{\overset{O}{\|}}{C}-CH_2-CH_2-CH_2-CHO$

Q50

$$Ph-\overset{\overset{O}{\|}}{C}-H + CH_3-NO_2 \xrightarrow[\Delta]{\ominus OH} E$$

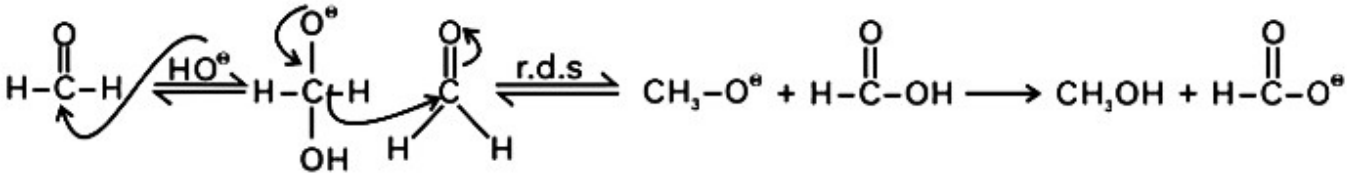
Sol. $Ph - CH = CH - NO_2$

Cannizaro Reaction:



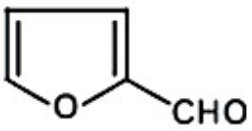
This reaction is given by aldehyde having no α- hydrogen in the presence of conc. NaOH/Δ or KOH/Δ.

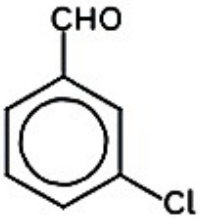
Mechanism

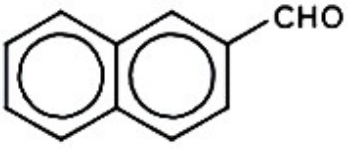


Order of reaction will be 3 and molecularity is also 3.

Q51 **Which of the following will not undergo Cannizaro reaction :**

(A) 

(B) 

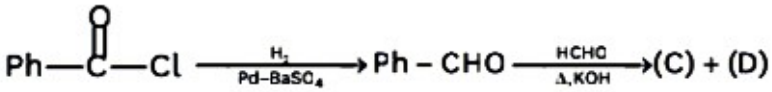
(C) 

(D) Cl_3C-CHO

Sol. (D)

As Cl_3C-CHO given chloroform

Q52



Product (C) and (D) are :

(A) $Ph-CO_2H$, $Ph-OH$ **(B) $Ph-CO_2^-$, HCO_2^-**

(C) $Ph-C_2OH$, $H-CO_2^-$ **(D) $Ph-CO_2^-$, CH_3OH**

Sol. (C)

In cross cannizzaro reactrion move reactive carbonyl ompound is oxidized and less in reduced.

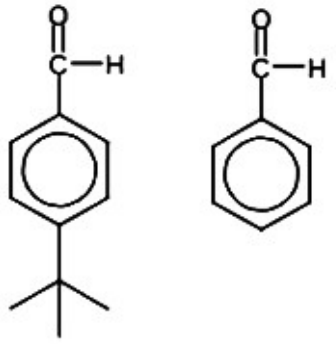
Some facts to know about Cannizzaro :

(i) In Cannizaro reaction, compound do not have α-hydrogen.

(ii) High con. Base is required so that reaction take place in forward direction.

(iii) H-shift is RDS, it means more electron donor hydride more the rate of reaction.

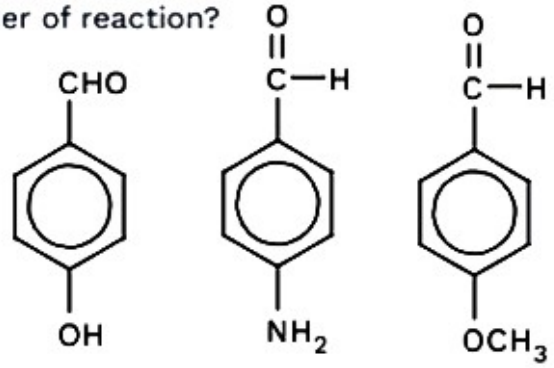
(a)



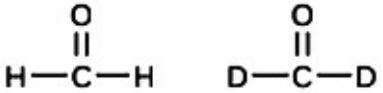
A > B ? B >A ?

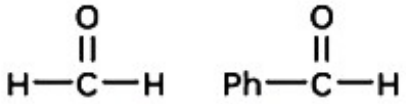
Order of reaction?

(b)

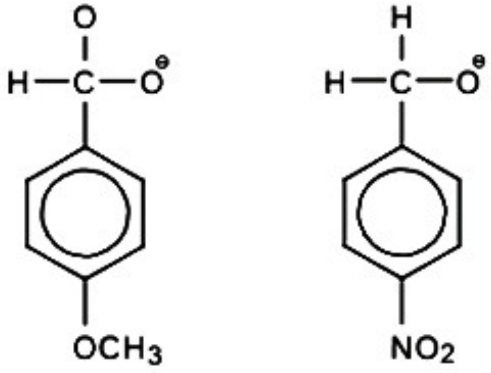


(C)



(D) 

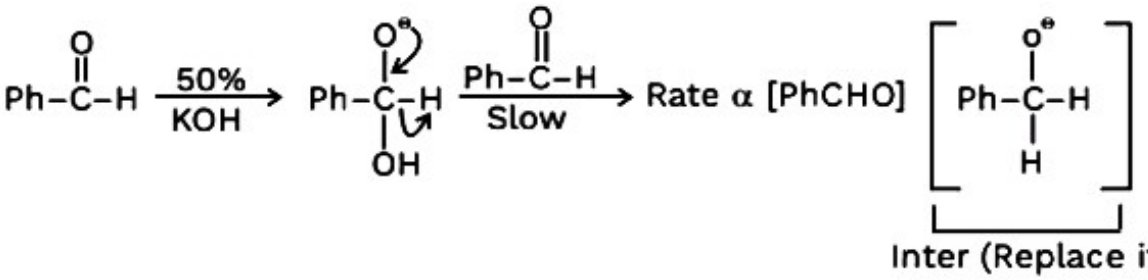
A > B (Less Steric)

(F) 

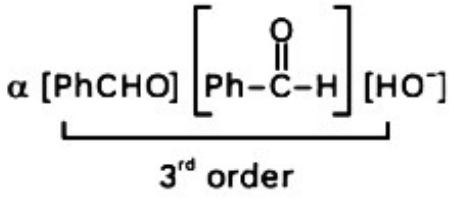
A > B why?

(iv) Cannizaro reaction is a bimolecular 3rd order but in presence of 60% or more it is interconvert into 4th order.

3rd order



Inter (Replace it)



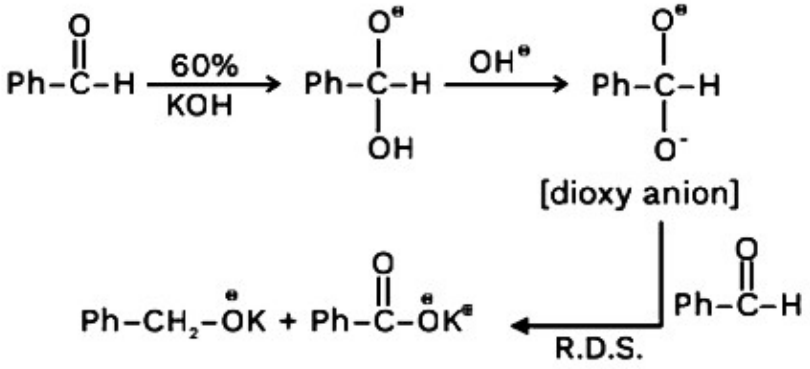
3rd order

Bimolecular

Question

What will be order of reaction when base is 60% or more concentrated ??

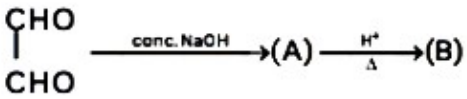
In the presence of a very strong concentration of alkali, aldehyde first forms a doubly charged anion (I) from which a hydride anion is transferred to the second molecule of the aldehyde to form acid and an alkoxide ion. Subsequently, the alkoxide ion acquires a proton from the solvent.



Rate α $[PhCHO]^2[OH^-]^2$

Derivation ?

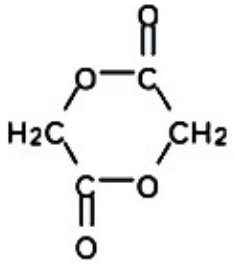
Q53



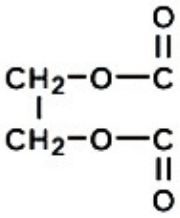
Product (B) is :

(A) $H_2C=CH-CO_2H$

(B)



(C)



(D) $H_2C=C=O$

Sol. (B)

Intramolecular cannizaro foltlowed by heating effect

Q54 $HCHO \xrightarrow[\Delta]{NaOD}$

Sol. $CH_3OD + HCOONa$

Q55 $DCHO \xrightarrow[\Delta]{NaOD}$

Sol. $DCH_2OD + DCOONa$

Q56 $PH-CHO \xrightarrow[\Delta]{^{\oplus}OD/DOD}$

Sol. $PhCH_2OD + PhCOONa$

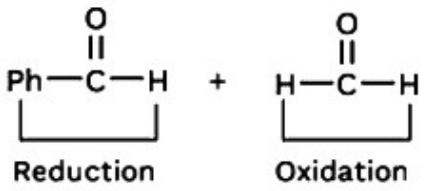
Q57 $Ph-CHO \xrightarrow[\Delta]{^{18}\overset{\oplus}{O}H}$

Sol. $PhCH_2OH + PhCO\overset{18}{O}Na$

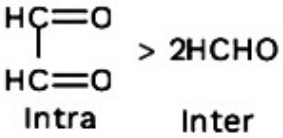
Redox Cannizaro or Cross Cannizaro

(v) It two different type of compounds then Cannizaro is a redox reaction.

Example:



* More reactive oxidation, less reactive reduction.



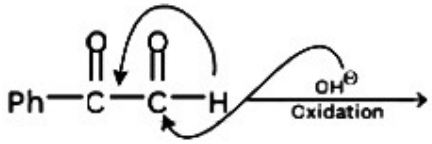
> 2HCHO

Inter

Q58 $Ph-CHO + HCHO \xrightarrow[\Delta]{KOH}$

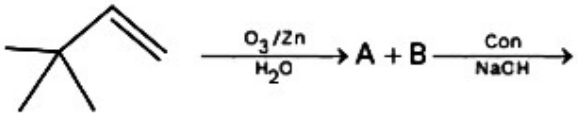
Sol. $Ph-CH_2OH + HCOOK$

INTRAMOLECULAR CANNIZARO

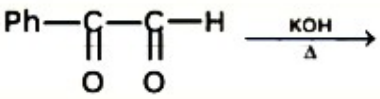
(vi) 

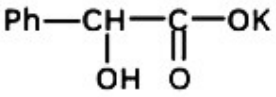
Intramolecular/2nd order

Question

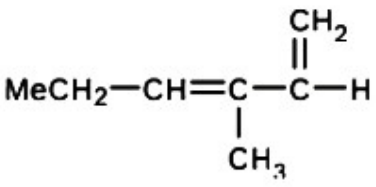


Solution

Q59 

Sol. 

Q60 $MeCH_2-CHO \xrightarrow[\Delta]{KCH}$

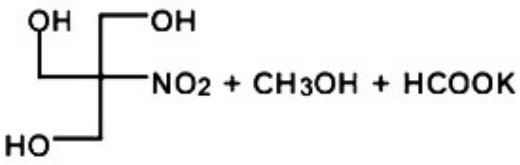
Sol. 

Q61 $MeCH_2-CHO \xrightarrow[\Delta]{conc.KOH}$

Sol. $Me_2CH-CH_2OH + Me_2CHCOOK$

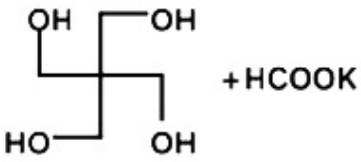
Q62 $Me-NO_2 + HCHO \xrightarrow[\text{(excess)}]{KOH}$

Sol.



Q63 $MeCHO-HCHO \xrightarrow[\text{(excess)}]{KOH}$

Sol.

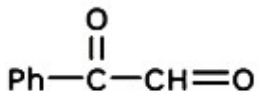


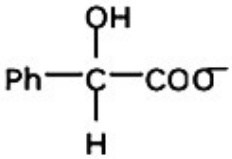
Q64 $PHCOCHBr_2 \xrightarrow{Aq.\bar{O}H} (A) \xrightarrow{Conc.\bar{O}H} (B) \xrightarrow{H^+}$; The compound 'C' is:

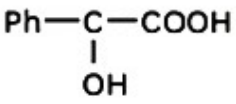
(A) PhCH(OH)CHO (B) PhCH(OH)COOH

(C) PhCOOH (D) None of these

Sol. (B)

(A) 

(B) 

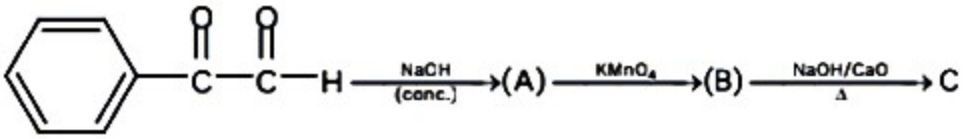
(C) 

Q65 Match the column :

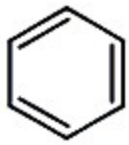
Column-I	**Column-II**
(A) HCHO + NaOD (conc.)	(P) $DCOO^- + CDH_2OH$
(B) DCHO + NaOH (conc.)	(Q) $DCOO^- + CD_3OH$
(C) DCDO + NaOH (conc.)	(R) $DCOO^- + CDH_2OD$
(D) DCHO + NaOD (conc.)	(S) $HCOO^- + CH_3OD$

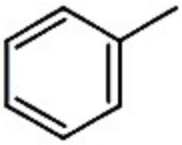
Sol. (A) → S; (B)→ P; (C) → Q; (D) → R

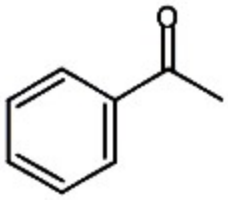
Q66

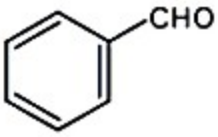


Product (C) is :

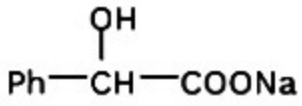
(A) 

(B) 

(C) 

(D) 

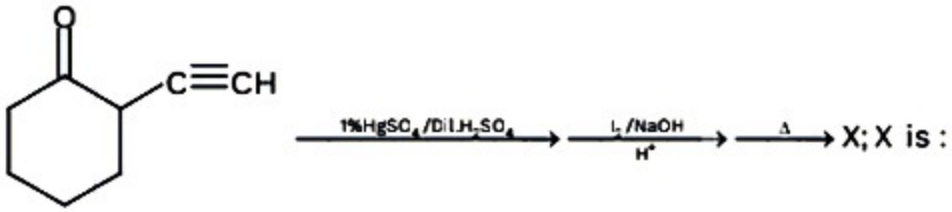
Sol. (A)

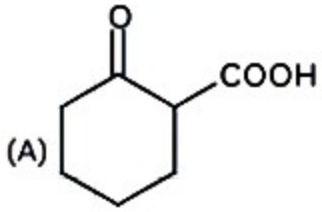
(A) 

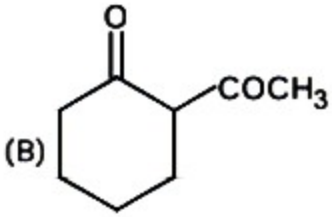
(B) PhCOOH

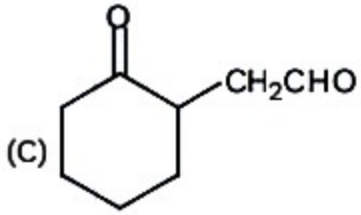
(C) 

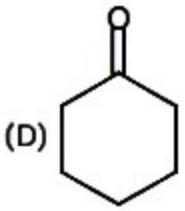
Q67



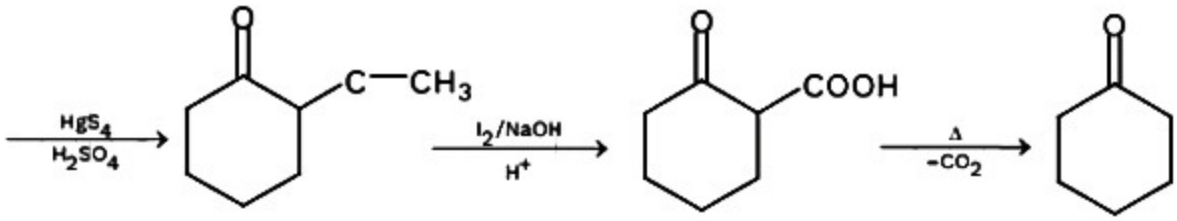
(A) 

(B) 

(C) 

(D) 

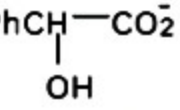
Sol. (D)



Q68

The cannizaro reaction of PhCOCHO from the product(s)

(A) $PhCOCH_2OH + PhCOCO_2^-$

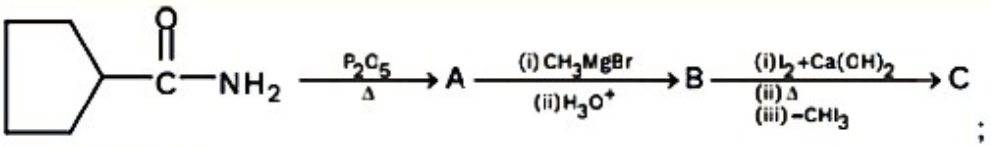
(B) 

(C) $PhCO_2^- + PhCOCH_2OH$

(D) Both (A) and (C)

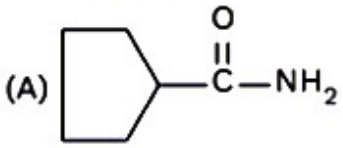
Sol. (B)

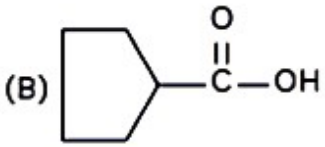
Q69

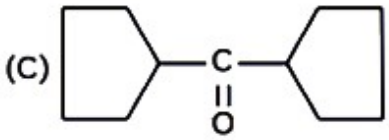


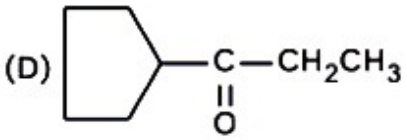
;

Product is :

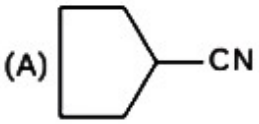
(A) 

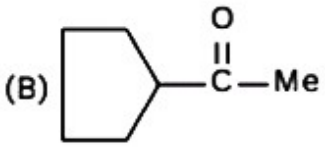
(B) 

(C) 

(D) 

Sol.

(A) 

(B) 

Q70



True about this reaction is/are

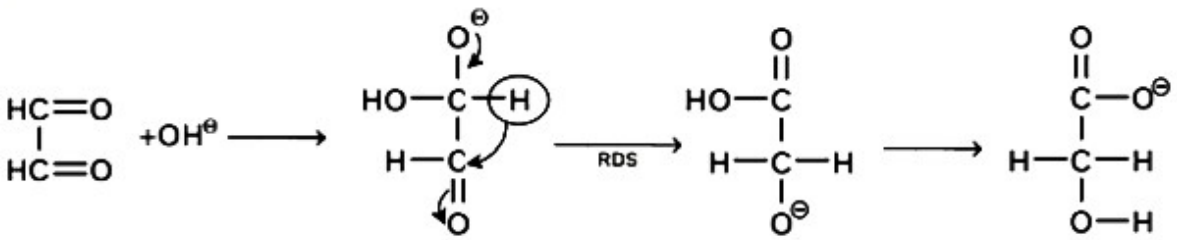
(A) Cannizaro name is associated with this reactoin

(B) It is a disproportion reaction

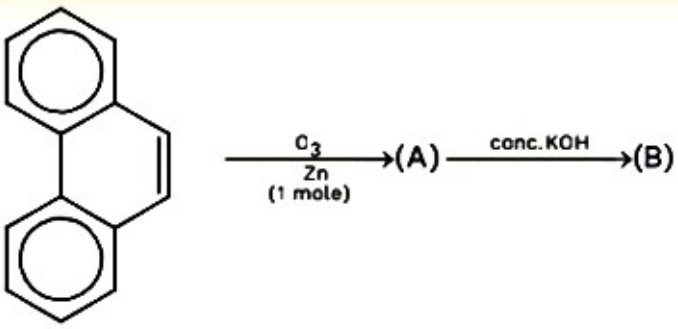
(C) It is a bimolecular reaction in r.d.s.

(D) All of these

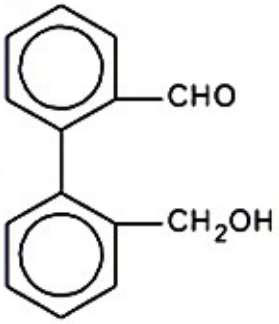
Sol. (A)

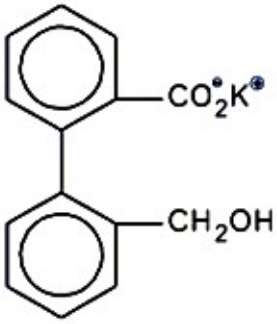


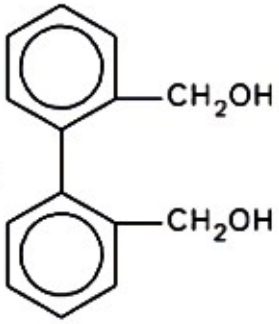
Q71

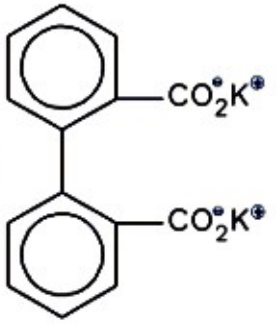


End product (B) 0f above reaction is :

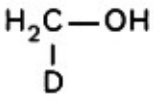
(A)


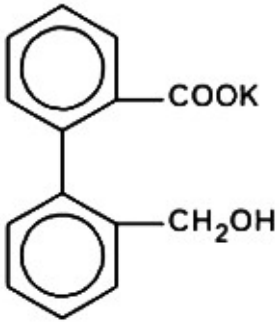
(B)


(C)


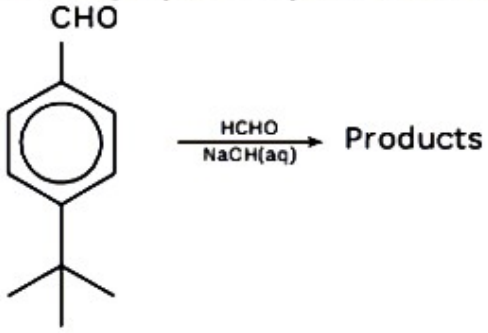
(D)


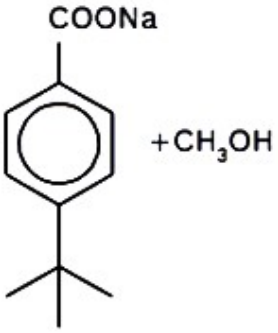
Sol.

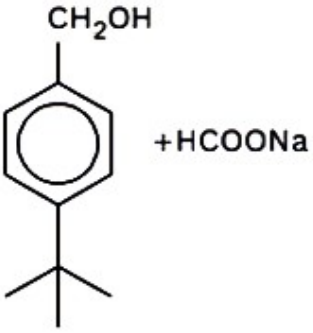
(A)


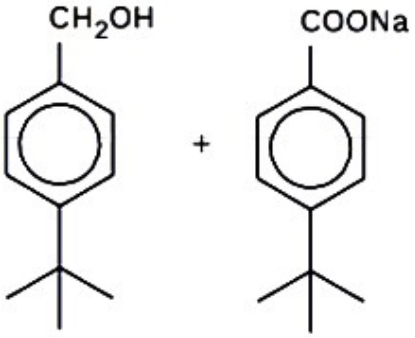
(B)


Q72 **The major product pair of the following reaction will be**



(A) 

(B) 

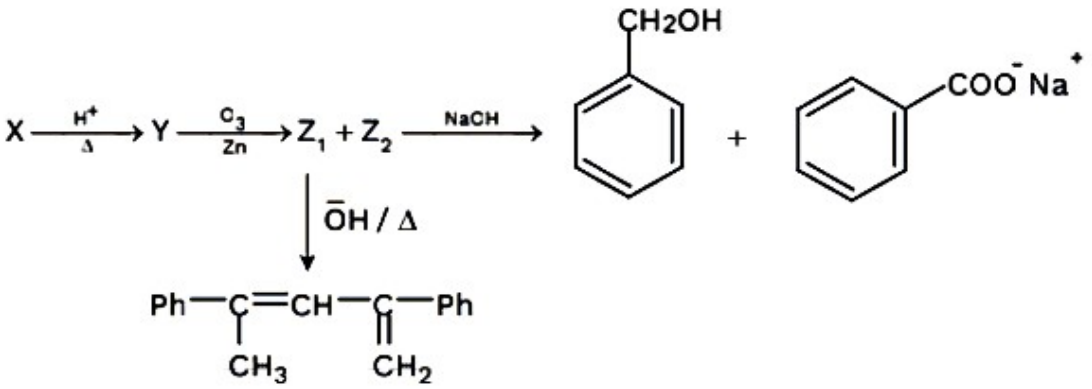
(C) 

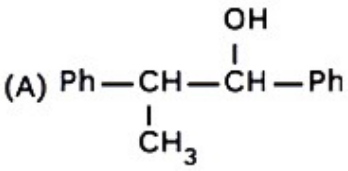
(D) CH_3OH + $HCOONa$

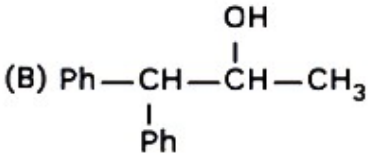
Sol. (B)

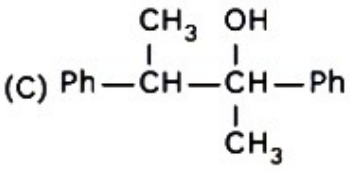
Cross cannizaro reaction

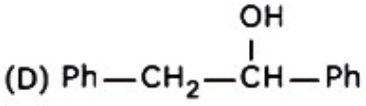
Q73 **Reactant 'X' will be :**



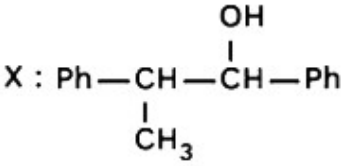
(A) 

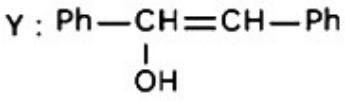
(B) 

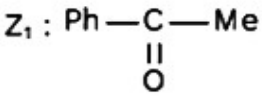
(C) 

(D) 

Sol. (A)

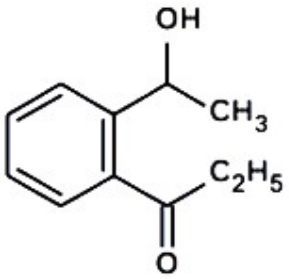
X : 

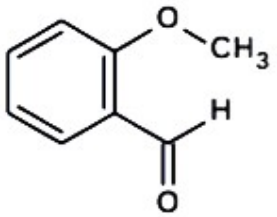
Y : 

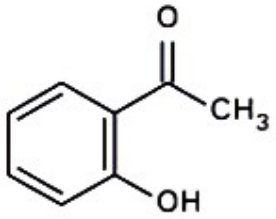
Z_1 : 

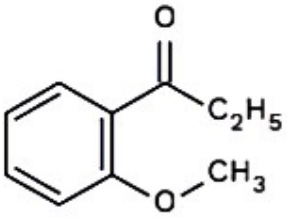
Z_2 : Ph–CH=O

Q74 **An organic compund neither reacts with neutral ferric chloride solution nor with Fehling soltuion, It however, reacts wih Grignard reagent and gives positive iodoform test. The compund is : [JEE-Main (April)-2019]**

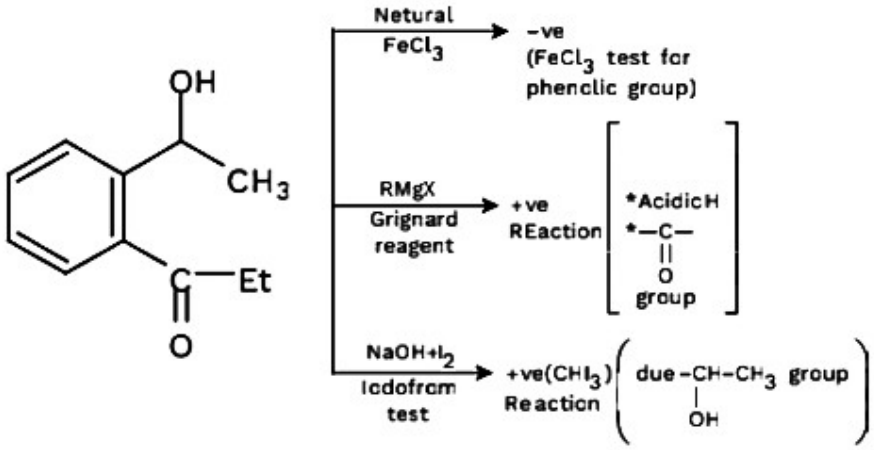
(A) 

(B) 

(C) 

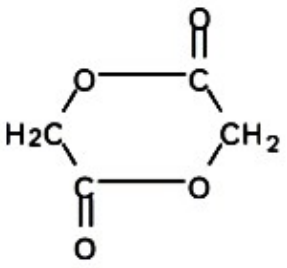
(D) 

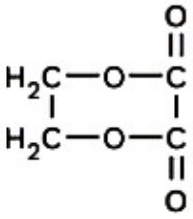
Sol. (A)



Q75 **Cannizaro reaction does not take place with**

(A) $H_2C=CH-CO_2H$

(B) 

(C) 

(D) $H_2C=C=O$

Sol. (D)

Q76 **In the reaction $(CH_3)_3CCHO + HCHO \xrightarrow[\text{heat}]{NaCH} A + B$. The products (A) and (B) are respectively.**

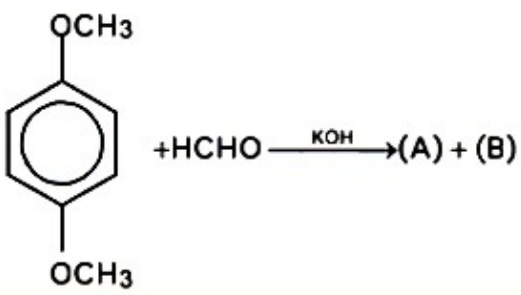
(A) $(CH_3)_3CCH_2OH$ and $HCOO^-Na^+$

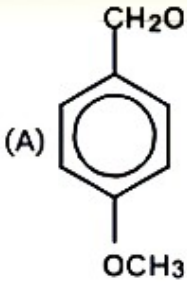
(B) $(CH_3)_3CCOONa$ and CH_3OH

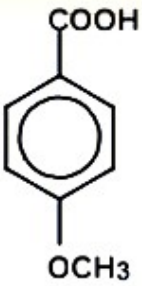
(C) $(CH_3)_3CCH_2OH$ and CH_3OH

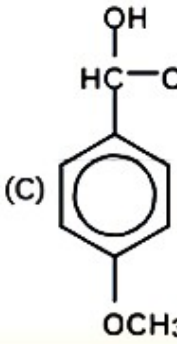
(D) $(CH_3)_3CCOONa$ and $HCOO^-Na^+$

Sol. (A)

Q77 

(A)  + $HCO\overset{\ominus}{O}\overset{\oplus}{K}$

(B)  + CH_3OH

(C)  + $HCO\overset{\ominus}{O}\overset{\oplus}{K}$

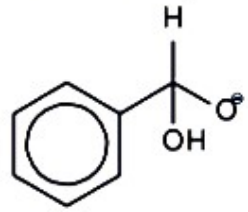
(D) (A) and (B) both

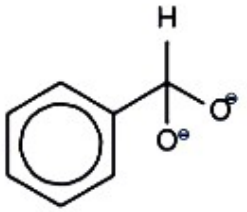
Sol. (A)

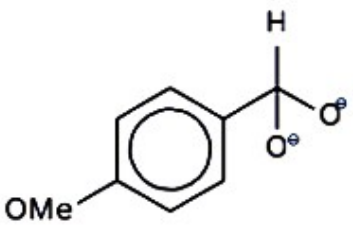
In the given Cannizzaro reaction :

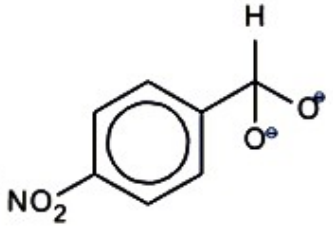
$$-2Ph-CHO \xrightarrow{\overset{\bullet}{O}H} Ph-CH_2OH + PhCO_2^-$$

The slowest step is :

(A) 

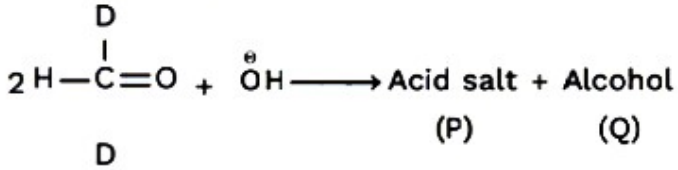
(B) 

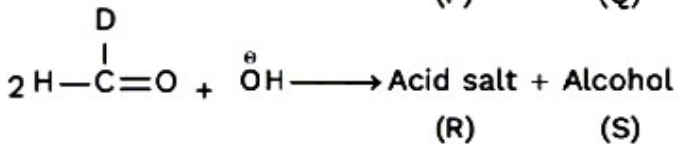
(C) 

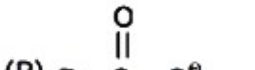
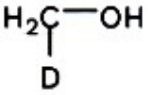
(D) 

(B)

Q79 Identify the products in the following disproportionation reaction :

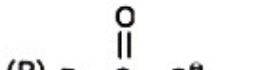




(A) (P)  (Q) $D-CH_2-OH$ (R)  (S) 

(B) (P)  (Q) CH_3OH (R)  (S) $-CH_2-OH$

(C) (P)  (Q) $D-CH_2-OH$ (R)  (S) $D-CH_2-OH$

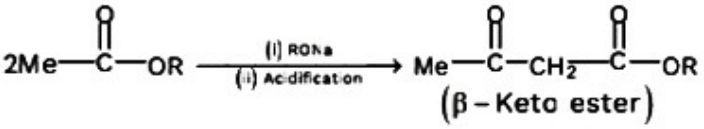
(D) (P)  (Q) $D-CH_2-OH$ (R)  (S) CD_3-OH

Sol. (D)

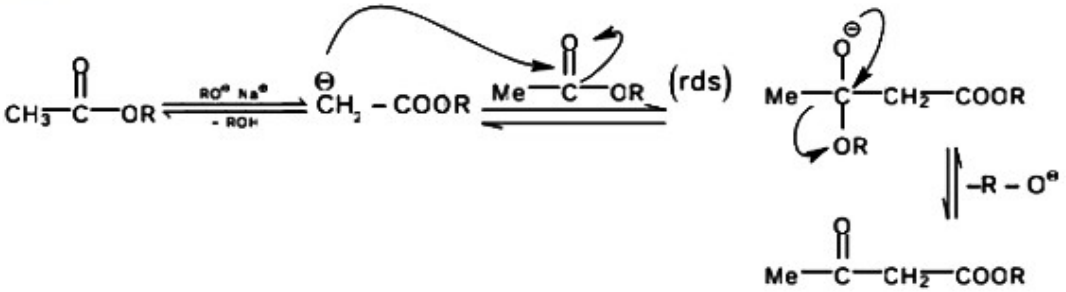
SOME IMPORTANT NAME REACTIONS OF CARBONYL COMPOUND

a) CLAISEN CONDENSATION:

Esters undergo S_NAE Reaction, when attacked by a Nu^- generated by the interaction of a base (usually base related to the Alkoxy anion of ester) with one of the molecules of ester and this Nu^- attacks on another molecule. The reaction over all is considered as condensation of esters known as claisen ester condensation.



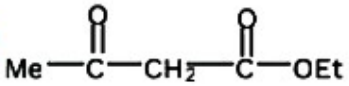
Mechanism



Some times, when two ester groups are present within the molecule then the condensation occurs intramolecular then cyclization caused thus is known as Dieckmann cyclization or Dieckmann's condensation.

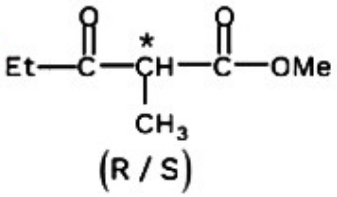
Q80 $MeCOOEt \xrightarrow[(ii)\ Acidification]{(i)\ EtOK} A$

Sol.



Q81 $EtCOOMe \xrightarrow[(ii)\ Acidification]{(i)\ MeOK} B$

Sol.



Q82 $MeCOOMe + EtCOOMe \xrightarrow[\text{(i) Acidification}]{\text{(i) MeOK}} C$

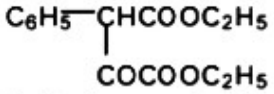
Sol. 6 products

Q83 $\underset{\text{Ethyl benzoate}}{C_6H_5COOC_2H_5} + CH_3COOC_2H_5 \xrightarrow[\text{(ii) Acidification}]{\text{(i) } C_2H_5ONa} D + D'$

Sol. $\underset{\text{Ethyl benzoylacetate}}{C_6H_5COCH_2COOC_2H_5}$, $CH_3COCH_2COOC_2H_5$

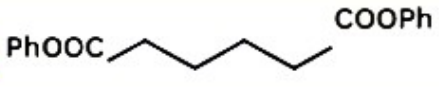
Q84 $\begin{matrix}COOC_2H_5\\ |\\ COOC_2H_5\end{matrix} + C_6H_5CH_2COOC_2H_5 \xrightarrow[\text{(ii) Acidification}]{\text{(i) } C_2H_5ONa} E$

Sol.



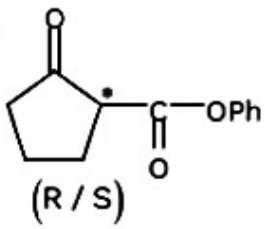
Diethyl-α oxalyphenyl acetate

Q85

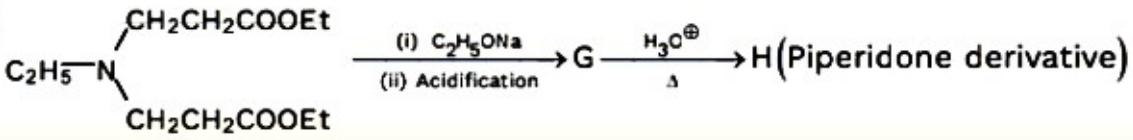


$\xrightarrow[\text{(ii) Acidification}]{\text{(i) PhONa}} F$

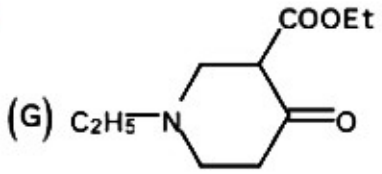
Sol.

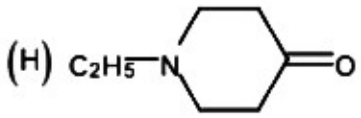


Q86



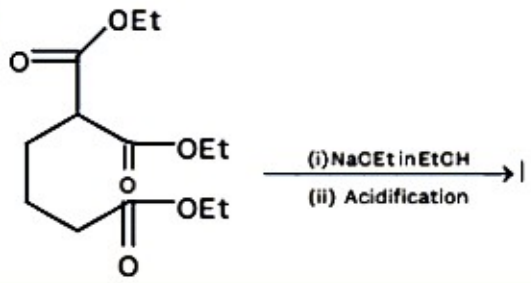
Sol.



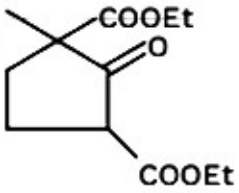


1 – Ethyl – 4 – piperiodone

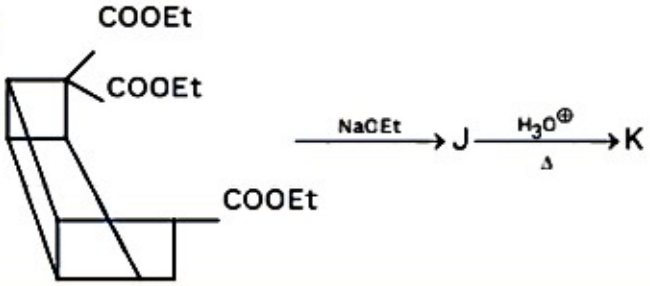
Q87



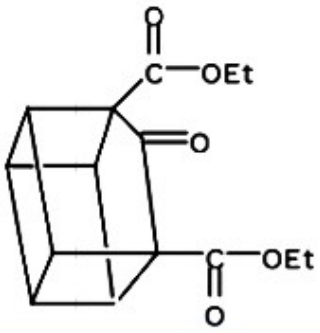
Sol.

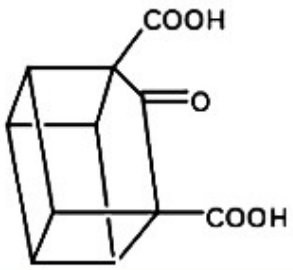


Q88

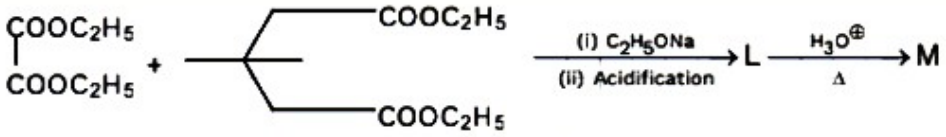


Sol.

(J) 

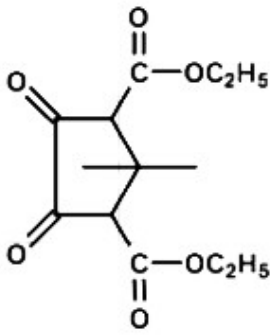
(K) 

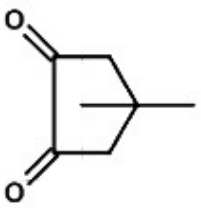
Q89

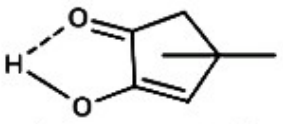


Diethyl – β,β – Dimethyl Glutarate

Sol.

(L) 

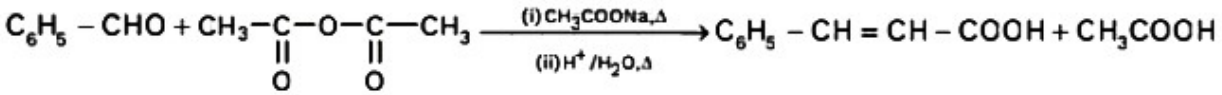
(M) 

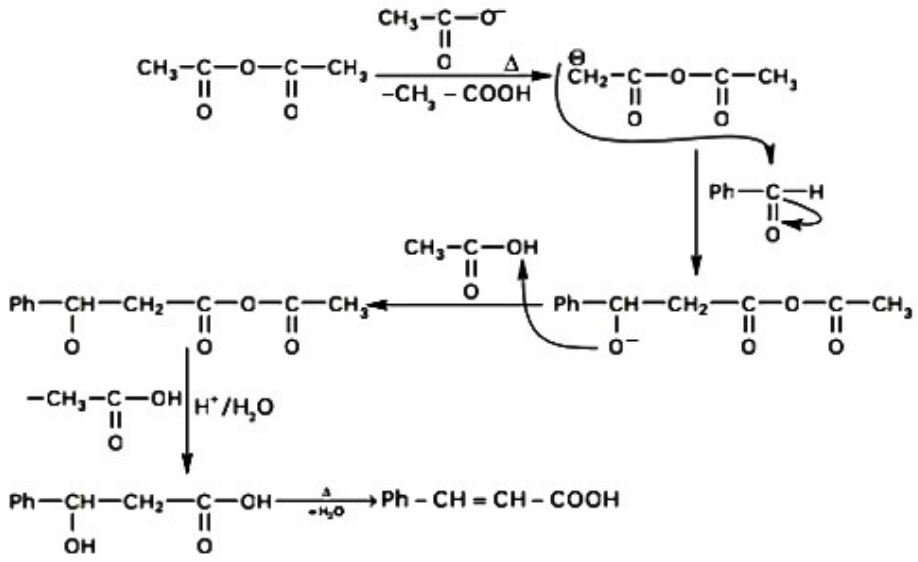
(N) 

enol is more stable

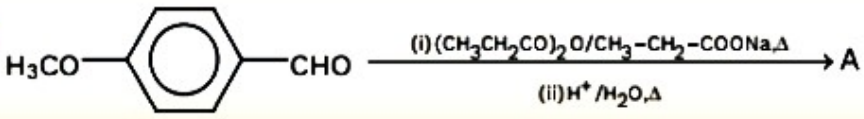
b) PERKIN CONDENSATION :

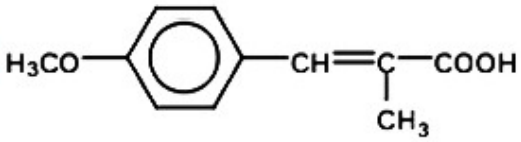
In Perkin reaction, condensation reaction takes place between aromatic aldehydes and aliphatic acid anhydrides in the presence of sodium or potassium salt of corresponding acid of that anhydride, to yield α, βunsaturated aromatic acids.

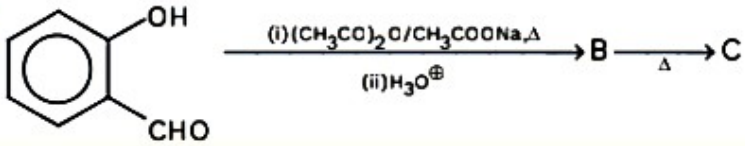


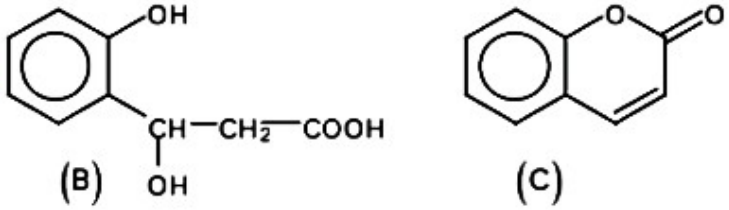


β – Hydroxy acid

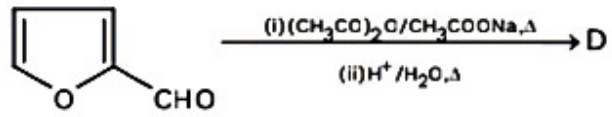
Q90 

Sol. 

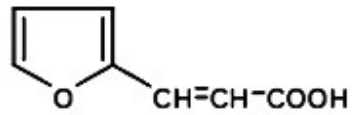
Q91 

Sol. 

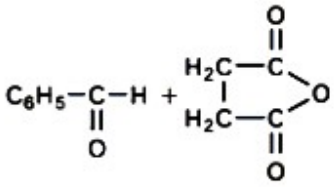
Q92



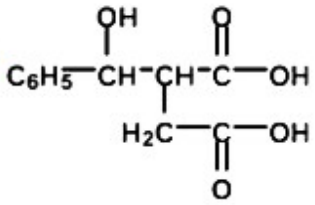
Sol.



Q93

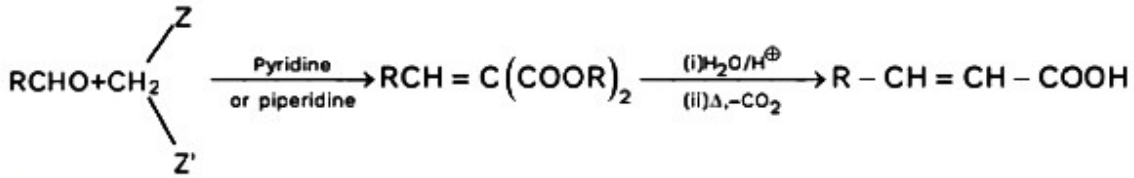


Sol.

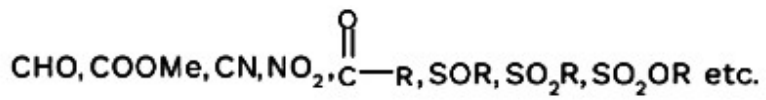


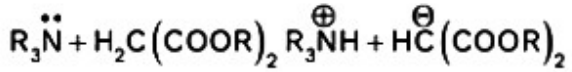
c) KNOEVENAGEL REACTION :

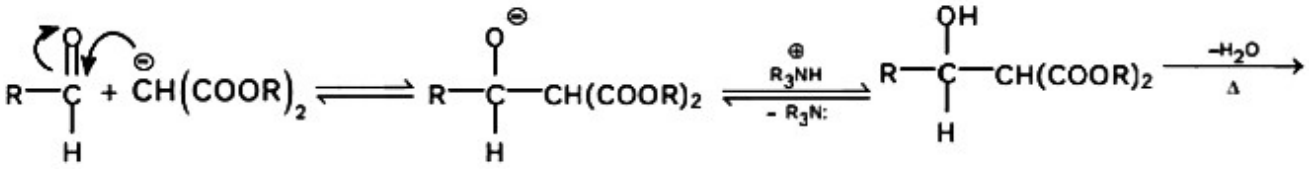
Reaction of active methylene group with aldehyde & Ketones is known as knoevenagel reaction.



Z can be







$$R-CH=C(COOR)_2 \xrightarrow[(ii)\Delta,-CO_2]{(i)H_2O/H^{\oplus}} R-CH=CHCOOH$$

High reactivity of the methylene group of the active methylene compound prevents self-condensation of the aldehyde.

Q94 $CH_3CHO + H_2C(COOR)_2 \xrightarrow[(ii)H_3O^{\oplus},\Delta]{(i)\,Pip} A$

Sol. $CH_3-CH=CH-COOH$
Crotonic acid

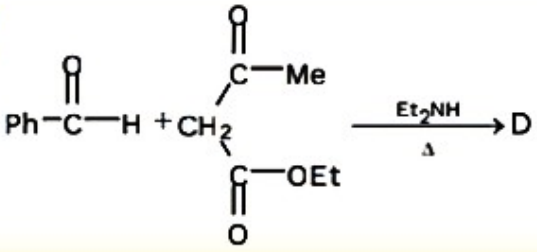
Q95 $C_6H_5CHO + H_2C(COOR)_2 \xrightarrow[(ii)H_3O^{\oplus},\Delta]{(i)\,Pip} B$

Sol. $C_6H_5-CH=CH-COOH$
Cinnamic acid

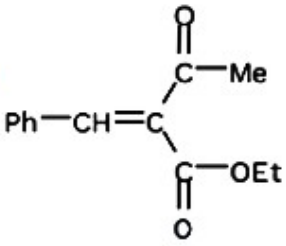
Q96 $\underset{\text{Glyoxalic acid}}{HOOC-CHO} + H_2C(COOR)_2 \xrightarrow[(ii)H_3O^{\oplus},\Delta]{(i)\,Pip} C$

Sol. $HOOC-CHO=CH-COOH$
Malec acid

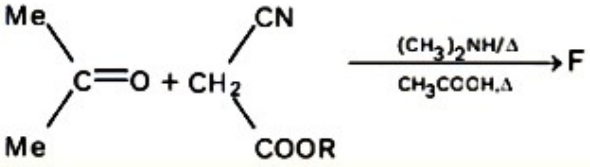
Q97



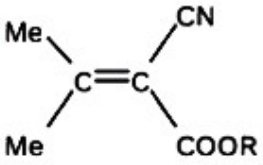
Sol.



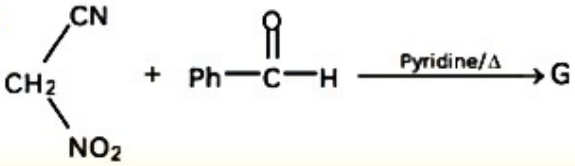
Q98



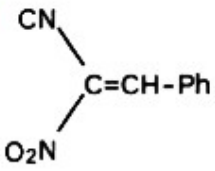
Sol.



Q99

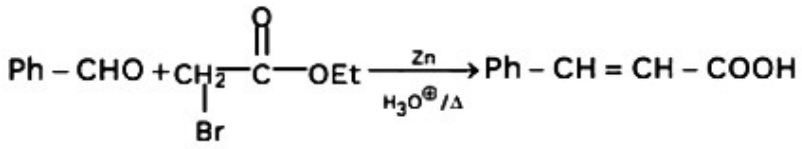


Sol.

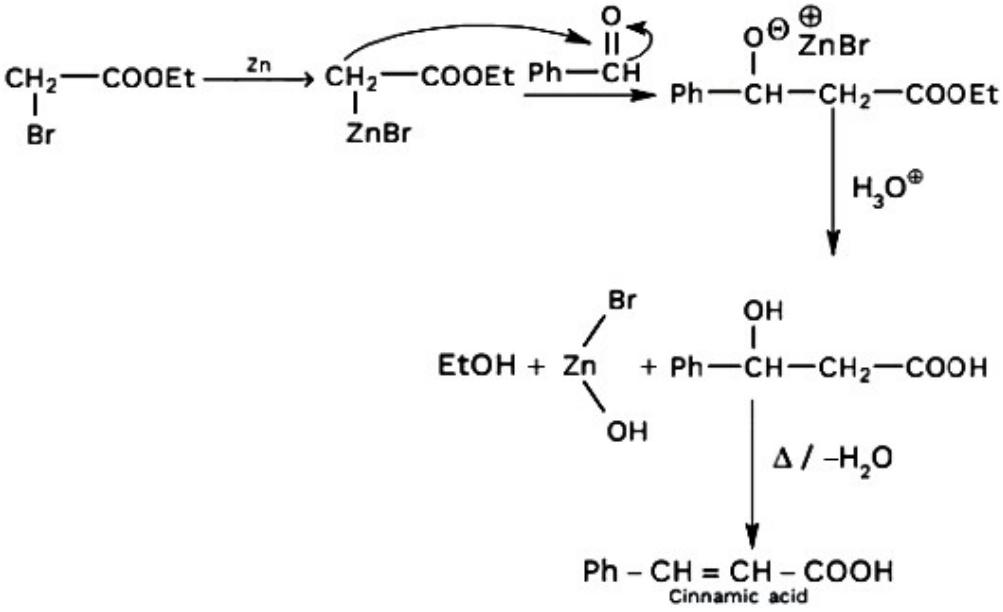


d) REFORMATSKY REACTION :

α-halo esters when treated with Zn it gives organometallic halo ester which provides the attacking $Nu^{\ominus}$ for the another reactant, which is a carbonyl compound. When $Nu^{\ominus}$ attacks on carbonyl compound it gives an intermediate which upon acidic hydrolysis followed by heating, results in formation of α, β-unsaturated acid. The overall reaction is known as Reformatsky reaction.



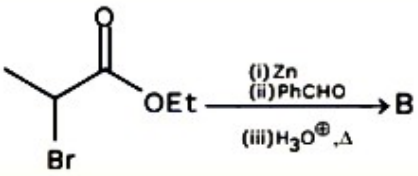
Benzaldehyde — Cinnamic acid

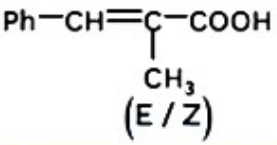


Q100 $C_6H_5COCH_3 + BrCH_2COOEt \xrightarrow[2.\ H_3O^{\oplus},\Delta]{1.\ Zn} A$

Sol. $C_6H_5-C(CH_3)=CH-COOH$ (E / Z)

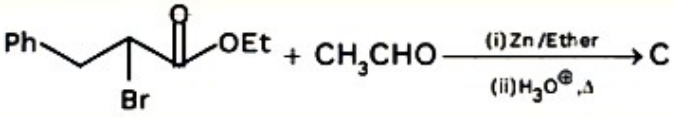
Q101



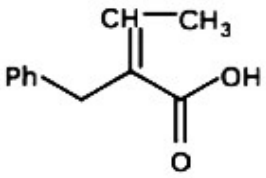
Sol. 

(E / Z)

Q102



Sol.

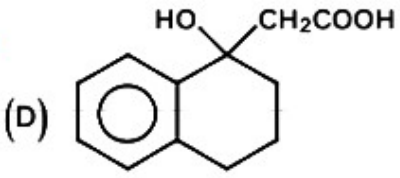


Q103



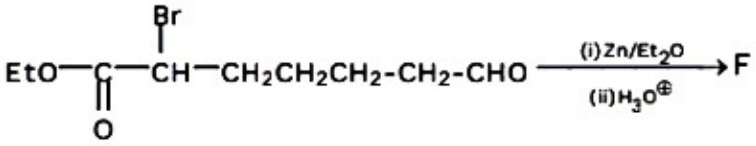
α – Tetralone

Sol.

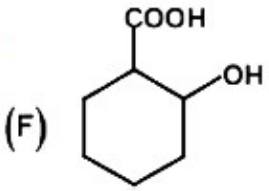


(D)

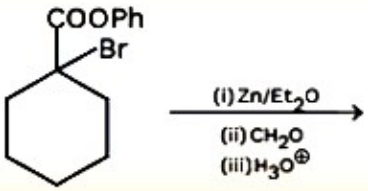
Q104



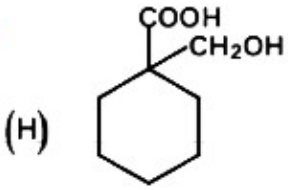
Sol.



Q105

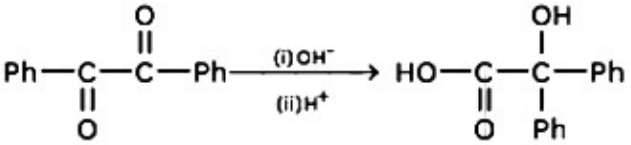


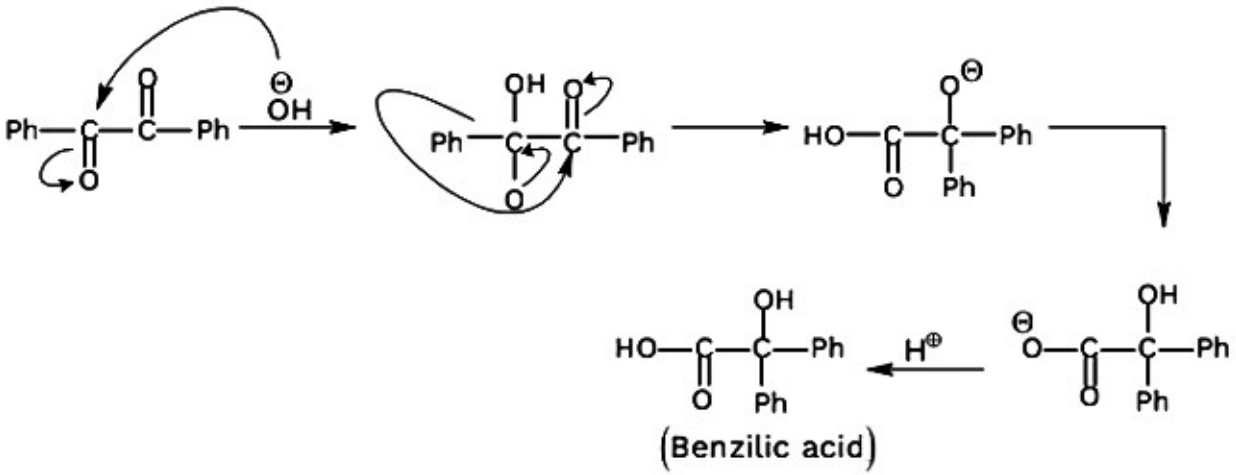
Sol.



e) BENZIL-BENZILIC REARRANGEMENT OR BENZILIC ACID REARRANGEMENT

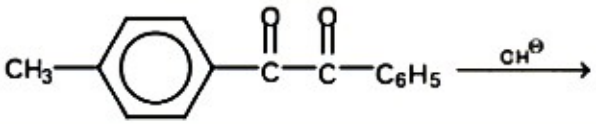
The base catalysed reaction of 1, 2-diketones to a salt of 2-hydroxy carboxylic acid is known as Benzilic acid rearrangement, this reaction generally take place when aryl group is present on both carbonyl carbons.



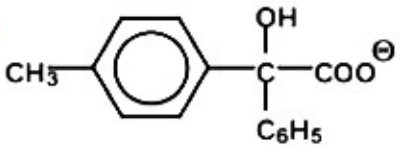


(Benzilic acid)

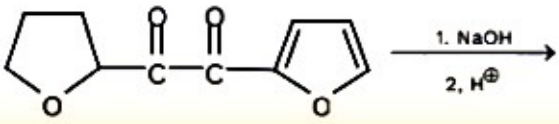
Q106



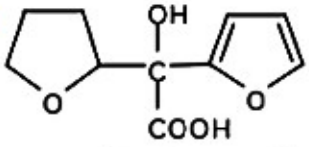
Sol.



Q107

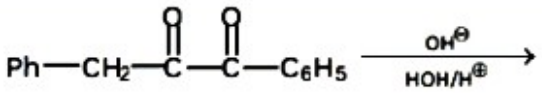


Sol.

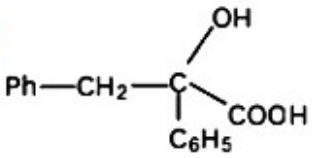


(Furilic acid)

Q108



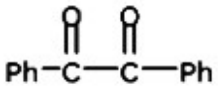
Sol.

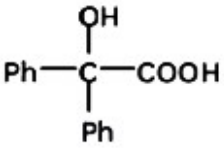


Q109

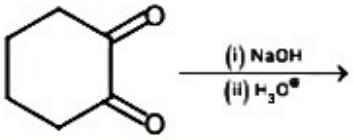


Sol.

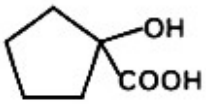
(A) 

(B) 

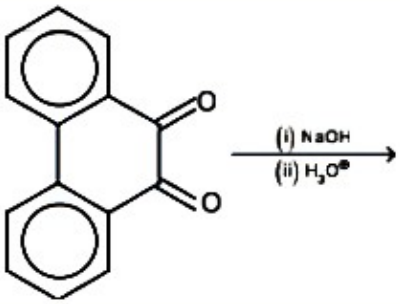
Q110



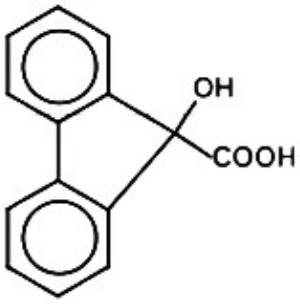
Sol.



Q111

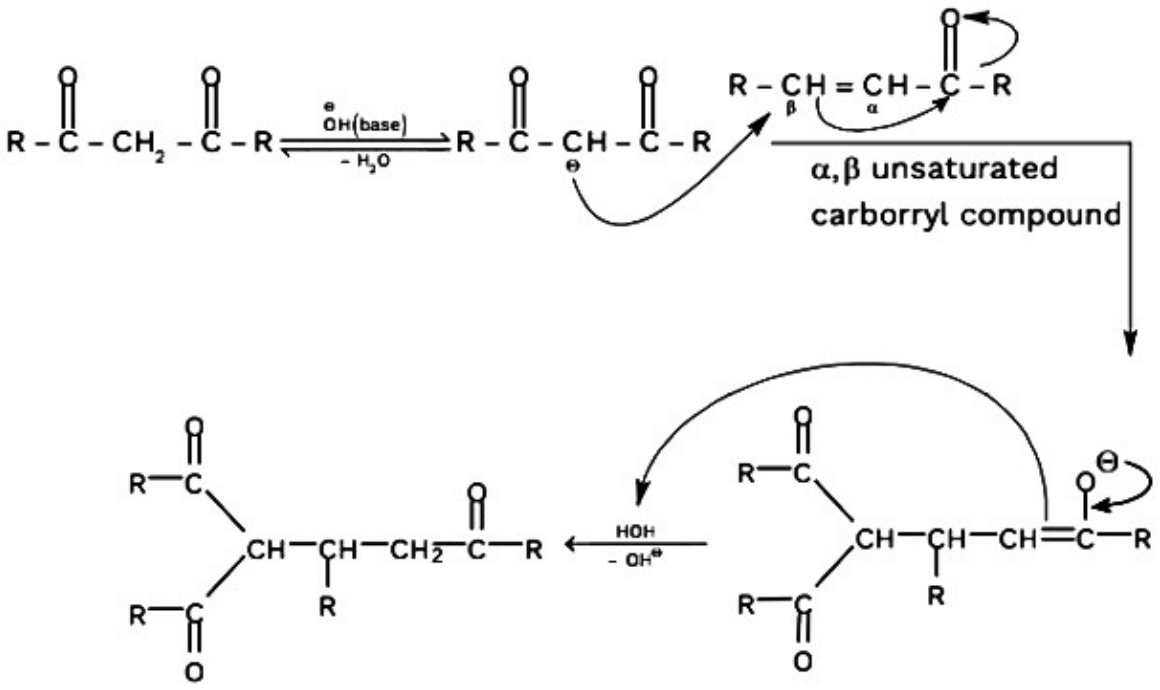


Sol.

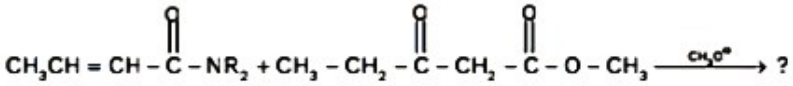


f) **MICHAEL ADDITION :**

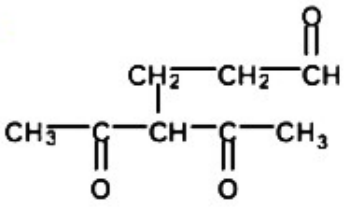
α-β unsaturated carbonyl compound undergo michael reaction with compounds having active methylene. Many different nucleophile can add to α, β unsaturated carbonyl compound. When the nucleophile is an enolate the addition reaction has a special name MICHAEL REACTION.



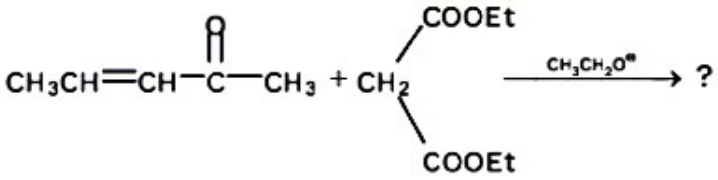
Q112



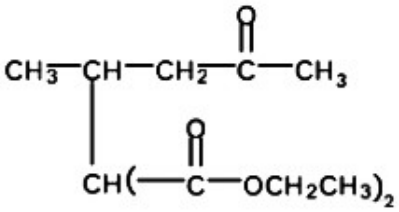
Sol.



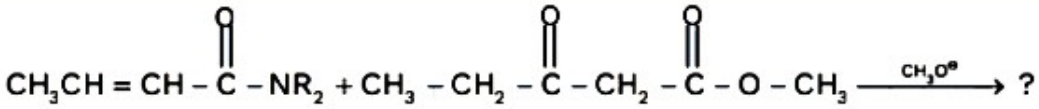
Q113



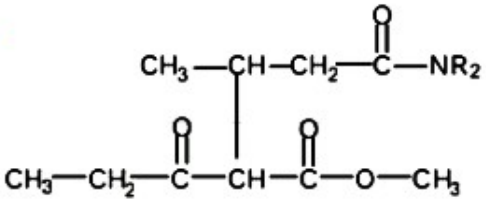
Sol.



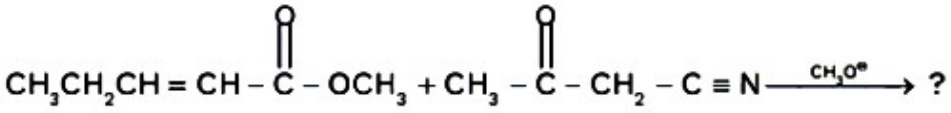
Q114



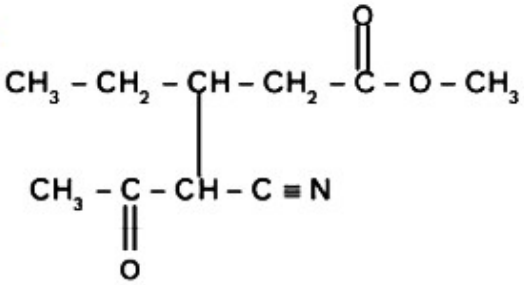
Sol.



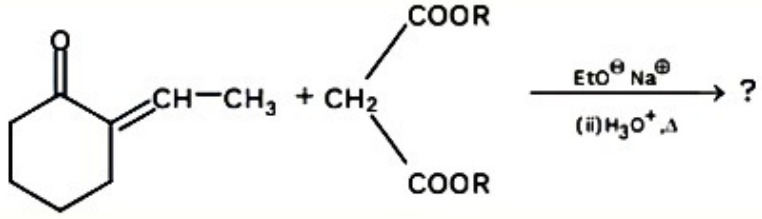
Q115



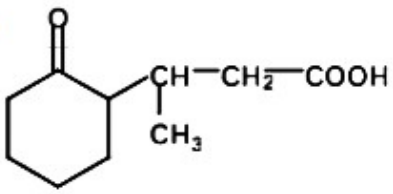
Sol.



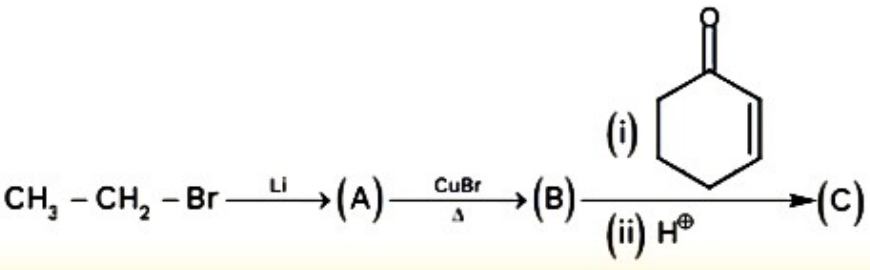
Q116



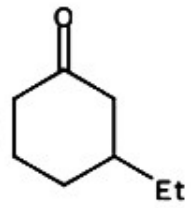
Sol.



Q117



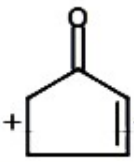
Sol.

(C) 

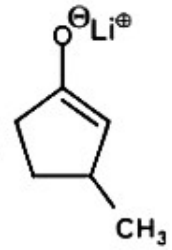
Q118 $(CH_3)_2CuLi + CH_3-CH\overset{O}{\diagdown\diagup}CH_2 \longrightarrow (A) \xrightarrow{H^{\oplus}} (B)$

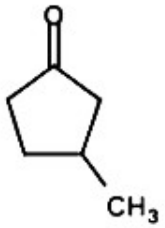
Sol. (A) $CH_3 - \underset{\underset{O^{\ominus}Li^{\oplus}}{|}}{CH} - CH_2 - CH_3$ (B) $CH_3 - \underset{\underset{OH}{|}}{CH} - CH_2 - CH_3$

Q119

$(CH_3)_2CuLi +$  $\longrightarrow (A) \xrightarrow{H^{\oplus}} (B)$

Sol.

(A) 

(B) 

Q120 $(CH_3)_2CuLi + CH_2=CH-\underset{\underset{O}{\|}}{C}-H \longrightarrow (A) \xrightarrow{H^{\oplus}} (B)$

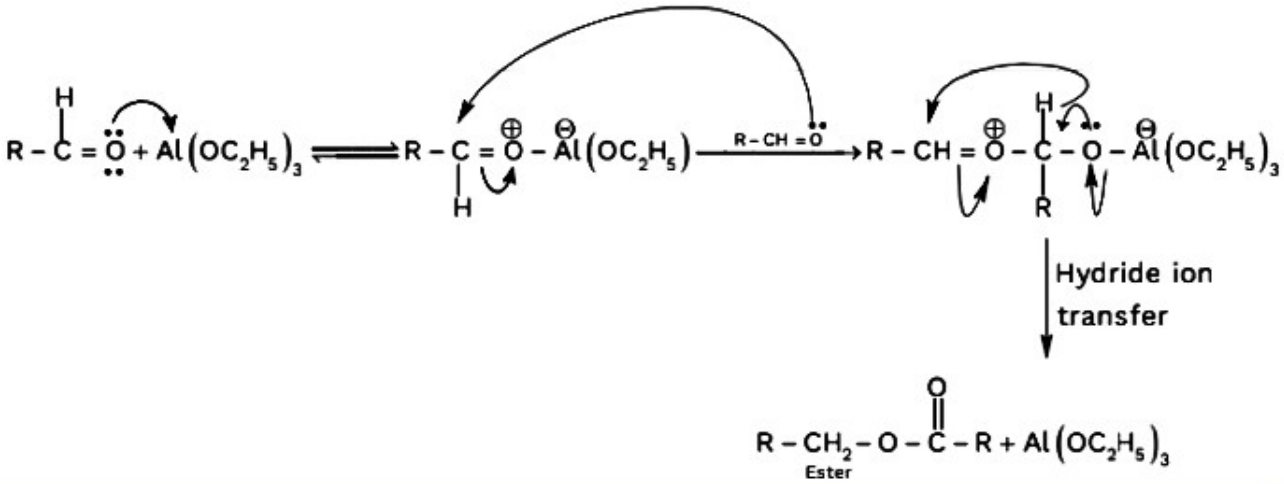
Sol. (A) $CH_3-CH_2-CH=\underset{\underset{{}^{\ominus}OLi^{\oplus}}{|}}{C}-H$ (B) $CH_3-CH_2-CH_2-\underset{\underset{O}{\|}}{C}-H$

g) TISCHENKO REACTION / TISCHENKO CONDENSATION :

(1) This reaction takes place between two molecules of aldehydes.
(2) Reaction is catalysed by aluminium ethoxide.
(3) This is a two-step reaction, i.e., redox reaction followed by ester formation.

Thus this reaction is extension of Cannizzaro reaction.

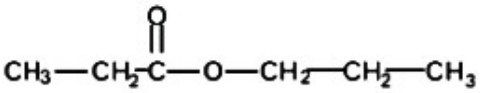
$$R-CHO + R-CHO \xrightarrow[\Delta]{(C_2H_5O)_3Al} R-CH_2-O-\overset{\overset{O}{\|}}{C}-R$$



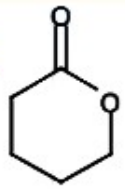
Q121 $CH_3-CHO \xrightarrow[\Delta]{(C_2H_5O)_3Al} ?$

Sol. $CH_3-\overset{\overset{O}{\|}}{C}-O-C_2H_5$

Q122 $CH_3 - CH_2 - CHO \xrightarrow[\Delta]{(C_2H_5O)_3Al} ?$

Sol. 

Q123 $OHC - CH_2 - CH_2 - CH_2 - CHO \xrightarrow[\Delta]{(C_2H_5O)_3Al} ?$

Sol. 

Q124 $CH_3 - CH_2 - CH_2 - CH_2 - CHO \xrightarrow[\Delta]{(C_2H_5O)_3Al} ?$

Sol. $CH_3 - CH_2 - CH_2 - CH_2 - \underset{\underset{O}{\|}}{C} - OCH_2 - CH_2 - CH_2 - CH_2 - CH_3$

Q125 **In given reaction which of the following products is / are possible ?**

$$Me - \underset{\underset{O}{\|}}{C} - H + Et - \underset{\underset{O}{\|}}{C} - H \xrightarrow{Al(OC_2H_5)_3}$$

(A) $Me - \underset{\underset{O}{\|}}{C} - O - Pr$

(B) $Me - \underset{\underset{O}{\|}}{C} - O - Me$

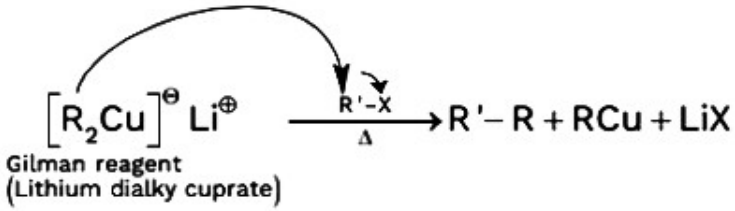
(C) $Et - \underset{\underset{O}{\|}}{C} - O - Et$

(D) $Et - \underset{\underset{O}{\|}}{C} - O - Pr$

Sol. A, C, D

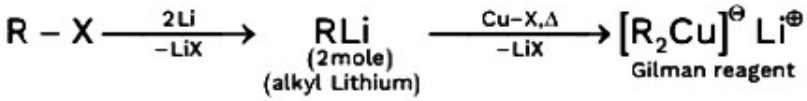
h) COREY HOUSE SYNTHESIS :

Reaction of Gilman's reagent with alkyl halide gives alkane as one of the product which is known as Corey House synthesis,



To obtain good yield of alkane, the alkyl halide R'–X must be either a methyl halide, a primary alkyl halide, or a secondary cycloalkyl halide.(Best CH_3X, if possible)

Preparation of Gilman reagent :



Complete the following reaction sequence :

Q126 $CH_3-I \xrightarrow[Et_2O]{Li} (A) \xrightarrow[\Delta]{CuI} (B) \xrightarrow[\Delta]{CH_3CH_2CH_2CH_2CH_2I} (C)\, CH_3CH_2CH_2CH_2Br \xrightarrow[Et_2O]{Li}$

$(D) \xrightarrow[\Delta]{CuI} (E) \xrightarrow[\Delta]{CH_3CH_2CH_2CH_2CH_2Br} (F)$

Sol. (A) CH_3Li,

(B) $(CH_3)_2CuLi$,

(C) $CH_3-CH_2CH_2CH_2CH_2CH_3$,

(D) $CH_3CH_2CH_2CH_2Li$,

(E) $(CH_3CH_2CH_2CH_2)_2CuLi$,

(F) $CH_3CH_2CH_2CH_2-\ CH_2CH_2CH_2CH_2CH_3$

Q127 $(A) \xrightarrow[h\upsilon]{Cl_2} (B) \xrightarrow{Li} (C) \xrightarrow[\Delta]{CuCl} (D) \xrightarrow[\Delta]{CH_3Cl} CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2-CH_3$

Sol. (A) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_3$

(B) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2-Cl$

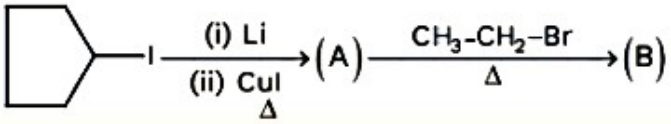
(C) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2Li$

(D) $\left[CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2\right]^{\ominus}_2 CuLi^{\oplus}$

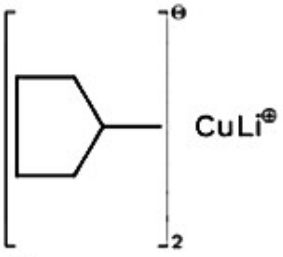
Q128 $CH_3-CH_2-Br \xrightarrow[(ii)\,CuBr,\,\Delta]{(i)\,Li} (A) \xrightarrow[\Delta]{CH_3-Cl} (B)$

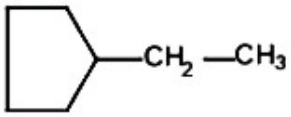
Sol. (A) $[CH_3CH_2]^{\ominus}_2CuLi^{\oplus}$

(B) $CH_3-CH_2-CH_3$

Q129 

Sol.

(A) 

(B) 

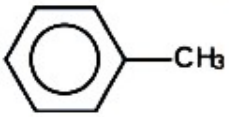
Q130 $CH_3-CH_2-Br \xrightarrow{Li} (A) \xrightarrow[\Delta]{CuBr} (B) \xrightarrow{CH_3-CH_2-Cl} C$

Sol. (A) $CH_3-\overset{\ominus\delta}{C}H_2\,\overset{\oplus\delta}{Li}$

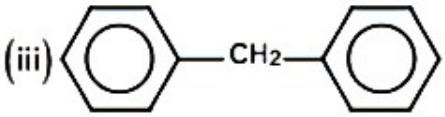
(B) $[CH_3CH_2]^{\ominus}_2CuLi^{\oplus}$

(C) $CH_3-CH_2-CH_2-CH_3$

Q131 **How can you prepare following compounds using corey house synthesis.**

(i) 

(ii) $CH_2 = CH - CH_2 - CH_3$

(iii) 

Sol. (i) $[Ph]_2^{\ominus}CuLi^{\oplus} \xrightarrow{CH_3-Cl} Ph - CH_3$

(ii) $\left[CH_2 = CH - CH_2\right]_2^{\ominus} CuLi \xrightarrow{CH_3 - Cl} CH_2 = CH - CH_2 - CH_3$

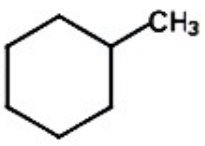
(iii) $\left[Ph\right]_2^{\ominus} CuLi^{\oplus} \xrightarrow{Ph - CH_2 - Cl} Ph - CH_2 - Ph$

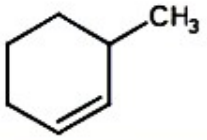
Q132 **Which of the combination of Gilman's reagent & alkylhalide will be most suitable for preparation of following compounds.**

(i) $CH_3 - CH_2 - CH_3$

(ii) $CH_3 - \underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}} - CH_3$

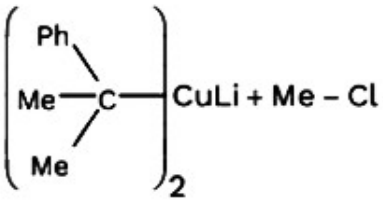
(iii) $CH_3 - \underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}} - Ph$

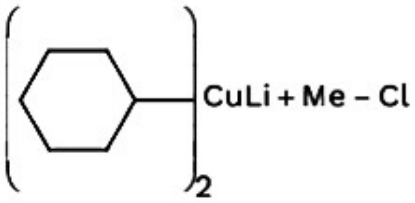
(iv) 

(v) 

Sol. (i) $[CH_3 - CH_2]_2^{\ominus}CuLi^{\oplus} + CH_3 - Cl$

(ii) $[Me_3C]_2^{\ominus}CuLi^{\oplus} + CH_3 - Cl$

(iii) 

(iv) 

(v) 